

BASE24

EPS-Net INTERCHANGE

INTERFACE MANUAL

RELEASE 6.0



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PREFACE

Throughout this manual, the term EPS-Net describes the network operated by MasterCard Europe for the transmission of transactions. Within EPS-Net, there are endpoints to which each transaction issuer or acquirer connects; these are known as European Modules (EMs).

The BASE24 EPS-Net Interchange Interface is a BASE24 product that provides an interface between a BASE24 acquirer or issuer system and EPS-Net. The term EPS-Net Interchange Interface or EPS-Net Interchange Interface is used to describe the BASE24 process that interfaces with an EM.

The BASE24 EPS-Net Interchange Prefix File (IPF) Refresh module, associated with the BASE24 EPS-Net Interchange Interface, creates a file containing all MasterCard Europe BINs, allowing an acquiring BASE24 system to route transaction messages to the MasterCard Europe network.

Intended Audience

This document is primarily intended for staff responsible for the configuration, operation or maintenance of the BASE24 EPS-Net Interchange Interface. Users are expected to be familiar with the principles of BASE24 configuration, its operation and certain functions. Users should consult the relevant MasterCard Europe, HP NonStop (formerly Tandem), or BASE24 documentation where appropriate.

This document can be used as an aid to training but readers are advised to attend the relevant training courses.

Software

This manual documents standard processing for the Release 6.0 BASE24 EPS-Net Interchange Interface.

Custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Location of Information

A list of section headings and related page references are given in the *Contents* page. The headings are function oriented.

Section 1, “Introduction,” provides an introduction to the purpose and functionality of the BASE24 EPS-Net Interchange Interface, including the MasterCard Europe Network.

Section 2, “Processing Overview,” provides an overview of the functionality of the EPS-Net Interchange Interface, plus the EPS-Net IPF Refresh module.

Section 3, “Configuration and Files Maintenance,” describes the configuration requirements specific to the EPS-Net Interchange Interface.

Section 4, “Startup and Control,” documents the network control commands supported by the EPS-Net Interchange Interface and describes the network management message flow.

Section 5, “Transaction Handling,” describes the different transaction message flows between the European Module and the BASE24 EPS-Net Interchange Interface and provides further information on transaction processing.

Section 6, “File Action Message Handling,” describes the file action message flows between an Issuer Host System and the EPS-Net Interchange.

Section 7, “Cutover and Reporting,” documents the procedures for settlement with MasterCard Europe and describes the configuration of standard BASE24 interchange reporting.

Section 8, “EMV Support,” describes the support within the BASE24 EPS-Net Interchange Interface for EMV transaction processing.

Section 9, “EPS-Net IPF Refresh,” describes the loading of the full positive list of all bank identification numbers (BINs) sent to EPS-Net for authorisation.

Appendix A, “Response Code Translation,” presents tables describing the translation of response codes between BASE24-pos and EPS-Net and between BASE24-atm and EPS-Net.

Appendix B, “File Formats,” define the data structures used by the EPS-Net Interchange Interface.

Appendix C, “Log Messages,” describes the messages logged by the BASE24 EPS-Net Interchange Interface and provides guidance on their causes and the appropriate action to take.

Appendix D, “Glossary,” explains the definitions of the terms used in this manual.

Related Documentation

BASE24 Documentation

- *NET24-XPNET Network Control Operations Guide*
- *NET24-XPNET Network Control Command Reference Manual*
- *BASE24 External Message Manual*
- *BASE24 Files and Record Structures Reference Manual*
- *BASE24 Files Maintenance Manual*
- *BASE24 Logical Network Configuration File Manual*
- *BASE24 Tokens Manual*
- *BASE24 User Security Manual*
- *BASE24 Transaction Security Manual*
- *BASE24 Transaction Security Services Processing Guide*
- *BASE24-atm Settlement and Reporting Manual*
- *BASE24-atm Transaction Processing Manual*
- *BASE24-atm EMV Support Manual*
- *BASE24-pos Settlement and Reporting Manual*
- *BASE24-pos Transaction Processing Manual*
- *BASE24-pos EMV Support Manual*

MasterCard Europe Documentation

- *V5 Interface Specifications, November 2007*
- *Authorisation System Release 07.2, Member Specifications, June 2007*
- *PIN Management Services, Service Description, July 2002*
- *Host-EM Programmer Specifications for UK Domestic Maestro, June 2007*
- *Authorisation System Release 07.2, Member Specifications for UK Domestic Maestro, May 2007*
- *UK Intracountry ATM Enhancements Service Description, November 2007*

Other Documentation

- *ISO Standard for Bank Card Originated Messages, ISO 8583-1987*
- *ISO Standard for Bank Card Originated Messages, ISO 8583-1993*
- *Codes for the Representation of Currencies and Funds, ISO 4217-1995*
- *EMV ICC Specifications for Payment Systems, version 4.1, May 2004*

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1 INTRODUCTION

The BASE24 EPS-Net Interchange Interface is a BASE24 switch product that provides an interface between a BASE24 acquirer or issuer system and EPS-Net, or more precisely, a European Module (EM).

The BASE24 EPS-Net Interchange Prefix File (IPF) Refresh module, associated with the BASE24 EPS-Net Interchange Interface, creates a file containing all MasterCard Europe BINs, allowing an acquiring BASE24 system to route transaction messages to the MasterCard Europe network.

1.1 BASE24 SYSTEM ARCHITECTURE

The BASE24 EPS-Net Interchange Interface connects a BASE24-atm or BASE24-pos system with the MasterCard Europe network.

The BASE24 EPS-Net Interchange Interface is compliant with the *V5 Interface Specifications*, up to and including the MasterCard Europe Authorisation System Release 07.2.

1.1.1 System Overview

The following figure provides an example of a BASE24 EPS-Net Interchange Interface configuration.

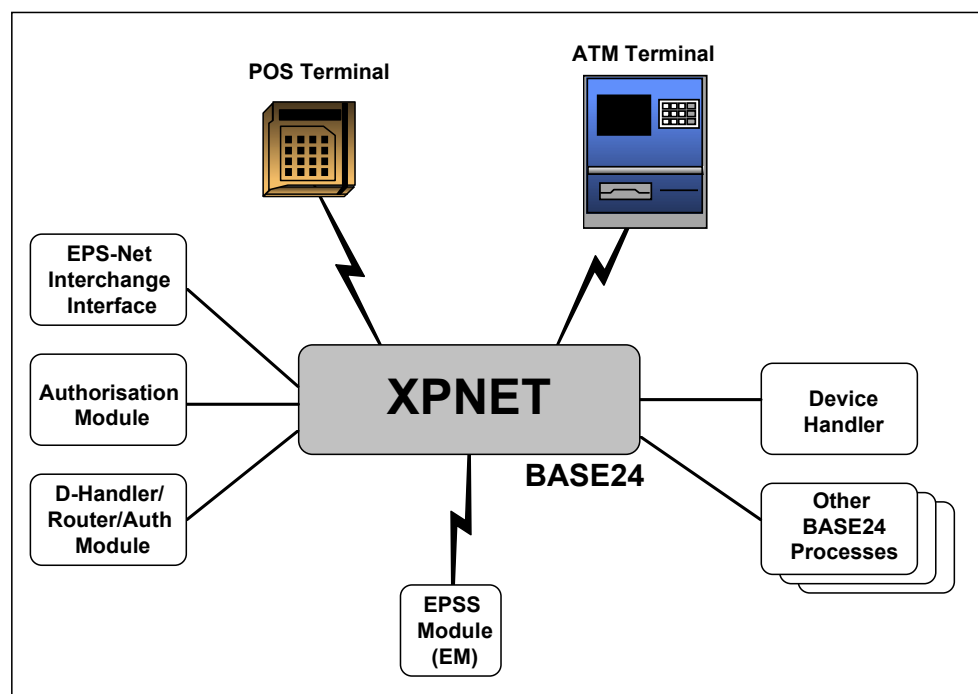


Figure 1-1 System Overview

1.1.2 Issuer And Acquirer Support

In the financial industry, the terms *issuer* and *acquirer* define the parties on opposite ends of the electronic message exchanges required to authorise and process transactions.

- *Issuers* are card-issuing or account-owning institutions or their agents. These institutions or agents are responsible for authorising cardholder transactions and keeping track of cardholder accounts.
- *Acquirers* are the parties initiating transactions on behalf of the card acceptors.

The BASE24 EPS-Net Interchange Interface can function in the role of an issuer, an acquirer, or both. When the BASE24 EPS-Net Interchange Interface sends financial transactions to EPS-Net for authorisation, the BASE24 EPS-Net Interchange Interface is representing the transaction acquirer, even though the transaction may not have originated at a BASE24-controlled terminal. When financial transactions are being sent to the BASE24 EPS-Net Interchange Interface for authorisation from EPS-Net, the BASE24 EPS-Net Interchange Interface is representing the card issuer, even though BASE24 may only forward the transaction to an issuing system for authorisation.

1.1.3 Transaction Routing

The ATM Authorisation process and the POS Router module use the SPROUTE file for routing transactions to the appropriate interchange interface. One of the files used to create SPROUTE records is the Interchange Prefix File (IPF), which contains card prefixes supported by individual authorisation networks.

The EPS-Net IPF Refresh module therefore allows an IPF to be created, containing all prefixes supported by the MasterCard Europe network. This file is used in turn by the ATM Authorisation process and the POS Router module to create the SPROUTE file.

To support POS advice messages to be sent under a 'bilateral' agreement, the EPS-Net acquirer must configure CPF/IDF routing for the EPS-Net issuer.

1.2 EPS-NET SYSTEM ARCHITECTURE

EPS-Net is a telecommunications network operated by MasterCard Europe. The network allows communications between MasterCard Europe Members for transaction authorisation and communications with the MasterCard networks for authorisation of European acquired MasterCard financial transactions.

1.2.1 Network Overview

The following figure provides an example of a configuration involving the MasterCard Europe network, BASE24 and a back-end host.

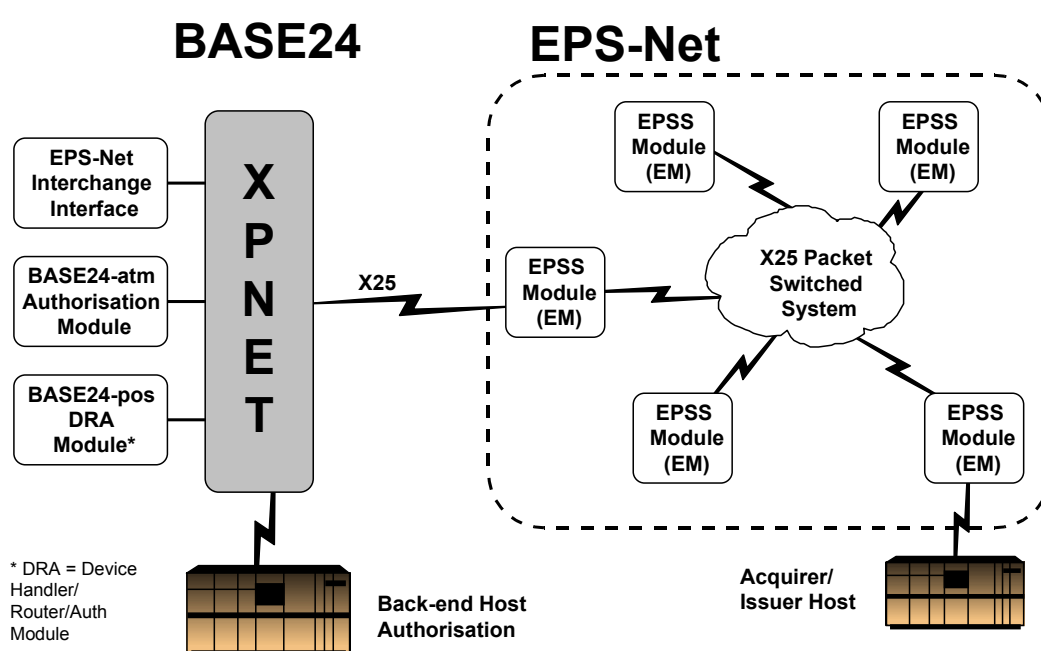


Figure 1-2 Network Overview

The data processing system operated by (or acting on behalf of) the acquirer (and providing these acquirer processing facilities) is referred to as the Acquirer Host System ('AHS'). The data processing system operated by (or acting on behalf of) the issuer (and providing these issuer processing facilities) is referred to as the Issuer Host System ('IHS'). Both the AHS and IHS are connected to EPS-Net via 'European Modules' (EMs). The online interface between the AHS and its EM, and between the IHS and its EM, is known as the 'V5 Interface'.

1.2.2 European Module

The EPS-Net network consists of multiple European Modules (EMs) connected using X.25 or TCP/IP. Each EM runs software provided by MasterCard Europe. The EM can be based at the Member's own site or, by agreement with MasterCard Europe, use can be made of an EM at another MasterCard Europe Member's site.

The EM is responsible for:

- Transaction routing
- Checking message integrity
- Providing signon/signoff access to the network
- Performing currency conversion
- Logging transactions for subsequent reporting and accounting

The EM also verifies settlement data provided from the Host Issuer or Acquirer system for onward transmission to the MasterCard Global Clearing Management System (GCMS).

The Host-EM Interface can provide alternate and down processing options if required by the MasterCard Europe Member:

- Alternate processing enables transactions to be re-routed by the EM to another pre-determined Issuer EM for authorisation.
- Down processing provides a method of authorisation on the EM using files held for checking activity limits, account ranges and various down processing options. Down processing is performed at the transaction acquirer's EM when the Issuer and alternate authoriser are both unavailable for transaction authorisation (assuming prior business agreements have been reached between the Issuer and Acquirer for down processing).

1.2.3 Authorisation and Settlement

Authorisation of transactions is performed using online authorisation messages through EPS-Net. An authorisation request is sent from the acquirer to the issuer, and the issuer returns an authorisation response to the acquirer. The Host-EM authorisation message protocol is based on the ***ISO Standard for Bank Card Originated Messages, ISO 8583-1987***.

Settlement of transactions is performed using the MasterCard Global Clearing Management System (GCMS). All the settlement transactions received before 00:00 (GMT) are settled the next bank working day.

1.3 MESSAGE PROCESSING

For a full description of the online (Host-EM) interface between a member's host system and the MasterCard Europe network, please refer to the *V5 Interface Specifications*.

1.3.1 Message Types

The Host-EM Interface supports the following message types:

- 0100 Authorisation request
- 0110 Authorisation response
- 0120/0121 Authorisation advice/Authorisation advice repeat
- 0130 Advice acknowledgment
- 0420/0421 Reversal advice/Reversal advice repeat
- 0430 Reversal acknowledgment
- 0800 Network management request
- 0810 Network management response
- 1304 File action request
- 1314 File action response

1.3.2 Transaction Types

The Host-EM Interface supports the following transaction types:

- Debit transaction; purchase of goods and services
- Debit transaction; ATM or debit card cash withdrawal
- Debit transaction; Purchase of goods and services with cashback
- Debit transaction; credit card cash advance
- Credit transaction; payment deposit
- Non-financial transaction; PIN unblock
- Non-financial transaction; PIN change
- Non-financial transaction; balance inquiry (domestic only)

1.3.3 Message Integrity

MasterCard Europe assigns the following identification numbers, so that MasterCard Europe and an Acquirer or Issuer can verify the identity of the other:

- **Acquiring Institution Identification Number (AID)**

Identifies the Acquiring Host System (AHS) that generated the transaction to the EM. It is sent in all transaction requests to the EM. The EPS-Net Interchange Interface obtains the value from the Interchange Configuration File (ICF) or the Eurocard Institution Identification File (ERIIF) or the BASE24 internal message.

- **Forwarding Institution Identification Number (FID)**

Identifies the forwarding institution (BASE24). This value is taken from the SWITCH ID field on Interchange Configuration File (ICF) Screen 1.

- **Receiving Institution Identification Number (RID)**

Identifies the receiving institution and is allocated by MasterCard Europe. It is held in the EPS-NET INSTITUTION ID field on ICF Screen 12.

MasterCard Europe requires all transactions to be uniquely identified in the system within a 24-hour period. All transactions are identified by a combination of the following fields:

- System Trace Audit Number (ISO data element 011)
- Acquiring Institution Identification Number (ISO data element 032)
- Forwarding Institution Identification Number (ISO data element 033)

1.3.4 Transaction Security

The Host-EM Interface requires the transaction acquirer to specify, in the authorisation request (0100), whether the transaction is PIN-based or signature-based. This allows additional flexibility at the EM for separate routing of PIN-based and signature-based transactions. The difference is indicated by the presence of a P or S in sub-field 20 of the ISO data element 048 (additional data).

The Issuer Host system communicates its preferred PIN Block Format to the EM. This is achieved using the 0800 network management sign on request (ISO data element 048, sub-field 08) to the EM. The BASE24 EPS-Net Interchange Interface supports ISO Format 0 only.

Three methods are used for the processing of PIN-based transactions between the host and EM, for both acquirer and issuer functions:

- Single encryption
- Double encryption
- Pre-validated PIN.

The BASE24 EPS-Net Interchange Interface supports single encryption PIN Block processing only.

The Host-EM Interface supports PIN blocks encrypted using Triple DES. The BASE24 EPS-Net Interchange Interface supports this method of encryption.

The Host-EM Interface does not support message authentication or dynamic key exchange.

1.3.5 EMV Support

The MasterCard Europe network supports the processing of transactions performed on EMV-compliant chip cards. The Host-EM Interface accommodates the transmission of EMV-specific transaction data in data element 055 within the ISO 8583 (1987) message format.

1.3.6 UK Domestic Maestro Support

The MasterCard Europe network supports the processing of transactions performed on United Kingdom (UK) domestic Maestro cards. The Host-EM Interface accommodates UK Domestic Maestro transactions by supporting additional transaction types and catering for differences in message formats.

1.4 IMPLEMENTATION CONSIDERATIONS

This section documents environmental issues to be considered when implementing the EPS-Net Interchange Interface.

1.4.1 Software Dependencies

HP NonStop Operating System D46 or higher is required in order to implement Format-2 files.

The EPS-Net Interchange Interface is compatible with XPNET version 3.0.

The product is a Switch 6.0 interface, and therefore operates in a Release 3.4/4.0, a Release 5.0, a Release 5.1, a Release 5.3, or a Release 6.0 BASE24 system.

The BASE24-atm product and/or the BASE24-pos product must be installed in order to process authorisation messages. The BASE24-fhm product must be installed in order to generate File Action request messages.

The BASE24 subvolume, SW60SWPM, contains the source file libraries required to support Transactional Interface Monitoring within an Interchange Interface. This subvolume will be required so that the EPS-Net Interchange Interface can support T.I.M.

The BASE24 subvolume, PSxxADDR, contains the source file required to support address verification within the POS Router/Authorisation process. This subvolume will be required to allow address data to be verified during authorisation processing.

To provide full support for the encryption of PIN blocks using Triple DES, the Transaction Security Services (TSS) must be installed and configured.

1.4.2 Disk File Requirements

The Interchange Log File (ILF) is impacted by changes to the Standard External Message (SEM) definition (and whether the transaction is an EMV or non-EMV transaction), because a subset of the expanded SEM structure is included in each ILF authorisation record.

Each authorisation record contains a fixed length portion (818 bytes), followed by the SEM structure, followed by any tokens configured for logging to the ILF. Currently, one or two ILF records can be written to a 4Kb disk block (depending on the number of tokens configured for logging). However, any significant increase in the length of the SEM definition will mean that only a single record per block may be logged (with a consequent impact on disk requirements). The amount of data that can be written to a 4Kb disk block depends on whether the ILF is created as a Format-1 file or a Format-2 file (both formats are supported by the EPS-Net Interchange Interface).

The length of the EPS-Net SEM for an EMV transaction is 1204 bytes, giving a minimum length (i.e., including no tokens) of 2022 (1204 plus 818) bytes for an authorisation record logged to the EPS-Net ILF. For a non-EMV transaction, the record length is 452 bytes less, i.e. 1570 bytes.

1.4.2.1 Format-1 ILF

The maximum length of an authorisation record on the ILF record that will still allow two ILF records to be included in a 4Kb disk block is 2034 bytes. For this to be achieved, the length of the SEM definition must not exceed 1216 (2034 minus 818) bytes.

Therefore, for EMV transactions, as long as the token data included in each ILF record does not exceed 12 (1216 minus 1204) bytes, a total of two ILF records will continue to be included in each 4Kb disk block. For non-EMV transactions, the corresponding amount of token data would be 464 (12 plus 452) bytes.

1.4.2.2 Format-2 ILF

The maximum length of an authorisation record on the ILF record that will still allow two ILF records to be included in a 4Kb disk block is 2022 bytes. For this to be achieved, the length of the SEM definition must not exceed 1204 (2034 minus 818) bytes.

Therefore, for EMV transactions, as long as no token data is included in each ILF record (the length of the EPS-Net SEM for an EMV transaction is 1204 bytes), a total of two ILF records will continue to be included in each 4Kb disk block. For non-EMV transactions, the token data included in each ILF record must not exceed 452 bytes for two ILF records to be included in each 4Kb disk block.

Any file converted to Format-2 with alternate key(s) requires more disk space for the alternate key file(s), as the key length for Format-2 files is greater. This is true when the primary file is Format-2, regardless of whether the alternate key file(s) is Format-2.

1.4.3 Security Requirements

The minimum device and firmware requirements for HSMs in order to support Triple DES in the Host-EM Interface are as follows:

- Thales e-security (formerly Racal) HSMs must be RG7000 with Revision 1.05/5.05 firmware or greater.
- Atalla HSMs must be an A9000 or A10000 with Revision 2.8.4 firmware.

1.4.4 EMV Support

To support EMV processing, the BASE24 EMV product must be licensed and installed.

1.4.5 UK Domestic Maestro Support

To support UK Domestic Maestro processing, the UK Domestic Maestro extension to the EPS-Net Interchange Interface product must be licensed and installed.

2 PROCESSING OVERVIEW

The EPS-Net Interchange Interface process is responsible for the online message transfer between BASE24 and the MasterCard Europe network.

This process performs the following functions:

- Manages communications between BASE24 and the MasterCard Europe network.

The EPS-Net Interchange Interface sends and receives network management messages to determine the status of the link between BASE24 and the MasterCard Europe network.

- Facilitates message exchange between BASE24 and the MasterCard Europe network.

When BASE24 receives a transaction from the MasterCard Europe network, the EPS-Net Interchange Interface translates the message into the appropriate internal message format and sends the message to a BASE24 Authorisation process. When BASE24 receives a transaction for the MasterCard Europe network to authorise, the EPS-Net Interchange Interface translates the message from the internal message format to the external message format and forwards the message to the MasterCard Europe network. The EPS-Net Interchange Interface supports communication with the MasterCard Europe network using the message formats defined in the *V5 Interface Specifications*. These formats are based on the ISO 8583 (1987) message standard and the ISO 8583 (1993) message standard.

- Provides store-and-forward capabilities.

When the link between BASE24 and the MasterCard Europe network is down, the EPS-Net Interchange Interface stores certain messages intended for EPS-Net in its Store-and-Forward File (SAF). When the link is restored, these messages are retrieved from the file and sent to the MasterCard Europe network.

- Facilitates settlement between BASE24 and the MasterCard Europe network.

The EPS-Net Interchange Interface produces a set of interchange reports to assist in reconciling transactions between BASE24 and the MasterCard Europe network. These reports allow the BASE24 participants to reconcile the settlement positions of the MasterCard Europe network and BASE24.

2.1 FILE USAGE

The EPS-Net Interchange Interface process uses a number of BASE24 files. These files, which can be divided into control files and transaction files, are identified below.

2.1.1 Control Files

Control files contain configuration information and parameters that determine how the EPS-Net Interchange Interface process operates. The control files that contain configuration information for the EPS-Net Interchange Interface include:

- Network Environment File (NEF)
- Logical Network Configuration File (LCONF)
- Interchange Configuration File (ICF)
- Key File (KEYF or KEY6)
- Station Name File (SNF)
- Switch Terminal File (STF)
- Eurocard Institution Identification File (ERIIF)

For information on how these files are used to configure the EPS-Net Interchange Interface process, refer to section 3.

For information on the switch-specific ICF data definition and the ERIIF data definition, refer to Appendix B.

2.1.2 Transaction Files

Transaction files are used to store information about individual transactions processed through the EPS-Net Interchange Interface. The transaction files used by the EPS-Net Interchange Interface include:

- Interchange Log File (ILF)
- Store-and-Forward File (SAF)
- EPS-Net Request Log File (ENRLF)

The ILF and SAF DDLs can be accessed using DDLDOC. For information on DDLDOC, refer to the ***BASE24 Files and Record Structures Reference Manual***.

For information on the ENRLF data definition, refer to Appendix B.

2.2 NETWORK MANAGEMENT

The BASE24 EPS-Net Interchange Interface connects a BASE24 system with the MasterCard Europe network. The communications between the networks are managed by ACI's XPNET software and application-level network management messages.

For further information on network management processing in the EPS-Net Interchange Interface process, refer to section 4.

2.2.1 XPNET

Stations provide the means by which communications takes place between BASE24 and the MasterCard Europe network. All messages sent to EPS-Net are actually sent to the stations defined for the MasterCard Europe network. Likewise, all messages received from EPS-Net are actually received from stations.

The XPNET process handles all protocol-level functions between BASE24 and the stations, using the communications control software, provided with the HP NonStop (formally Tandem) hardware. The EPS-Net Interchange Interface process sends and receives messages through the XPNET process. It is the responsibility of the XPNET process to maintain the communications links with the stations according to the protocols under which the stations are defined.

Configuration of an XPNET system includes defining attributes for each XPNET node and all of its associated processes, links, lines, stations, and devices. All objects, including the XPNET node itself, can be added, changed, and deleted online without stopping the XPNET system. Configuration information relating to each XPNET node is held in the node's Network Environment File (NEF). There is one NEF for each XPNET node in the system. Each NEF is managed by the XPNET process in that XPNET node.

No two XPNET systems are configured exactly alike. In addition to being composed of different applications, processes, devices, and stations, there are a number of settings that depend entirely on how you want your system to work.

For complete information on the syntax of the commands, attributes, and directives used to configure an XPNET system, see the *NET24-XPNET Network Control Command Reference Manual*.

2.3 TRANSACTION PROCESSING

The BASE24 EPS-Net Interchange Interface connects a BASE24-atm or BASE24-pos network with the MasterCard Europe network. It allows a MasterCard Europe cardholder to initiate a transaction at a terminal belonging to a network not operated by the card issuer.

For further information on transaction processing in the EPS-Net Interchange Interface process, refer to section 5.

2.3.1 Message Formats

This section defines the external and internal message formats, and the translation between the two message types.

2.3.1.1 External Message Format

The structure and content of the external message used by the EPS-Net Interchange Interface process to communicate with the MasterCard Europe network is specified in the *V5 Interface Specifications*.

This message format is based on a standard (8583–1987) developed by the International Organisation for Standardisation (ISO). It is a variable-length, variable-content message that contains different information, based on the type of message being sent.

2.3.1.2 Internal Message Format

The internal message format is used whenever information is routed between the EPS-Net Interchange Interface process and the rest of BASE24. Depending on the BASE24 product receiving the message, the internal message is commonly referred to as the BASE24-atm Standard Internal Message (STM) or the BASE24-pos Standard Internal Message (PSTM).

For more information on the STM, see the *BASE24-atm Transaction Processing Manual*.

For more information on the PSTM, see the *BASE24-pos Transaction Processing Manual*.

2.3.1.3 Message Translation

The EPS-Net Interchange Interface process translates messages it receives from EPS-Net into the appropriate internal message format, and then sends the internal message to the appropriate BASE24 Authorisation process. When the message has been processed, the authorizing entity sends an internal response to the EPS-Net Interchange Interface process. Using information from the internal message, the EPS-Net Interchange Interface process formats an external response message and sends it to the MasterCard Europe network.

The EPS-Net Interchange Interface process translates the internal message it receives from BASE24 into the external message format, and then sends the external message to the interchange for authorisation. When the message has been processed, EPS-Net sends an external response to the EPS-Net Interchange Interface process. Using information from the external message, the EPS-Net Interchange Interface process formats an internal response message and sends it to BASE24.

2.3.2 EPS-Net Security

The transaction acquirer passes the PIN data to the card issuer in ISO data element 052.

The BASE24 EPS-Net Interchange Interface supports only the single encryption security format and the Triple DES algorithm for processing PIN data.

When using the single encryption security format, the value for ISO data element (DE) 052 is obtained by first transforming the PIN into a PIN Block, using ISO Format 0. This PIN Block is then encrypted under a Zone Master Key (ZMK) and it is this encrypted value that is carried in data element (DE) 052. The PIN block is encrypted using Triple DES (double length ZMK).

Both Inbound and Outbound Zone Master Keys are manually exchanged between the BASE24 EPS-Net Interchange Interface and the EM.

2.3.3 Supported Transaction Types

The EPS-Net Interchange Interface forwards the following BASE24-atm transactions to EPS-Net for authorisation:

- Cash Withdrawal
- PIN Change
- EMV PIN Unblock
- Balance Inquiry

Other BASE24-atm transaction types may be routed to an alternate authorising destination, (e.g., the ATM Authorisation process).

The EPS-Net Interchange Interface supports BASE24-pos transactions originating from any source, including attended POS devices, Cardholder Activated Terminals (CATs), the internet (e-commerce), recurring payments, and voice authorisations.

The EPS-Net Interchange Interface forwards the following BASE24-pos transactions to EPS-Net for authorisation:

- Normal Purchase
- Pre-Authorisation Request
- Mail/Phone Order
- Cash Advance
- Purchase with Cash Back
- Payment Deposit

Card verification transactions that contain Address Verification Service (AVS) data are forwarded to EPS-Net as “AVS only” requests.

Electronic commerce (e-commerce) transactions are identified by a POS Entry Mode (data element 022) value of ‘81x’ (where *x* indicates the PIN capability of the terminal) in the Host-EM message. Any transaction type is valid for e-commerce, with the exception of those involving the disbursement of cash.

BASE24-pos adjustments may be mapped to reversal messages before being sent to EPS-Net.

Other BASE24-pos transaction types may be authorised “blind”, or routed to an alternate authorising destination, (e.g., the POS Router/Authorisation process).

2.3.4 Cash Withdrawals From Bank Branch Terminals

Traditionally, MasterCard Europe has supported two processing codes for cash transactions: '01' for ATM withdrawals and '17' for POS cash advances. However, the distinction has become blurred because cash withdrawals (processing code '01') are allowed from bank branch terminals. This type of transaction is identified by a merchant category code of '6010'. A PIN is required and only debit cards can be used for cash withdrawals from bank branch terminals.

Cash withdrawals from terminals in bank branches can be routed to EPS-Net through BASE24-pos. The EPS-Net Interchange Interface uses the card type (debit) and the SIC Code ('6010') in the PSTM to identify these transactions as cash withdrawals from bank branch terminals.

Cash withdrawals from terminals in bank branches can be routed to either BASE24-atm or BASE24-pos when received from EPS-Net. The EPS-Net Interchange Interface supports a configuration option to route these transactions either to BASE24-atm or to BASE24-pos.

2.3.5 Non-Currency Dispense Transactions

Some ATMs are able to dispense items other than cash, e.g. gift vouchers. This type of transaction is known as Non-Currency Dispense (NCD), and is identified by a merchant category code of '6012'.

Non-currency dispense (NCD) transactions originating from an ATM (including mobile top-ups) are forwarded to EPS-Net as either normal purchase requests (if the NCD segment is present in the ICF) or cash withdrawal requests (if the NCD segment is not present in the ICF).

Normal purchase requests with a merchant category code of '6012' may be routed to either BASE24-atm or BASE24-pos when received from EPS-Net. The EPS-Net Interchange Interface routes these transactions to BASE24-atm if the NCD segment is present in the ICF, and routes these transactions to BASE24-pos if the NCD segment is not present in the ICF.

2.3.6 Transaction Logging

The EPS-Net Interchange Interface logs the following types of messages to the Interchange Log File (ILF):

- Requests containing error conditions
- Responses
- Reversals
- Force posts

Whenever a reversal occurs, the EPS-Net Interchange Interface attempts to update the ILF record already logged for the transaction.

2.3.7 Store-and-Forward Function

A store-and-forward function is incorporated into the EPS-Net Interchange Interface process to collect transaction messages during periods when the MasterCard Europe network is unavailable. When EPS-Net is unavailable, messages required as notification to the MasterCard Europe network are stored in the Store-and-Forward File (SAF). These messages are then transmitted later when communications are restored.

2.3.8 Triple DES Support

Triple DES is supported fully only in Release 6.0 (the full solution). To facilitate a phased migration, BASE24 Switch 6.0 interchange interfaces can be used in a 5.x environment to provide limited Triple DES support (the interim solution).

2.3.8.1 Triple DES “Full” Solution

With the advent of Triple DES, ACI has introduced a new architecture for managing transaction security. In BASE24 Release 6.0, all cryptographic functions, such as PIN translation, can be managed through the *Transaction Security Services*. A full Triple DES environment is based on BASE24 6.0 using the Transaction Security Services.

In the traditional BASE24 environment, the application processes (device handlers, authorisation modules and interface processes) manage their own cryptographic services. This includes storing and retrieving keys, and making calls to a Hardware Security Module (HSM). In an environment using Transaction Security Services (TSS), the applications call the TSS process rather than the HSM. TSS manages the storing and retrieving of keys and making calls to the HSM.

When using BASE24 Transaction Security Services, the application processes use “key locators” in place of actual keys. Key locators serve as tags, or indexes, representing a key. The actual keys are stored in new key files managed exclusively by Transaction Security Services. The TSS uses the key locators to retrieve the actual keys, before formatting and sending commands to the HSM.

The full solution impacts the EPS-Net Interchange Interface because of the new configuration requirements. The Key File (KEYF) stores key locators to the TSS database instead of the actual PIN block encryption keys themselves.

Further details of the TSS are given in the *BASE24 Transaction Security Services Processing Guide*.

2.3.8.2 Triple DES “Interim” Solution

Transaction Security Services is not supported prior to Release 6.0, however, the same version of BASE24 interchange interfaces can run in all releases of BASE24. To enable Release 6.0 interchange interfaces to run in all releases of BASE24, modifications have been made to interchanges interfaces to support Triple DES when the interface is running in a BASE24 Release 5.x network. The interchange interfaces must support translating a PIN block in transactions originating at the interchange from Triple DES encryption to Single DES encryption. The interchange interface must support translating a PIN block in transactions originating in BASE24 or a host from Single DES encryption to Triple DES encryption. To accomplish this, the interchange interfaces call utilities added to support double length PIN encryption keys in PIN block translation calls to an HSM. An LCONF parameter indicates when the special PIN block translation occurs.

This interim solution impacts the EPS-Net Interchange Interface in the following ways:

- Support for the new KEY6 file, replacing the Key File (KEYF). This new file stores double length PIN block encryption keys instead of single length PIN block encryption keys.
- Support for translating PIN blocks encrypted using Single DES (received from the BASE24-atm Authorisation process) to encryption using Triple DES (for transmission to MasterCard Europe).
- Support for translating PIN blocks encrypted using Triple DES (received from MasterCard Europe) to encryption using Single DES (for transmission to the BASE24-atm Authorisation process and the BASE24-pos Router/Authorisation process).

An additional call to the HSM (to convert a PIN Block from Triple DES encryption to Single DES encryption) will be required to support PIN Blocks received from MasterCard Europe if the Triple DES “interim” solution is implemented.

2.3.9 UK Domestic Maestro Support

POS transactions performed on UK Domestic Maestro cards are sent to the issuer from the acquirer via EPS-Net. In order to support functions specific to the UK Domestic Maestro card scheme, the message format and contents differ to that used for “international” transactions.

For authorisation systems such as BASE24, this involves supporting additional data elements in the Host-EM message and supporting additional values in existing data fields.

UK Domestic Maestro support is provided by a separately licensable module, which is built on top of the Switch 6.0 version of the EPS-Net Interchange Interface.

Full details of the UK Domestic Maestro requirements are provided in the *Host-EM Programmer Specifications for UK Domestic Maestro*.

2.4 FILE ACTION MESSAGE PROCESSING

The BASE24 EPS-Net Interchange Interface connects a BASE24-atm or BASE24-pos network with the MasterCard Europe network. It provides a card issuer with online access to MasterCard Europe’s Account Management System (AMS).

For further information on file action message processing in the EPS-Net Interchange Interface process, refer to section 6.

2.4.1 Message Formats

File Action messages are not initiated by BASE24. A File Action request is sent to EPS-Net only when a File Action message has been received from a Host system, via a BASE24 ISO Host Interface (HISO) process.

File Action messages are passed from the Host Interface to the EPS-Net Interchange Interface in a format used within BASE24 for file action messages. This format is commonly referred to as a PATHWAY (PATH) Request message.

The structure and content of the external message used by the EPS-Net Interchange Interface process to communicate with the MasterCard Europe network is specified in the *V5 Interface Specifications*. For File Action messages, the format is based on a standard (8583–1993) developed by the International Organisation for Standardisation (ISO).

The EPS-Net Interchange Interface process translates the PATH request message it receives from BASE24 into the external message format, and then sends the external request (message type 1304) to the interchange. When the message has been processed, the interchange sends an external response (message type 1314) to the EPS-Net Interchange Interface process.

2.4.2 Supported Transaction Types

File Action messages are used to indicate which action should be taken on the AMS with respect to a specific card. The EPS-Net Interchange Interface supports the following issuer-specific file action functions:

- Add a new record;
- Modify an existing record;
- Delete an existing record.

Information from File Action request messages sent to EPS-Net and File Action response messages received from EPS-Net is logged to the EPS-Net Request Log File (ENRLF).

2.5 INTERCHANGE REPORTING

The EPS-Net Interchange Interface produces two reports: SETL01 and STAT09. The purpose of these reports is to facilitate settlement between BASE24 and the MasterCard Europe network.

For information on these reports, refer to section 7.

2.6 CHIP-BASED DEBIT/CREDIT SUPPORT

MasterCard Europe is committed to introducing chip-based debit and credit functionality, based on the *EMV ICC Specifications for Payment Systems*.

For authorisation systems such as BASE24, this involves supporting additional data elements in the Host-EM message and supporting additional values in existing data fields.

EMV support is provided by a separately licensable module, which is built on top of the Switch 6.0 version of the EPS-Net Interchange Interface.

The Release 6.0 EPS-Net Interchange Interface is NOT backwards compatible with the Release 5.1.2 ICC product, i.e. it is not possible to bind the Release 6.0 interface with the Release 5.1.2 EPS-Net ICC extension module.

For information on EPS-Net EMV processing, refer to section 8.

Full details of EMV support in BASE24 are provided in the *BASE24-atm EMV Support Manual* and the *BASE24-pos EMV Support Manual*.

2.7 IPF REFRESH

The EPS-Net IPF Refresh module is a BASE24 switch product that runs as a stand-alone batch process. It creates a standard BASE24 Interchange Prefix File (IPF), using the MasterCard Europe Optimised Current Parameters File as input.

For information on IPF Refresh processing, refer to section 9.

2.8 FUTURE REQUIREMENTS

Support for the following features may be added in a future release of the EPS-Net Interchange Interface:

- Optional functions introduced in Authorization System Release 07.1:
 - ⇒ Purchase Approval Only for Purchase with Cash Back Transactions;
 - ⇒ Prepaid Card Support;
 - ⇒ Partial Approvals.
- The BASE24 Release 6.0 ‘Transactions Allowed’ enhancement, including the Enhanced Interchange Configuration File (ICFE).

3 CONFIGURATION AND FILES MAINTENANCE

This section describes the configuration files that must be set up or verified in order that the EPS-Net Interchange Interface process can be configured into the BASE24 network.

This section also provides details of the BASE24 files maintenance screens and field descriptions that have been enhanced to support the BASE24 EPS-Net Interchange Interface module. Function keys that operate differently from the standard function keys are also documented.

For full details of the screen layouts, standard BASE24 function keys and online help screens, refer to the *BASE24 Base Files Maintenance Manual*.

3.1 XPNET PROCESS

The BASE24 Interchange Interface process supports the following data communications protocols, as defined in the *V5 Interface Specifications*:

- X.25
- TCP/IP

A single BASE24 EPS-Net Interchange Interface process can support a maximum of 16 stations with the EM.

This section provides a brief explanation of the way in which BASE24 lines, stations and subdevices are defined for the EPS-Net Interchange Interface process.

See the *NET24-XPNET Network Control Operations Guide* for descriptions of the XPNET network control functions. This manual provides descriptions of the various functions used in configuring, monitoring, and controlling an XPNET system.

See the *NET24-XPNET Network Control Command Reference Manual* for descriptions of the XPNET network control commands. This manual provides comprehensive command descriptions of all network control commands used to monitor and control an XPNET system.

3.1.1 Terminology of Lines, Stations and Subdevices

All BASE24 communication to external devices takes place using logical entities known as stations and lines. For asynchronous and byte synchronous protocols, a line is synonymous with a physical communications line and there can be one or more stations per line, depending upon the protocol. For X.25, however, a line is equivalent to an X.25 virtual circuit and only one station can be defined per line.

3.1.2 Network Management

This section presents sample BASE24 process, station, line, and device objects definitions for the EPS-Net Interchange Interface.

3.1.2.1 Process Definition

One process definition must exist for each EPS-Net Interchange Interface process. An example of a process definition for the EPS-Net Interchange Interface process is given below.

```
SET PROCESS PROGRAM $DATA.SW60OBJ.EURO
SET PROCESS CPU 1
SET PROCESS BCPU 0
SET PROCESS LIBRARY $DATA.XPNET.SKELB
SET PROCESS PRIORITY 149
SET PROCESS PPD $P1E1
SET PROCESS STARTUP AUTOMATIC
SET PROCESS QAT 20
SET PROCESS QMI 85
SET PROCESS QMT 150
SET PROCESS SWAP $DATA
ADD PROCESS \SYS1.P1A^NODE.P1A^EPS1
```

3.1.2.2 Device Declarations

The BASE24 Network definition must contain one device declaration for each generic station type.

An example of a device declaration for an X.25 device is given below.

```
SET DEVICE RECVLEN 4095
SET DEVICE XMITLEN 4095
SET DEVICE ADD DESTSTRING 1, P1A^DPROC, 2, %HA0
SET DEVICE ADD DESTSTRING 2, P1A^EPROC, 2, %HA1
SET DEVICE ADD PROTOCOLSTRING 1, ITI, 10, %HF1
SET DEVICE ADD PROTOCOLSTRING 2, ATALLA, 10, %HF2
SET DEVICE ADD PROTOCOLSTRING 3, DATAFONO, 10, "?"
ADD DEVICE \SYS1.P1A^NODE.D1A^X251
```

An example of a device declaration for a TCP/IP device is given below.

```
RESET DEVICE
SET DEVICE INPUTFORMAT TXT
SET DEVICE OUTPUTFORMAT TXT
SET DEVICE POLLFORMAT NUL
SET DEVICE RECVLEN 4096
SET DEVICE SELECTFORMAT NUL
SET DEVICE XMITLEN 4096
ADD DEVICE \SYS1.P1A^NODE.EURO^CTS^DEV
```

3.1.2.3 Station Declarations

There must be a station declaration for each station in the network. For the X.25 protocol, there must be one station declaration for each switched virtual circuit used by the EPS-Net Interchange Interface process.

An example of an X.25 station declaration is given below.

```
RESET STATION
SET STATION DEVICENAME \SYS1.P1A^NODE.D1A^X251
SET STATION LINENAME \SYS1.P1A^NODE.L1A^X251
SET STATION DESTINATION \SYS1.P1A^NODE.P1A^DPROC
SET STATION UCRATE X25STARATE
SET STATION UCSTATE X25STASTATE
SET STATION UCQUEUE X25STAQUEUE
SET STATION CLASS GROUP1
ADD STATION \SYS1.P1A^NODE.S1AX251
```

An example of a TCP/IP station declaration is given below.

```
RESET STATION
SET STATION DEVICENAME EURO^CTS^DEV
SET STATION LINENAME \SYS1.P1A^NODE.L1A^CTS^EURO
SET STATION DESTINATION P1A^EURO
SET STATION RETRY 3
SET STATION ERRORACT REINEVERY
SET STATION RADDR 6201:172.16.146.32
SET STATION USERDATA "-5,L2,TB,PF2%H00%H00 -K+,-B+"
ADD STATION \SYS1.P1A^NODE.S1AEURO
```

3.1.2.4 Line Declarations

There must be one line declaration for each communication line used by the EPS-Net Interchange Interface process. For the X.25 protocol, there must be one line declaration for each X.25 switched virtual circuit used by the EPS-Net Interchange Interface process. The EPS-Net Interchange Interface process supports two X.25 switched virtual circuits to communicate with EPS-Net.

An example of an X.25 line declaration is given below.

```
RESET LINE
SET LINE PORT $AX10
SET LINE PROTOCOL X.25
SET LINE ETE NONE
SET LINE WFCALL ON
SET LINE DUPLEX FULL
SET LINE ATIMER 1
SET LINE VERDTE OFF
SET LINE UCRATE X25LINRATE
SET LINE UCSTATE X25LINSTATE
SET LINE UCQUEUE X25LINQUEUE
SET LINE CLASS GROUP1
ADD LINE \SYS1.P1A^NODE.L1A^X251
```

An example of a TCP/IP line declaration is given below.

```
RESET LINE
==
== The PORT name must match the PPD name of the CTS/E Client
== process
==
SET LINE PORT $EURO
SET LINE PROTOCOL CTS
SET LINE ERANALYSIS 100
SET LINE RETRY 3
SET LINE CINIT IMMEDIATE
SET LINE CMETHOD ALLOCATE
SET LINE DUPLEX FULL
SET LINE EXPDATA ROUTE
SET LINE FMM ALL
SET LINE SENDCONFIRM REQUEST
ADD LINE \SYS1.P1A^NODE.L1A^EURO
```

3.2 LOGICAL NETWORK CONFIGURATION FILE

The Logical Network Configuration File (LCONF) is the central location of file information and parameters for the BASE24 Logical Network; one LCONF exists for each logical network.

The LCONF contains assigns that link the physical names of objects in the logical network to names used in the BASE24 code and parameters (params) that define processing values at the logical network level.

The LCONF assigns and params listed in this subsection are required by the EPS-Net Interchange Interface process to initialise and process correctly.

For more details, refer to the *BASE24 Logical Network Configuration File Manual*.

3.2.1 LCONF Assigns

Unless otherwise stated, the assigns specified below are optional.

Assign	Description
ENRLF	The name of the EPS-Net Request Log File (ENRLF). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides. This assign is required if the interface process is required to support File Action messages.
ERIIF	The name of the EPS-Net Institution Identification File (ERIIF). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
ICF (Required)	The name of the Interchange Configuration File (ICF). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
ILF (Required)	The name of the Interchange Log File (ILF). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides. The ILF assign must be associated with the symbolic name of the Interface process. A template file name is required as the EPS-Net Interchange Interface process creates the ILF periodically.

Assign	Description
KEYF (Required unless the Triple DES “interim” solution is supported)	The name of the Key File (KEYF). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
KEY6 (Required if the Triple DES “interim” solution is supported)	The name of the KEY File (KEY6). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
LMCF	The name of the Log Message Configuration File (LMCF) that is used by all BASE24 processes. If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
RPT	The name of the BASE24 reports file that contains BASE24 and Interchange reports. A template file is required as the EPS-Net Interchange Interface process creates the Interchange reports periodically.
RPT-ERROR-FILE	The name of the file to which the Interchange report programs route the report error log messages.
SAF (Required)	The name of the Store-and-Forward File. If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides. The SAF assign must be associated with the symbolic name of the EPS-Net Interchange Interface process. A template file is required because the EPS-Net Interchange Interface process creates the SAF periodically.
SNF	The name of the Station Name File (SNF). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
STF	The name of the Switch Terminal File (STF). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
SW-RPT01-PROG-NAME	The name of the executable object code that generates the Interchange Clearings Statement Report (RP01SWI). The symbolic name of the EPS-Net Interchange Interface process name is required for the assign. The format is: \$volume.subvolume.RP01SWI

Assign	Description
SW-RPT09-PROG-NAME	The name of the executable object code that generates the Interchange Transaction Detail Report (RP09SWI). The symbolic name of the EPS-Net Interchange Interface process name is required for the assign. The format is: <i>\$volume.subvolume.RP09SWI</i>
SW-RPT-OBEY-FILE	The name of the EPS-Net Interchange Interface report obey file. The symbolic name of the EPS-Net Interchange Interface process is required for the assign. The format is: <i>\$volume.subvolume.Swyymmdd</i>
TKN (Required)	The name of the Token File (TKN). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.

For assigns required for Hardware Security Module (HSM) use, refer to the ***BASE24 Transaction Security Manual***.

3.2.2 LCONF Parameters

Unless otherwise stated, the parameters specified below are optional.

Parameter	Description
ACCT-NUM-TYP	A value indicating whether account numbers are stored in binary or decimal form. The binary form allows the account number to be up to 28 digits and the decimal form allows an account number of 19 digits. This parameter is required for report processing. Valid values are: B = Binary format D = Decimal format (default)
AFT-PAN-DIGITS	This parameter contains a 3-digit numeric value, and specifies how to mask PANs displayed on the ENRLF screen. The first digit specifies the maximum number of leftmost digits to be displayed unmasked. The second digit specifies the minimum number of digits to be masked. The third digit specifies the number of rightmost digits to be displayed unmasked. Valid values are any numeric digits. The default value is 094.
ATM-LN-CUTOVER (Required)	The time of day the entire BASE24-atm Logical Network is cut over. The 24-hour clock format is used. Valid values are 00:00–23:59.
AVS-IND	A value indicating the contents of DE 094 sent in 0800 messages by issuers participating in the AVS service. Valid values are: 0 = Do not send DE 094 (default) 1 = Issuer receives complete address data 2 = Issuer receives format 2 address data 3 = Issuer receives format 3 address data
BAL-ENQ-SUPPORT	A value indicating the maximum number of account balances that can be returned by an issuer in a response message to EPS-Net. Valid values are 0 (default), 1, or 2.

Parameter	Description
BASE24-REL	A code indicating whether the BASE24 logical network is currently running release 3.4/4.0 or release 5.x or release 6.0. The Release 6.0 EPS-Net Interchange Interface uses this parameter to determine the processing required in different releases of a BASE24 system. Valid values are '40', '50', or '60' (default).
CASH-WDL-AUTH-PROD	This parameter specifies whether cash withdrawals from terminals in bank branches should be routed to BASE24-atm (current processing) or BASE24-pos for authorisation. Valid values are: 'ATM' (default) or 'POS'.
DATE-DISPLAY-TYPE	The format of the month and day portion of the dates on reports. The parameter does not affect dates that include the year (i.e., dates in the format YYMMDD). Valid values are: 1 = DDMM 2 = MMDD (default)
DEBIT-CARD-TYPES	This parameter specifies up to 23 card types, identifying debit card products, e.g. Maestro (but NOT Debit MasterCards), separated by a semi-colon character. (Note that the maximum length of LCONF parameter text is 70 bytes.) Card types will be validated to be two characters in length. A numeric or space character will only be allowed for the second byte of any card type. A semi-colon should not terminate the list. For example: AA;BB;C ;DD
DEBIT-MC-CARD-TYPES	This parameter specifies up to 23 card types, identifying Debit MasterCards, separated by a semi-colon character. (Note that the maximum length of LCONF parameter text is 70 bytes.) Card types will be validated to be two characters in length. A numeric or space character will only be allowed for the second byte of any card type. A semi-colon should not terminate the list. For example: AA;BB;C ;DD

Parameter	Description
DENY-MC-E-COMM-TXN	This parameter indicates if the issuer wants to deny e-commerce transactions received from an acquirer not participating in the MasterCard Electronic program. Valid values are: Y = Yes, deny transactions (default) N = No, do not deny transactions
DES-TRIPLE-SINGLE	This parameter indicates if Triple DES support in the interface is provided using the “interim” solution, i.e. with the KEY6 (Transaction Security Services not used). Valid values are: Y = Yes, KEY6 is used N = No, KEYF is used (default)
ENRLF-NOTIFY-INTERVAL	The number of writes made to the ENRLF between reading and calculating the percentage file full. Once the file reaches the percent full specified in the ENRLF-NOTIFY-PERCENTAGE param, a message is logged indicating that the file is becoming full. Valid values are 10–32,766. The default value is 50.
ENRLF-NOTIFY-PERCENTAGE	The percentage of file full at which the EPS-Net Interchange Interface process logs a message indicating that the ENRLF is becoming full. Valid values are 10–99. The default value is 80.
EURO-STAND-IN-RCS	This parameter specifies up to 23 response codes for which alternate routing will be invoked. The response codes are separated by a semi-colon character, for example 31;91;96. A semi-colon should not terminate the list.
FALLBACK-ALWD	This parameter indicates whether Fallback transactions are permitted for ATM transactions, cash withdrawals from bank branch terminals, and transactions initiated at CAT Level 1 and CAT Level 2 devices. Valid values are: Y = EPS-Net processing permits Fallback transactions N = EPS-Net processing does not permit Fallback transactions (default)
FIRST-EURO-TRACE-NUM	This optional parameter specifies the initial numeric value to be used for the STAN within the BASE24 system.

Parameter	Description
FRWD-DCLN-ADVC	This parameter indicates whether the interface is to forward an advice message to BASE24 when the transaction has been declined. Valid values are: Y = Forward declined advice messages N = Drop declined advice messages (default)
FRWD-UNMATCHED-RVSL	This parameter indicates whether the interface is to format an internal reversal message from an external reversal message, and forward the message to BASE24, when the original transaction cannot be found on the ILF. Valid values are: Y = Format internal reversal messages and forward the messages N = Drop unmatched external reversal messages (default)
ILF-NOTIFY-INTERVAL	The number of writes made to the ILF between reading and calculating the percentage file full. Once the file reaches the percent full specified in the ILF-NOTIFY-PERCENTAGE param, a message is logged indicating that the file is becoming full. Valid values are 10–32,766. The default value is 50.
ILF-NOTIFY-PERCENTAGE	The percentage of file full at which the EPS-Net Interchange Interface process logs a message indicating that the ILF is becoming full. Valid values are 10–99. The default value is 80.
INCR-EURO-TRACE-NUM-BY	This optional parameter specifies the value used to increment the STAN within this process. It must be equivalent to the number of EPS-Net processes within the BASE24 system.
MAX-STF-RECORDS	This optional parameter allows an acquirer to specify the maximum number of STF records that can be held in extended memory. Valid values are 0–32,766. The default value is 0.
NCD-DFLT-HOPR-CONTENTS	This optional parameter specifies the default setting for the ATM hopper contents in a Non-Currency Dispense (NCD) transaction. Valid values are 1–99. The default value is 10.

Parameter	Description
NO-MANUAL-ENTRY-CARD-TYPES	The list of card types for which key entry is not permitted at a POS device. There may be up to 20 card types defined. The format is: CT;CT;CT;CT where: ‘CT’ is the card type ‘C’ must be an alphabetic character or ‘*’ (wild card) ‘T’ (if present) must be an alphanumeric character, a space, or ‘*’. A match between the Card Type field in the PSTM and one of the values listed will cause the EPS-Net Interchange Interface to check that the transaction data has not been key-entered.
POS-LN-CUTOVER (Required)	The time of day the entire BASE24-pos Logical Network is cut over. The 24-hour clock format is used. Valid values are 00:00–23:59.
SAF-NOTIFY-INTERVAL	The number of writes made to the SAF between reading and calculating the percentage file full. Once the file reaches the percent full specified in the SAF-NOTIFY-PERCENTAGE parameter, a message is logged indicating that the file is becoming full. Valid values are 10–32,766. The default value is 50.
SAF-NOTIFY-PERCENTAGE	The percentage of file full at which the EPS-Net Interchange Interface process logs a message indicating that the SAF is becoming full. Valid values are 10–99. The default value is 80.
SW-ILF-FILE-RETENTION	The number of days an ILF remains on the system, after which the EPS-Net Interchange Interface process purges the ILF from the system. Valid values are 3–99. The default value is 7.
SW-RPT-STARTUP	The time the EPS-Net Interchange Interface process runs the report obey file. The 24-hour clock format is used. Valid values are 00:00–23:59.

Parameter	Description
SWI-ACQ-AVS-DATA-SUPPRS	A value indicating whether Address Verification (AVS) data will be suppressed before logging acquirer transactions to the ILF. Valid values are: Y = Suppress AVS data before logging N = Do not suppress AVS data (default)
SWI-ACQ-CVD2-DATA-SUPPRS	A value indicating whether manual Card Verification (CVC2) data will be suppressed before logging acquirer transactions to the ILF. Valid values are: Y = Suppress CVC2 data before logging N = Do not suppress CVC2 data (default)
SWI-ACQ-DISCR-DATA-SUPPRS	A value indicating whether discretionary data from Track-1 and/or Track-2 will be suppressed before logging acquirer transactions to the ILF. Valid values are: Y = Suppress discretionary data before logging N = Do not suppress discretionary data (default)
SWI-ISS-AVS-DATA-SUPPRS	A value indicating whether Address Verification (AVS) data will be suppressed before logging issuer transactions to the ILF. Valid values are: Y = Suppress AVS data before logging N = Do not suppress AVS data (default)
SWI-ISS-CVD2-DATA-SUPPRS	A value indicating whether manual Card Verification (CVC2) data will be suppressed before logging issuer transactions to the ILF. Valid values are: Y = Suppress CVC2 data before logging N = Do not suppress CVC2 data (default)
SWI-ISS-DISCR-DATA-SUPPRS	A value indicating whether discretionary data from Track-1 and/or Track-2 will be suppressed before logging issuer transactions to the ILF. Valid values are: Y = Suppress discretionary data before logging N = Do not suppress discretionary data (default)

For parameters required for Hardware Security Module (HSM) use, refer to the *BASE24 Transaction Security Manual*.

The additional parameters specified below are used in BASE24-pos Release 3.4 and BASE24-atm Release 4.0 systems only.

Parameter	Description
HOST-COMPLETE-TRACK2	This parameter indicates whether transactions acquired from a Host contain complete and unaltered Track-2 data. Valid values are: Y = Yes (default) N = No
INCOMPLETE-TRACK2-TERM-TYPES	The list of terminal types, as specified in the ATM Terminal Data File (TDF) and POS Terminal Data File (PTDF), for which incomplete Track-2 data is transmitted from the device, or for which the device or device handler can edit the Track-2. The format is: TT;TT;TT;TT where 'TT' is the terminal type and must be two alphanumeric characters. There can be up to 20 terminal types defined. A semi-colon should not terminate the list.
SWITCH-COMPLETE-TRACK2	A value indicating whether transactions acquired from an interchange (other than EPS-Net) contain complete and unaltered Track-2 data. Valid values are: Y = Yes (default) N = No

3.3 INTERCHANGE CONFIGURATION FILE

The EPS-Net Interchange Interface process accesses data maintained in the Interchange Configuration File (ICF) to determine the variable parameters relevant to the MasterCard Europe network. These parameters define the processing of transactions between the EPS-Net Interchange Interface and the EM.

For the EPS-Net Interchange Interface ICF record, the data used is configured on the following screens:

- Screen 1
- Screen 2
- Screen 3
- Screen 6
- Screen 7
- Screen 8
- Screen 9
- Screen 10
- Screen 11
- Screen 12
- Menu Screen

The fields on these screens used by the EPS-Net Interchange Interface are described in the sections below.

Operator access to the ICF is permitted in accordance with the security access levels assigned in the BASE24 CRT access system. For more information on the BASE24 CRT access system, refer to the ***BASE24 User Security Manual***.

For more details on the ICF, refer to the ***BASE24 Base Files Maintenance Manual***.

3.3.1 ICF Screen 1

The ICF screen 1 (Figure 3–1) identifies the Interchange and provides general configuration information for the Interchange Interface process.

BASE24-BASEINTERCHANGE CONFIG

EME399/08/1813:4001 OF 13

INTERCHANGE FIID:PROCESS:

SWITCH TYPE:EURO

INTERCHANGE LOGICAL NET:

REPORTING NAME:

INSTITUTION ID:

SWITCH ID:

STATION 1:

STATION 2:

SIC CODE:

CURRENCY CODE:826(UK)

DEFAULT TERM NUM:

DEFAULT ACQUIRER ID NUM:00000000000

CUSTOMER BALANCE DISPLAY:0(DON'T DISPLAY OR PRINT)

***** BASE24 *****

NEW PAGE:FILE DESTINATION:NEW LOGICAL NETWORK ID:

F12-HELP

Figure 3–1 ICF Screen 1

INTERCHANGE FIID4 (A) M

This field is the institution identifier that uniquely distinguishes the Interchange. It must match the Logical Network ID given to the Interchange. This field and the Process field form the primary key of the file.

PROCESS16 (A) M

This field specifies the name of the EPS-Net Interchange Interface process defined in the Network Environment File (NEF). The Process and Interchange FIID fields form the primary key of the ICF.

SWITCH TYPE4 (A) M

This field identifies the Interchange and allows access to the Interchange Interface specific screens. For the EPS-Net Interchange Interface the field must be set to EURO.

INTERCHANGE LOGICAL NET**4 (A) M**

This field is the name of the Logical Network to which the EPS-Net Interchange Interface process belongs as defined in the process definition in the Network Environment File (NEF). It is used by the RP01SWI Interchange reports to calculate the net settlement position.

REPORTING NAME**20 (A)**

This field contains the name of the EPS-Net Interchange that is to appear as the heading in the EPS-Net Interchange reports.

INSTITUTION ID**15 (A)**

This field contains the MasterCard Member ID (used as the Acquiring Institution Identifier) for the institution. The value assigned by MasterCard Europe must be specified in this field.

If the EPS-Net Interchange Interface processes transactions for more than one MasterCard Member, or if the MasterCard Member supports more than one Member ID, then this field must contain spaces. The appropriate MasterCard Member ID for a transaction will instead be retrieved from either the ERIIF or the (P)STM.

SWITCH ID**15 (A)**

This field is the Forwarding Institution Identifier. The value assigned by MasterCard Europe must be specified in this field.

STATION 1**16 (A)**

This field specifies the symbolic name of one of the stations that will be used to communicate with the EM. This name must match the name of a station in the NEF for the EPS-Net Interchange Interface process. If there is only one station configured to the EM, then that station name must be specified in this field. If there is a Station Name File (SNF) record configured for the process, then this field is not used.

STATION 2

16 (A)

This field specifies the symbolic name of one of the stations that will be used to communicate with the EM. This name must match the name of a station in the NEF for the EPS-Net Interchange Interface process. If there is only one station configured to the EM, then this field is left blank. If there is a Station Name File (SNF) record configured for the process, then this field is not used.

SIC CODE

4 (A)

The EPS-Net Interchange Interface process does not use this field.

CURRENCY CODE

3 (N) M

This field identifies the currency code for transactions that the EPS-Net Interchange Interface receives. Valid values are listed in the ***Codes for the Representation of Currencies and Funds***, ISO 4217-1981. For the United Kingdom, a value of 826 (Sterling pound) is specified. The value entered here should match the value entered for the currency code in the BASE24 IDF, unless the Multiple Currencies product has been licensed and installed.

Default Value: The default value depends upon configuration variables.

DEFAULT TERM NUM

16 (A)

The EPS-Net Interchange Interface process does not use this field.

DEFAULT ACQUIRER ID NUM

11 (N)

The EPS-Net Interchange Interface process does not use this field.

CUSTOMER BALANCE DISPLAY

1 (N) M

Indicates the interchange preference concerning balance presentation for transactions with incoming responses. The valid values are:

- 0 = Do not display or print
- 1 = Display only
- 2 = Print only
- 3 = Display and print

Default Value: 0

3.3.2 ICF Screen 2

The ICF screen 2 (Figure 3–2) displays the Interchange settlement information as well as holiday dates.

```

BASE24-BASE  INTERCHANGE CONFIG      EME3      99/08/18  13:40  02 OF 13
INTERCHANGE FIID:      _____  PROCESS: _____

                SETTLEMENT INFORMATION
SETTLEMENT HOUR: 23      SETTLEMENT MINUTE: 59
SETTLEMENT DAYS: 1  (PROCESS 7 DAYS)
REPORT PRIORITY: 100      REPORT CPU: 0

                SWITCH POSTING DATE (YYMMDD)

                HOLIDAY DATES  (YYMMDD)

                REPORT DATA MASKING PARAMETERS
DATA MASK FLAG: Y  (MASK SENSITIVE DATA)      RIGHT UNMASKED DIGITS: 4
MIN MASKED DIGITS: 9      MAX LEFT UNMASKED DIGITS: 0

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP

```

Figure 3–2 ICF Screen 2

SETTLEMENT INFORMATION

The following fields contain Interchange settlement information.

SETTLEMENT HOUR

2 (N) M

This field indicates the current hour of settlement (local time) of the Interchange and is used to calculate the Interchange posting date (at warmboot or initialisation). It is recommended that the field be set to 23 for the EPS-Net Interchange Interface.

Valid values are 00–23.

Default Value: 00

SETTLEMENT MINUTE

2 (N) M

This field indicates the current minute of settlement (local time) of the Interchange and is used to calculate the Interchange posting date (at warmboot or initialisation). It is recommended that the field be set to 59 for the EPS-Net Interchange Interface.

Valid values are 00–59.

Default Value: 00

SETTLEMENT DAYS

1 (N) M

This field indicates the processing schedule of the Interchange Interface.

The EPS-Net Interchange processes 7 days a week.

The valid values are:

0 = Processes 5 days a week

1 = Processes 7 days a week

Default Value: 1 (7 days a week). However, if this field is blank-filled, the default value is 0 (5 days a week).

REPORT PRIORITY

3 (N) M

This field determines the priority for reports started by the EPS-Net Interchange Interface process.

The value entered in this field must be less than or equal to 255. It is recommended that this field be set to 100 for the EPS-Net Interchange Interface.

Default Value: 100

REPORT CPU

1 (N) M

This field determines the CPU used for reports started by the EPS-Net Interchange Interface.

Default Value: 0

SWITCH POSTING DATE (YYMMDD) 6 (N)

This field displays the current Interchange posting date, in the format YYMMDD.

This date is updated whenever the Interchange cuts over to a new processing date and at initialisation.

The posting date cannot be processed unless six numeric characters representing a valid date are entered.

HOLIDAY DATES (YYMMDD)

The EPS-Net Interchange Interface process does not use these fields.

REPORT DATA MASKING PARAMETERS

The values in the following fields specify the report data masking parameters. These parameters support the masking of sensitive information in accordance with PCI data security standards.

DATA MASK FLAG 1 (A) M

A code indicating whether sensitive data should be masked or unmasked. Valid values are as follows:

Y = Mask sensitive data

N = Do not mask sensitive data.

Default Value: Y

RIGHT UNMASKED DIGITS 1 (N) M

A code defining the number of rightmost digits to be displayed unmasked. Valid values are as follows:

0–9 = Number of rightmost digits to be displayed unmasked.

Default Value: 4

MIN MASKED DIGITS

1 (N) M

A code defining the minimum number of digits to be masked. Valid values are as follows:

0–9 = Minimum number of digits to be masked.

Default Value: 9

MAX LEFT UNMASKED DIGITS

1 (N) M

A code defining the maximum number of leftmost digits to be displayed unmasked. Valid values are as follows:

0–9 = Maximum number of leftmost digits to be displayed unmasked.

Default Value: 0

3.3.3 ICF Screen 3

The ICF screen 3 (Figure 3–3) defines the timers and processing parameters.

BASE24-BASE INTERCHANGE CONFIG		EME3	99/08/18	13:40	03 OF 13
INTERCHANGE FIID: _____		PROCESS: _____			
TIMER LIMITS			PROCESSING OPTIONS		
NETWORK MANAGEMENT:	30 (SEC)	ACQUIRER:	N (Y/N)		
EXTENDED NETWORK:	30 (SEC)	ISSUER:	N (Y/N)		
WAIT FOR TRAFFIC:	600 (SEC)	PROCESSING MODE:			
PERFORMANCE PERIOD:	20 (MIN)	AUTO SIGNON START:	N (Y/N)		
MAXIMUM OUTSTANDING TRANSACTIONS		MAXIMUM TIMEOUTS:	2		
OUTBOUND:	30	MAX SAF RETRY:	5		
INBOUND:	30	ACK TO SWITCH:	Y (Y/N)		
		ACK FROM SWITCH:	N (Y/N)		
NETWORK MANAGEMENT MESSAGE ENABLED: Y (Y = ENABLED, N = DISABLED)					
TYPE OF INTERCHANGE REPORTS: 0 (ATM/DETAIL AND SETTLEMENT)					
ILF EXTRACT NUMBER: 3					
***** BASE24 *****					
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:			
	F12-HELP				

Figure 3–3 ICF Screen 3

TIMER LIMITS

The following timer fields are used to control specific message activity between the BASE24 network and the Interchange. The values entered in the fields must be checked against the Interchange timer values to ensure the integrity of message traffic.

For more details on the use of these timers in session management processing, refer to the *V5 Interface Specifications*.

NETWORK MANAGEMENT

4 (N) M

The maximum number of seconds that the EPS-Net Interchange Interface waits for a response to a network management message; for example, sign on or sign off.

For the EPS-Net Interchange Interface, it is recommended that this field be set to 15 seconds, to comply with the session management requirements specified in the *V5 Interface Specifications*.

Default Value: 30 (SEC)

EXTENDED NETWORK

4 (N) M

For the EPS-Net Interchange Interface, this timer is used to enable the Interface to sign on again in certain situations; for example, the Interface has been signed off or the station has been reconnected.

For the EPS-Net Interchange Interface, it is recommended that this field be set to 30 seconds, to comply with the session management requirements specified in the *V5 Interface Specifications*.

Default Value: 60 (SEC)

WAIT FOR TRAFFIC

4 (N) M

For the EPS-Net Interchange Interface, this timer is used to enable the Interface to send a “repeat sign on” message to the EM. Repeat sign on messages are used to ensure that the application session with the EM remains open when there has been no traffic on the line.

For the EPS-Net Interchange Interface, it is recommended that this field be set to 600 seconds, to comply with the session management requirements specified in the *V5 Interface Specifications*.

Default Value: 60 (SEC)

PERFORMANCE PERIOD

4 (N) M

This field specifies the duration (in minutes) of an EPS-Net Interchange performance monitoring period. For the EPS-Net Interchange Interface, the timer is set to 20 minutes.

Default Value: 20 (MIN)

MAXIMUM OUTSTANDING TRANSACTIONS

The following two fields specify the maximum number of outstanding transactions inbound from and outbound to the EM. The combined value determines the number of extended memory pages that will be required for the transactions in process queue.

OUTBOUND

4 (N) M

This field specifies the maximum number of outbound transactions to the EM waiting to be processed. This field is used where the EPS-Net Interchange Interface is a transaction acquirer. For the EPS-Net Interchange Interface, the field is set to 30.

Default Value: 15

INBOUND

4 (N) M

This field specifies the maximum number of inbound transactions from the EM waiting to be processed. This field is used where the EPS-Net Interchange Interface is a transaction issuer. For the EPS-Net Interchange Interface, the field is set to 30.

Default Value: 15

PROCESSING OPTIONS

The following fields define the processing options for the EPS-Net Interchange Interface.

ACQUIRER

1 (A) M

This field indicates whether BASE24 will act as an acquirer of EPS-Net transactions. The valid values are:

Y = Yes, send transactions to the Interchange.

N = No, do not send transactions to the Interchange.

Default Value: N

ISSUER

1 (A) M

This field indicates whether BASE24 will act as issuer for EPS-Net transactions. The valid values are:

Y = Yes, accept transactions from the Interchange.

N = No, do not accept transactions from the Interchange.

Default Value: N

PROCESSING MODE

1 (A)

This field is Interchange Interface dependant and is used by the EPS-Net Interchange Interface to indicate whether the process supports UK Domestic Maestro transactions. If this field is set to 'D', then the EPS-Net Interchange Interface process supports UK Domestic Maestro transactions; otherwise, the EPS-Net Interchange Interface process does not support UK Domestic Maestro transactions.

AUTO SIGNON START

1 (A) M

This field specifies whether the EPS-Net Interchange Interface process will automatically sign on to the EM at startup. For the EPS-Net Interchange Interface, it is recommended that for normal operation, the field be set to Y. Valid values are:

Y = Yes, start up is set to automatically sign on to the Interchange.

N = No, start up is not set to automatically sign on to the Interchange.

Default Value: Y

MAXIMUM TIMEOUTS

4 (N) M

This field specifies the maximum number of consecutive timeouts allowed before network management measures are taken to determine the status of the link. For the EPS-Net Interchange Interface, it is recommended that this field be set to 2.

Default Value: 2

MAX SAF RETRY**4 (N) M**

This field specifies the maximum number of times a transaction from the SAF is transmitted to the EM before the record is written to the hard-copy log and deleted from the file. For the EPS-Net Interchange Interface, it is recommended that the field be set to 5, as defined in the *V5 Interface Specifications*.

Default Value: 0 (Transactions are transmitted indefinitely until the transmission is successful.)

ACK TO SWITCH**1 (A) M**

This field specifies whether the EPS-Net Interchange Interface process acknowledges advice messages from the Interchange. For EPS-Net, this field must be set to Y. Valid values are:

Y = Yes, acknowledges advices.

N = No, does not acknowledge advices.

Default Value: Y

ACK FROM SWITCH**1 (A) M**

The EPS-Net Interchange Interface process does not use this field.

NETWORK MANAGEMENT MESSAGE ENABLED**1 (A) M**

This field specifies whether network management messages between the EPS-Net Interchange Interface and the EM are enabled or disabled. For the EPS-Net Interchange Interface, this field must be set to Y. Valid values are:

Y = Enabled

N = Disabled

Default Value: N

TYPE OF INTERCHANGE REPORTS

1 (N)

This field specifies the types of reports the EPS-Net Interchange requires.
Valid values are:

- 0 = ATM (Detail and settlement reports)
- 1 = POS (Detail and settlement reports)
- 2 = POS (Detail report)
- 3 = POS (Debit side detail & settlement reports, including credit side detail report)

Default Value: 0

ILF EXTRACT NUMBER

1 (N)

This field defines the number of ILFs extracted for host reporting and is used by the Super Extract process to indicate the number of ILFs to extract.

Default Value: 3

Related Field: ECF RPT-EXTRACT

3.3.4 ICF Screen 6

The ICF screen 6 (Figure 3–4) defines parameters for Non–Currency Dispense transactions.

```

BASE24-NCD      INTERCHANGE CONFIG      EME3      99/08/18  13:40  06 OF 13
INTERCHANGE FIID:      _____      PROCESS:      _____

                                NCD ICF DATA
                                NOT ON US
NON-CRNCY DISP      0      TRANSACTIONS ALLOWED TO SWITCH
NON-CRNCY DISP CC  0      NON-CRNCY DISP      N
                                NON-CRNCY DISP CC  N

                                DEFAULT MERCHANT TYPE: _____

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                                F12-HELP

```

Figure 3–4 ICF Screen 6

NCD ICF DATA

The following fields contain information relevant to Non–Currency Dispense transactions.

NOT ON US

1 (N) M

Codes defining the circumstances in which incoming transactions from the EM are allowed (that is, where BASE24-atm cardholders use ATMs connected to and controlled by institutions serviced by an interchange).
Valid values are:

- 0 = Not allowed
- 1 = Allowed within the county
- 2 = Allowed within the state
- 3 = Allowed nationally
- 4 = Allowed internationally

Default Value: 0

TRANSACTIONS ALLOWED TO SWITCH

1 (A) M

The fields represent the transaction types that are supported by BASE24-atm and can be sent to MasterCard Europe. A value of Y is entered against these transactions; all others are set to N. Valid values are:

Y = Allowed

N = Not allowed

Default Value: N

DEFAULT MERCHANT TYPE

4 (A)

The default merchant type that is sent in a Non-Currency Dispense transaction.

3.3.5 ICF Screen 7

The ICF screen 7 (Figure 3–5) defines the BASE24-atm allowable transactions.

```

BASE24-ATM    INTERCHANGE CONFIG    EME3    99/08/18  13:40  07 OF 13
INTERCHANGE FIID:    _____    PROCESS:    _____

                                ATM ICF DATA

      NOT ON US                                TRANSACTIONS ALLOWED TO SWITCH
WITHDRAWAL                0                WITHDRAWAL                N
WITHDRAWAL CC              0                WITHDRAWAL CC              N
DEPOSIT                    0                DEPOSIT                    N
INQUIRY                    0                INQUIRY                    N
TRANSFER                   0                TRANSFER                   N
ELECTRONIC PAYMENT         0                ELECTRONIC PAYMENT         N
PAYMENT ENCLOSED           0                PAYMENT ENCLOSED           N
CASH CHECK                 0                CASH CHECK                 N
MSG TO INSTITUTION         0                MSG TO INSTITUTION         N
PIN CHANGE                 0                PIN CHANGE                 N
SPLIT DEPOSIT              0                SPLIT DEPOSIT              N
LOAD VALUE                 0                LOAD VALUE                 N
VALUES FOR NOT ON US ARE 'O' THRU '4', TRANS ALLOWED TO SWITCH ARE 'Y' OR 'N'
0 = NOT ALLOWED  1 = INTRACOUNTY  2 = INTRASTATE  3 = INTERSTATE  4 = INTNAT'L
***** BASE24 *****
NEW PAGE:                FILE DESTINATION:                NEW LOGICAL NETWORK ID:
                                F12-HELP
  
```

Figure 3–5 ICF Screen 7

ATM ICF DATA

The following fields define the BASE24-atm allowable transactions.

NOT ON US

1 (N) M

Codes defining the circumstances in which incoming transactions from the EM are allowed (that is, where BASE24-atm cardholders use ATMs connected to and controlled by institutions serviced by an interchange). For BASE24-atm users with the EPS-Net Interchange Interface, only Withdrawal, Withdrawal CC, and PIN Change transactions are currently allowed internationally. A value of 4 is entered against these transactions. To support domestic balance inquiries, a value of 3 is entered against the Inquiry transaction; all others are set to 0. Valid values are:

- 0 = Not allowed
- 1 = Allowed within the county
- 2 = Allowed within the state
- 3 = Allowed nationally
- 4 = Allowed internationally

Default Value: 0

TRANSACTIONS ALLOWED TO SWITCH

1 (A) M

The fields represent the transaction types that are supported by BASE24-atm and can be sent to MasterCard Europe. For BASE24-atm users with the EPS-Net Interchange Interface, the transactions listed in section 5.5.1 are currently allowed. A value of Y is entered against these transactions; all others are set to N. Valid values are:

Y = Allowed

N = Not allowed

Default Value: N

3.3.6 ICF Screen 8

The ICF screen 8 (Figure 3–6) defines the BASE24-atm sharing groups and the business traffic timers.

```

BASE24-ATM    INTERCHANGE CONFIG    EME3    99/08/18  13:40  08 OF 13
INTERCHANGE FIID:    _____    PROCESS:    _____

                                ATM ICF DATA

                                AUTH PROCESS: GATEWAY
                                DEFAULT ROUTING GROUP: 000000000000

                                TIMER LIMITS
                                STORE AND FORWARD: 15 (SEC)
                                OUTBOUND: 30 (SEC)
                                INBOUND: 30 (SEC)
                                COMPLETION: 60 (SEC)
                                COMPLETION ACK: 30 (SEC)

SHARING GROUPS:

***** BASE24 *****
NEW PAGE:    FILE DESTINATION:    NEW LOGICAL NETWORK ID:
                                F12-HELP
  
```

Figure 3–6 ICF Screen 8

ATM ICF DATA

The following fields contain information relevant to BASE24-atm sharing groups and the business traffic timers.

AUTH PROCESS 16 (A)

This field defines the Authorisation process to which the EPS-Net Interchange Interface process routes ATM transactions if it is servicing only one BASE24 logical network. If the Interchange is servicing more than one logical network, then the field should contain the word *GATEWAY*, so as to enable the EPS-Net Interchange Interface process to route the transaction.

DEFAULT ROUTING GROUP 11 (A)

This field contains the ATM routing group identification.

Default Value: 0

TIMER LIMITS

The following fields set the time limits for BASE24-atm transactions.

STORE AND FORWARD

4 (N) M

This field specifies the maximum number of seconds the EPS-Net Interchange Interface will wait for a response to a store-and-forward advice transaction. The value specified in the field is used in the first attempt to send the store-and-forward advice. For the EPS-Net Interchange Interface, the recommended minimum value is 15 seconds.

Default Value: 30 (SEC)

OUTBOUND

4 (N) M

This timer is used if the institution is an ATM transaction acquirer. It specifies the time limit associated with transaction responses from the EM. Responses must be received from the EM before the timer expires. For the EPS-Net Interchange Interface, the recommended minimum value is 30 seconds. If this value is set too low then the EM will not respond to the EPS-Net Interchange Interface within the allowed time period.

Default Value: 15 (SEC)

INBOUND

4 (N) M

This timer is used if the institution is an ATM transaction issuer. It specifies the time limit associated with transaction authorisation responses from a BASE24-atm Authorisation process (or a back-end host). The EPS-Net Interchange Interface process must receive responses before the timer expires. For the EPS-Net Interchange Interface, it is recommended that the timer be set to 30 seconds.

Default Value: 15 (SEC)

COMPLETION

4 (N) M

The EPS-Net Interchange Interface process does not use this field.

COMPLETION ACK

4 (N) M

The EPS-Net Interchange Interface process does not use this field.

SHARING GROUPS**1 (A)**

A maximum of 24 fields is allowed. Each field specifies a one-character code to identify Interchange sharing groups for both incoming and outgoing transactions. Zero is not a valid entry. Codes are not separated by blank spaces; however, unused portions to the right remain blank.

For outgoing transactions to MasterCard Europe from BASE24, if the acquiring terminal belongs to a sharing group, as indicated in the ATD (BASE24-atm) record, one or more of the ATD entries must match the sharing group codes entered in the ICF; otherwise the transaction will be denied.

For incoming transaction requests from MasterCard Europe, the sharing groups entered in the ICF are passed to the Authorisation process to check that one of the codes matches the sharing groups specified in the IDF record associated with the cardholder's institution. Again, the transaction will be declined if there is not at least one matching sharing group.

Related Fields: IDF SHARING GROUPS (for BASE24-atm users only)
 ATD SHARING GROUPS (for BASE24-atm users
 only)

3.3.7 ICF Screen 9

The ICF screen 9 (Figure 3–7) displays the POS transactions allowed to the Interchange.

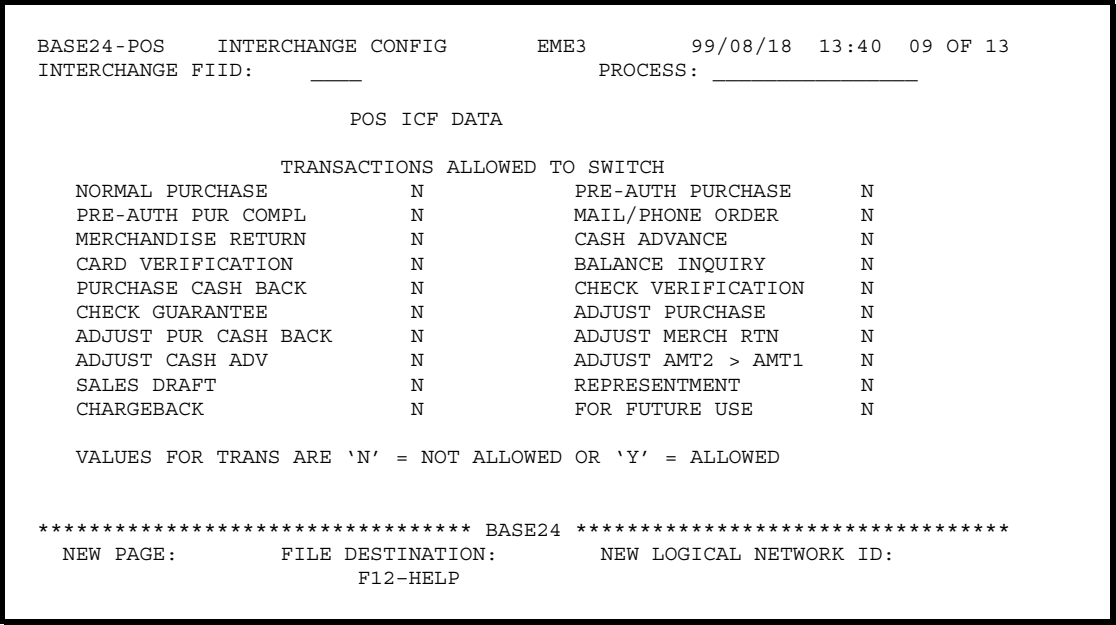


Figure 3–7 ICF Screen 9

POS ICF DATA

The following fields contain information relevant to BASE24-pos.

TRANSACTIONS ALLOWED TO SWITCH 1 (A) M

The fields displayed on the screen represent the transaction types that are supported by BASE24-pos and can be allowed or disallowed to MasterCard Europe. For BASE24-pos users with the EPS-Net Interchange Interface, the transactions listed in section 5.5.2 are currently allowed. These transactions are set to Y; all others are set to N. Valid values are:

- Y = Allowed
- N = Not allowed
- Default Value: N

3.3.8 ICF Screen 10

The ICF screen 10 (Figure 3–8) displays the POS transaction message timer limits, POS Authorisation process name, referral phone number, POS default retailer identifier, and POS timeout action.

```

BASE24-POS      INTERCHANGE CONFIG      EME3      99/08/18  13:40  10 OF 13
INTERCHANGE FIID:      _____      PROCESS:      _____

                                POS ICF DATA

                                AUTH PROCESS: GATEWAY
                                REFERRAL PHONE NUMBER: 000000000000000000
                                RETAILER ID DEFAULT: _____
                                TIMEOUT ACTION: 0 (DECLINE)
                                SETTLE ENTITY: 0 (NO DRAFT CAPTURE)

                                TIMER LIMITS

                                STORE AND FORWARD: 15 (SEC)
                                OUTBOUND: 30 (SEC)
                                INBOUND: 30 (SEC)
                                COMPLETION: 60 (SEC)
                                COMPLETION ACK: 30 (SEC)

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                                F12-HELP

```

Figure 3–8 ICF Screen 10

POS ICF DATA

The following fields contain information relevant to BASE24-pos.

AUTH PROCESS

16 (A)

This field defines the Authorisation process to which the EPS-Net Interchange Interface process routes BASE24-pos transactions if it is servicing only one BASE24 logical network. If the Interchange is servicing more than one logical network then the field should contain the word *GATEWAY*, so as to enable the EPS-Net Interchange Interface process to route the transaction.

REFERRAL PHONE NUMBER

16 (A)

This field defines the referral phone number that the EPS-Net Interchange Interface process places in all internal PSTM messages.

RETAILER ID DEFAULT

19 (A)

This field defines the default retailer identifier that the EPS-Net Interchange Interface process places in all internal PSTM messages.

Default Value: Blanks

TIMEOUT ACTION

1 (N) M

This field defines the action the EPS-Net Interchange Interface process takes when a timer waiting for a response from the EM expires before the response is received. The valid values are:

- 0 = Decline
- 1 = Default action
- 2 = Alternate route

Default Value: 0

SETTLE ENTITY

1 (N) M

A code specifying the type of settlement that should be performed for incoming POS requests. The PSTM field from TRAN.DFT-CAPTURE-FLG is set from this field. For EPS-Net, the recommended value is 0 because transactions are non-draft capture. The valid values are:

- 0 = No draft capture
- 1 = Draft capture

Default Value: 0

TIMER LIMITS

The following fields set the time limits for BASE24-pos transactions.

STORE AND FORWARD

4 (N) M

This field specifies the maximum number of seconds the EPS-Net Interchange Interface waits for a response to a store-and-forward advice transaction. The value specified in the field is used in the first attempt to send the store-and-forward advice. For the EPS-Net Interchange Interface, the recommended minimum value is 15 seconds.

Default Value: 30 (SEC)

OUTBOUND**4 (N) M**

This timer is used if the institution is a BASE24-pos transaction acquirer. It specifies the time limit associated with transaction responses from MasterCard Europe for POS transactions. Responses from MasterCard Europe for transactions acquired by BASE24-pos must be received before the timer expires. For the EPS-Net Interchange Interface, the recommended minimum value is 30 seconds.

Default Value: 15 (SEC)

INBOUND**4 (N) M**

This timer is used if the institution is a BASE24-pos transaction issuer. It specifies the time limit associated with transaction responses from a BASE24-pos Authorisation process (or a back-end host). If the EPS-Net Interchange Interface process has not received an authorisation response before the timer expires then the transaction is declined. The recommended value for the EPS-Net Interchange Interface is 30 seconds.

Default Value: 15 (SEC)

COMPLETION**4 (N) M**

The EPS-Net Interchange Interface process does not use this field.

COMPLETION ACK**4 (N) M**

The EPS-Net Interchange Interface process does not use this field.

3.3.9 ICF Screen 11

The ICF screen 11 (Figure 3–9) displays the POS pre-authorisation default time and amount, approval code length expected by the Interchange, and services allowed to the Interchange.

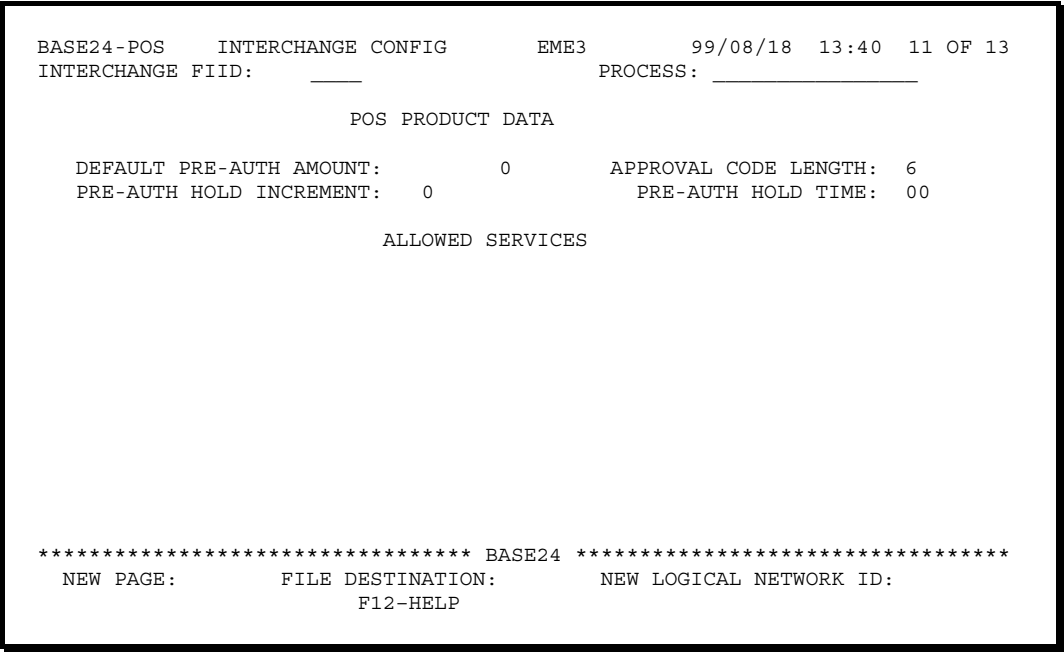


Figure 3–9 ICF Screen 11

POS PRODUCT DATA

The following fields contain information relevant to BASE24-pos.

DEFAULT PRE-AUTH AMOUNT 5 (N)

The EPS-Net Interchange Interface process does not use this field.

APPROVAL CODE LENGTH 1 (N)

The EPS-Net Interchange Interface process does not use this field. If ISO data element 027 (Authorisation Identification Response Length) is received in a request from EPS-Net, this is passed through BASE24 in PSTM field TRAN.APPRV-CDE-LGTH; otherwise, a default of 6 is used. The value indicates the number of digits that an interchange interface response message can handle in an authorisation identification response code.

PRE-AUTH HOLD INCREMENT **1 (N)**

The EPS-Net Interchange Interface process does not use this field.

PRE-AUTH HOLD TIME **2 (N)**

The EPS-Net Interchange Interface process does not use this field.

ALLOWED SERVICES **2 (A)**

A maximum of 30 fields is allowed. These fields define the card schemes (card types) supported between the EPS-Net Interchange Interface and MasterCard Europe.

3.3.10 ICF Screen 12

The ICF screen 12 (Figure 3–10) displays the EPS-Net Interchange Interface specific processing parameters.

```

BASE24-BASE   ICF FILE MAINTENANCE   EME3           99/08/18  13:40  12 OF 12
INTERCHANGE FIID: _____ PROCESS: _____

                EPS-NET SWITCH SPECIFIC DATA

    AUTHORIZATION ONLY: Y (Y/N)                TEST: N (Y/N)

STATION 1 - ACQUIRER: N (Y/N)                ISSUER: N (Y/N)
STATION 2 - ACQUIRER: N (Y/N)                ISSUER: N (Y/N)

                EBCDIC: N (Y/N)    VERSION NUMBER: 30

                EPS-NET INSTITUTION ID: _____

                ILF HEADER MAX TRAN UPDATE: 0200

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
F8-GO TO EPS-NET MENU      F12-HELP
  
```

Figure 3–10 ICF Screen 12

AUTHORIZATION ONLY

1(A)

The EPS-Net Interchange Interface process does not use this field.

TEST

1 (A) M

This field specifies whether the EPS-Net Interchange Interface process is in test or production mode. This field is set to Y during testing. At all other times the field must be set to N. Valid values are:

Y = Test mode

N = Not in test mode

Default Value: N

STATION 1: ACQUIRER**1 (A)**

This field specifies whether Station 1 for the EPS-Net Interchange Interface process is in acquirer mode. The station name is defined on the ICF screen 1. If there is a Station Name File (SNF) record configured for the process, then this field is not used. Valid values are:

Y = Interface is an acquirer

N = Interface is not an acquirer

Default Value: N

STATION 1: ISSUER**1 (A)**

This field specifies whether Station 1 for the EPS-Net Interchange Interface process is in issuer mode. The station name is defined on ICF screen 1. If there is a Station Name File (SNF) record configured for the process, then this field is not used. Valid values are:

Y = Interface is an issuer

N = Interface is not an issuer

Default Value: N

STATION 2: ACQUIRER**1 (A)**

This field specifies whether Station 2 for the EPS-Net Interchange Interface process is in acquirer mode. The station name is defined on ICF screen 1. If there is a Station Name File (SNF) record configured for the process, then this field is not used. Valid values are:

Y = Interface is an acquirer

N = Interface is not an acquirer

Default Value: N

STATION 2: ISSUER**1 (A)**

This field specifies whether Station 2 for the EPS-Net Interchange Interface process is in issuer mode. The station name is defined on the ICF screen 1. If there is a Station Name File (SNF) record configured for the process, then this field is not used. Valid values are:

Y = Interface is an issuer

N = Interface is not an issuer

Default Value: N

EBCDIC

1 (A) M

This field determines whether the external message to the EM is in ASCII or EBCDIC. For the EPS-Net Interchange Interface, it is recommended that the value be set to N. Valid values are:

Y = EBCDIC

N = ASCII

Default Value: Y

VERSION NUMBER

2 (N) M

This field indicates the version of the Host-EM Interface.

Default Value: 30

EPS-NET INSTITUTION ID

11 (A) M

This field defines the Receiving Institution Identification Number (ISO field DE 100). MasterCard Europe assigns this number.

Default Value: Spaces

ILF HEADER MAX TRAN UPDATE

4 (N) M

This field specifies the number of transactions after which the ILF header record is updated. The ILF header record is used during process initialisation to determine the current trace number and authorisation code used in the EPS-Net external messages. The ILF header enables the EPS-Net Interchange Interface to determine the values without reading multiple ILF files in their entirety.

Default Value: 0200

3.3.11 ICF Menu Screen

The ICF menu screen (Figure 3–11) provides access to files specific to the EPS-Net Interchange Interface.

```
BASE24-BASE   ICF FILE MAINTENANCE   EME3       99/08/18  13:40  12 OF 12
INTERCHANGE FIID: _____          PROCESS: _____

- STATION NAME FILE (SNF)
- EUROCARD INSTITUTION IDENTIFICATION FILE (ERIIF)
- EPS-NET REQUEST LOG FILE (ENRLF)

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP
```

Figure 3–11 ICF Menu Screen

EPS-NET FILE MENU

The desired screen is selected by placing the cursor to the left of the entry and pressing function key F1.

3.4 KEY FILE

The KEYF contains the information and parameters required by the BASE24 Interchange Interface processes for PIN encryption, PIN translation, message authentication and dynamic key management.

The ISO Host Interface process and Interchange Interface processes use the KEYF instead of the KEY6 when the LCONF param DES-TRIPLE-SINGLE is either set to a value of N or not present. Otherwise these processes use the KEY6.

The KEY File (KEYF) contains one record for each:

- DPC/Host Interface process
- Interchange Interface process

The key to records in the KEYF is a combination of the values in the DPC/FIID and INTERFACE PROCESS fields.

The following screens are used to access data in the KEYF:

- **KEYF screen 1** contains PIN and MAC processing parameters, intermediate keys, and PIN and MAC key exchange keys.
- **KEYF screen 2** contains inbound and outbound keys for PINs and MACs.
- **KEYF screen 3** contains dynamic key management parameters.
- **KEYF screen 4** contains message encryption parameters.

Screens 3 and 4 are not covered here, as the EPS-Net Interchange Interface does not use dynamic key management or message encryption.

If the BASE24 system uses the TSS, then the security keys held on the KEYF are replaced by key locators (to the actual security keys held on the TSS database).

For more information on the usage of the KEYF, refer to the following:

- *BASE24 Base Files Maintenance Manual*
- *BASE24 Transaction Security Manual*

3.4.1 KEYF Screen 1

The KEYF screen 1 (Figure 3–12) contains PIN and MAC processing parameters, intermediate keys, and PIN and MAC key exchange keys.

```

BASE24-BASE      KEY FILE      EME3      99/08/18  13:40  01 OF 04
DPC/FIID: _____ INTERFACE PROCESS: _____

    ENCRYPT TYPE: 0 (NO ENCRYPTION)  BASE24 ENCRYPT TYPE: 0 (NO ENCRYPTION)
PIN BLOCK FORMAT: 0 (CLEAR)          ANSI PAN FORMAT: 0 (12 RIGHT/NO CHK)
MAC ENCRYPT TYPE: 0 (NO PROCESSING)   PIN PAD CHARACTER: F
    MAC DATA TYPE: 0 (ASCII)          NUMBER OF KEYS: 1 (COMBINED)
    KEY LENGTH: 1 (SINGLE)              FULL MESSAGE MAC: N (SELECTED FIELDS)

                                INTERMEDIATE KEYS

    CLEAR: 000000000000000000000000    CHECK DIGITS: 0000
    ENCRYPTED: 000000000000000000000000

                                EXCHANGE KEYS

    PIN KEY: 000000000000000000000000  000000000000000000000000  CHECK DIGITS: 0000
    MAC KEY: 000000000000000000000000  000000000000000000000000  CHECK DIGITS: 0000

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

Figure 3–12 KEYF Screen 1

DPC/FIID

4 (AN) M

The FIID of the interchange using this KEYF record. The value in this field matches the value in the INTERCHANGE FIID field on ICF screen 1.

INTERFACE PROCESS

16 (A) M

The name of the interface process associated with the interchange identified in the DPC/FIID field. The value in this field matches the value in the PROCESS field on ICF screen 1.

ENCRYPT TYPE

1 (N) M

Specifies the type of PIN management used by the interchange. The recommended value for PIN-based transactions is 1. Valid values are:

- 0 = Clear PINs
- 1 = Security module PIN management
- 2 = Software DES PIN management

Default Value: 0

B24 ENCRYPT TYPE

1 (N) M

Specifies the type of internal PIN encryption between the EPS-Net Interchange and BASE24. For transactions outbound to the EM, the type of internal PIN encryption is specified in the Standard Internal Message (STM or PSTM) in the PIN format field. The recommended value for PIN-based transactions is 1. Valid values are:

- 0 = Clear PINs
- 1 = Security module PIN management

Default Value: 0

PIN BLOCK FORMAT

1 (N) M

Specifies the PIN block format of the PIN in the inbound and outbound external messages. The recommended value for PIN-based transactions is 1. Valid values are:

- 0 = Clear PINs
- 1 = ANSI (PIN/PAN) PIN block
- 3 = PIN/PAD PIN block

While option 1 is called the ANSI PIN block, it is also known as the PIN/PAN PIN block because it includes three PAN formats, only one of which is part of the ANSI standard. The PAN format being used is specified in the ANSI PAN FORMAT field on this screen.

Default Value: 0

ANSI PAN FORMAT**1 (N) M**

If the value in the PIN BLOCK FORMAT field is 1 (ANSI), the value in this field specifies which PAN digits are used in the formation of the PIN/PAN PIN block in the external message. The recommended value for PIN-based transactions is 0. Valid values are:

- 0 = 12 right-most digits of the PAN, excluding the check digit (ANSI standard)
- 1 = 12 right-most digits of the PAN, including the check digit
- 2 = 12 left-most digits of the PAN

The ANSI standard is one of three PAN formats available with PIN/PAN PIN blocks. While this field is called the ANSI PAN FORMAT, it also includes two other PAN formats that are not part of the ANSI standard. The ANSI standard is also known as ISO Format 0.

Default Value: 0

MAC ENCRYPT TYPE**1 (N)**

The EPS-Net Interchange Interface process does not use this field. This field should be set to 0 because EPS-Net does not currently support message authentication.

PIN PAD CHAR**1 (A) M**

If the value in the PIN BLOCK FORMAT field is 3 (PIN/PAD), the value in this field specifies the PAD character used in the formation of the external message's PIN block. The valid values are A–F.

Default Value: F

MAC DATA TYPE**1 (N)**

The EPS-Net Interchange Interface process does not use this field.

NUMBER OF KEYS**1 (N)**

The EPS-Net Interchange Interface process does not use this field.

KEY LENGTH **1 (N)**

The EPS-Net Interchange Interface process does not use this field. Key Exchange Keys (KEKs) will always be single length.

FULL MESSAGE MAC **1 (A)**

The EPS-Net Interchange Interface process does not use this field.

INTERMEDIATE KEYS

Intermediate keys are used whenever a PIN that is encrypted under a secure key (security module PIN management) must be decrypted into the clear.

CLEAR **16 hexadecimal characters**

The EPS-Net Interchange Interface process does not use this field.

CHECK DIGITS **4 hexadecimal characters**

The EPS-Net Interchange Interface process does not use this field.

ENCRYPTED **16 hexadecimal characters M**

Security module encrypted intermediate key. A PIN that has been encrypted under a secure key will be translated to be encrypted under the intermediate key at the security module. The valid values for each character in this field are 0–9 and A–F. This field must contain 16 valid characters.

Default Value: 0000000000000000

EXCHANGE KEYS

When PIN or MAC keys are automatically exchanged between the Interchange Interface and the interchange, they are to be encrypted under this key for the interchange. The exchange key must be manually exchanged between the two entities.

PIN KEY **2 sets of 16 hexadecimal characters** **M**

The EPS-Net Interchange Interface process does not use this field.

CHECK DIGITS **4 hexadecimal characters** **M**

The EPS-Net Interchange Interface process does not use this field.

MAC KEY **2 sets of 16 hexadecimal characters**

The EPS-Net Interchange Interface process does not use this field.

CHECK DIGITS **4 hexadecimal characters**

The EPS-Net Interchange Interface process does not use this field.

3.4.2 KEYF Screen 2

The KEYF screen 2 (Figure 3–13) contains inbound and outbound keys for PINs and MACs.

```

BASE24-BASE      KEY FILE      EME3      99/08/18  13:40  02 OF 04
DPC/FIID: _____ INTERFACE PROCESS: _____

                                PIN KEY INFORMATION

OUTBOUND KEYS:
  PIN KEY 1: 0000000000000000 CHECK DIGITS1: 0000 CURRENT INDEX: 1
  PIN KEY 2: 0000000000000000 CHECK DIGITS2: 0000 KEY COUNTER: 000000
INBOUND KEYS:
  PIN KEY 1: 0000000000000000 CHECK DIGITS1: 0000 CURRENT INDEX: 1
  PIN KEY 2: 0000000000000000 CHECK DIGITS2: 0000 KEY COUNTER: 000000

                                MAC KEY INFORMATION

OUTBOUND KEYS:
  PIN KEY 1: 0000000000000000 CHECK DIGITS1: 0000 CURRENT INDEX: 1
  PIN KEY 2: 0000000000000000 CHECK DIGITS2: 0000 KEY COUNTER: 000000
INBOUND KEYS:
  PIN KEY 1: 0000000000000000 CHECK DIGITS1: 0000 CURRENT INDEX: 1
  PIN KEY 2: 0000000000000000 CHECK DIGITS2: 0000 KEY COUNTER: 000000

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                  F12-HELP

```

Figure 3–13 KEYF Screen 2

PIN KEY INFORMATION

The following fields contain working keys for PINs.

OUTBOUND KEYS

The following fields contain the PIN keys utilised for transactions sent from the interface to the interchange.

If the external PIN management is security module PIN management, the keys should be security module encrypted.

PIN KEY1**16 hexadecimal characters M**

Used for the encryption or translation of PINs in transactions sent to the interchange. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. The valid values for each character in this field are 0–9 and A–F. This field must contain 16 valid characters.

This value is known as the Zone Master Key (ZMK) and is manually exchanged between the member and MasterCard Europe.

For users of Thales e-security (formerly Racal) HSMs this key is encrypted under Local Master Key (LMK) pair 06-07.

For users of Atalla HSMs this key is encrypted under variant 1 of the Master File Key (MFK).

Default Value: 0000000000000000

CHECK DIGITS1**4 hexadecimal characters**

The EPS-Net Interchange Interface process does not use this field.

CURRENT INDEX**1 (N) M**

Indicates which outbound PIN encryption key is currently being used by the interface. Valid values are:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

Default Value: 1

PIN KEY2**16 hexadecimal characters M**

Used for the encryption or translation of PINs in transactions sent to the interchange. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. The valid values for each character in this field are 0–9 and A–F. This field must contain 16 valid characters.

This value is known as the Zone Master Key (ZMK) and is manually exchanged between the member and MasterCard Europe.

For users of Thales e-security (formerly Racal) HSMs this key is encrypted under Local Master Key (LMK) pair 06-07.

For users of Atalla HSMs this key is encrypted under variant 1 of the Master File Key (MFK).

Default Value: 0000000000000000

CHECK DIGITS2

4 hexadecimal characters

The EPS-Net Interchange Interface process does not use this field.

KEY COUNTER

6 hexadecimal characters

The EPS-Net Interchange Interface process does not use this field.

INBOUND KEYS

The following fields contain the PIN keys used in the transactions sent by the interchange to the interface.

If the external PIN management is security module PIN management, the keys should be security module encrypted.

PIN KEY1

16 hexadecimal characters M

Used for the decryption or translation of PINs in transactions sent from the interchange. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. The valid values for each character in this field are 0-9 and A-F. This field must contain 16 valid characters.

This value is known as the Zone Master Key (ZMK) and is manually exchanged between the member and MasterCard Europe.

For users of Thales e-security (formerly Racal) HSMs this key is encrypted under Local Master Key (LMK) pair 06-07.

For users of Atalla HSMs this key is encrypted under variant 1 of the Master File Key (MFK).

Default Value: 0000000000000000

CHECK DIGITS1

4 hexadecimal characters

The EPS-Net Interchange Interface process does not use this field.

CURRENT INDEX**1 (N) M**

Indicates which inbound PIN encryption key is currently being used by the interface. Valid values are:

1 = Value in the PIN KEY1 field

2 = Value in the PIN KEY2 field

Default Value: 1

PIN KEY2**16 hexadecimal characters M**

Used for the decryption or translation of PINs in transactions sent from the interchange. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. The valid values for each character in this field are 0-9 and A-F. This field must contain 16 valid characters.

This value is known as the Zone Master Key (ZMK) and is manually exchanged between the member and MasterCard Europe.

For users of Thales e-security (formerly Racal) HSMs this key is encrypted under Local Master Key (LMK) pair 06-07.

For users of Atalla HSMs this key is encrypted under variant 1 of the Master File Key (MFK).

Default Value: 0000000000000000

CHECK DIGITS2**4 hexadecimal characters**

The EPS-Net Interchange Interface process does not use this field.

KEY COUNTER**6 hexadecimal characters**

The EPS-Net Interchange Interface process does not use this field.

3.5 KEY6 FILE

The Key File (KEY6) contains the information and parameters required by release 6.0 BASE24 Host Interface and Interchange Interface processes running in release 5.x or 6.0 networks for PIN encryption, PIN translation, message authentication and dynamic key management.

The KEY6 allows for double length encryption keys which are required when translating PINS from encryption under a double length key using the Triple DES algorithm to encryption under a single length key using the Single DES algorithm and vice versa.

The ISO Host Interface process and Interchange Interface processes use the KEY6 instead of the KEYF when the LCONF param DES-TRIPLE-SINGLE is set to a value of Y. Otherwise, these processes use the KEYF.

The KEY File (KEY6) contains one record for each:

- DPC/Host Interface process
- Interchange Interface process

The key to records in the KEY6 is a combination of the values in the DPC/FIID and INTERFACE PROCESS fields.

The following screens are used to access data in the KEY6:

- **Screen 1** contains PIN and MAC processing parameters, intermediate keys, and PIN and MAC key exchange keys.
- **Screen 2** contains inbound and outbound keys for PINs.
- **Screen 3** contains inbound and outbound keys for MACs.
- **Screen 4** contains dynamic key management parameters.

Screens 3 and 4 are not covered here, as the EPS-Net Interchange Interface does not use message authentication or dynamic key management.

For more information on the usage of the KEY6, refer to the following:

- *BASE24 BASE Files Maintenance Manual*
- *BASE24 Transaction Security Manual*

ENCRYPT TYPE

1 (N) M

Specifies the type of PIN management used by the interchange. The recommended value for PIN-based transactions is 1. Valid values are:

- 0 = Clear PINs
- 1 = Security module PIN management
- 2 = Software DES PIN management

Default Value: 0

BASE24 ENCRYPT TYPE

1 (N) M

Specifies the type of internal PIN encryption between the EPS-Net Interchange and BASE24. For transactions outbound to the EM, the type of internal PIN encryption is specified in the Standard Internal Message (STM or PSTM) in the PIN format field. The recommended value for PIN-based transactions is 1. Valid values are:

- 0 = Clear PINs
- 1 = Security module PIN management

Default Value: 0

PIN BLOCK FORMAT

1 (N) M

Specifies the PIN block format of the PIN in the inbound and outbound external messages. The recommended value for PIN-based transactions is 1. Valid values are:

- 0 = Clear PINs
- 1 = ANSI (PIN/PAN) PIN block
- 3 = PIN/PAD PIN block

While option 1 is called the ANSI PIN block, it is also known as the PIN/PAN PIN block because it includes three PAN formats, only one of which is part of the ANSI standard. The PAN format being used is specified in the ANSI PAN FORMAT field on this screen.

Default Value: 0

ANSI PAN FORMAT**1 (N) M**

If the value in the PIN BLOCK FORMAT field is 1 (ANSI), the value in this field specifies which PAN digits are used in the formation of the PIN/PAN PIN block in the external message. The recommended value for PIN-based transactions is 0. Valid values are:

- 0 = 12 right-most digits of the PAN, excluding the check digit (ANSI standard)
- 1 = 12 right-most digits of the PAN, including the check digit
- 2 = 12 left-most digits of the PAN

The ANSI standard is one of three PAN formats available with PIN/PAN PIN blocks. While this field is called the ANSI PAN FORMAT, it also includes two other PAN formats that are not part of the ANSI standard.

Default Value: 0

MAC ENCRYPT TYPE**1 (N)M**

The EPS-Net Interchange Interface process does not use this field. This field should be set to 0 because EPS-Net does not currently support message authentication.

PIN PAD CHARACTER**1 (A) M**

If the value in the PIN BLOCK FORMAT field is 3 (PIN/PAD), the value in this field specifies the PAD character used in the formation of the external message's PIN block. The valid values are A–F.

Default Value: F

MAC DATA TYPE**1 (N)M**

The EPS-Net Interchange Interface process does not use this field.

NUMBER OF KEYS**1 (N)M**

The EPS-Net Interchange Interface process does not use this field.

KEY LENGTH **1 (N)M**

The EPS-Net Interchange Interface process does not use this field. Key Exchange Keys (KEKs) will always be single length.

Default Value: 1 (single length).

FULL MESSAGE MAC **1 (A)M**

The EPS-Net Interchange Interface process does not use this field.

Default Value: N (the full message is not authenticated).

MAC KEY LENGTH **1 (N)M**

The EPS-Net Interchange Interface process does not use this field.

Default Value: 1 (single length MAC keys).

INTERMEDIATE KEYS

Intermediate keys are used whenever a PIN that is encrypted under a secure key (security module PIN management) must be decrypted into the clear.

CLEAR **16 hexadecimal characters**

The EPS-Net Interchange Interface process does not use this field.

Default Value: 0000000000000000

This field is mandatory.

CHECK DIGITS **4 hexadecimal characters**

The EPS-Net Interchange Interface process does not use this field.

Default Value: 0000

This field is mandatory.

ENCRYPTED**16 hexadecimal characters**

Security module encrypted intermediate key. This key is being used for the interim Triple DES solution when a PIN block encrypted using a double-length key is received from an interchange. A PIN block that has been encrypted under a secure key is translated to be encrypted under the intermediate key at the security module. For Thales e-security (formerly Racal) HSMs, the keys stored in this field need to be encrypted under LMK pair 06/07 with no variant applied. For Atalla HSMs, the keys stored here need to be encrypted under MFK variant 1. Valid values for each character in this field are 0–9 and A–F. This field must contain 16 valid characters.

Default Value: 0000000000000000

This field is mandatory.

EXCHANGE KEYS

When PIN or MAC keys are automatically exchanged between the Interchange Interface and the interchange, they are to be encrypted under this key for the interchange. The exchange key must be manually exchanged between the two entities.

PIN KEY**2 sets of 16 hexadecimal characters M**

The EPS-Net Interchange Interface process does not use this field.

Default Value: 0000000000000000

CHECK DIGITS**4 hexadecimal characters M**

The EPS-Net Interchange Interface process does not use this field.

Default Value: 0000

MAC KEY**2 sets of 16 hexadecimal characters M**

The EPS-Net Interchange Interface process does not use this field.

Default Value: 0000000000000000

CHECK DIGITS**4 hexadecimal characters M**

The EPS-Net Interchange Interface process does not use this field.

Default Value: 0000

3.5.2 KEY6 Screen 2

The KEY6 screen 2 (Figure 3–15) contains inbound and outbound keys for PINs.

BASE24-BASE	KEY6 FILE	LLLL	YY/MM/DD	HH:MM	02 OF 04
DPC/FIID: _____		INTERFACE PROCESS: _____			
PIN KEY INFORMATION					
OUTBOUND KEYS:					
PIN KEY 1:	0000000000000000	CHECK DIGITS1:	0000	CURRENT INDEX:	1
	0000000000000000				
PIN KEY 2:	0000000000000000	CHECK DIGITS2:	0000	KEY COUNTER:	000000
	0000000000000000				
INBOUND KEYS:					
PIN KEY 1:	0000000000000000	CHECK DIGITS1:	0000	CURRENT INDEX:	1
	0000000000000000				
PIN KEY 2:	0000000000000000	CHECK DIGITS2:	0000	KEY COUNTER:	000000
	0000000000000000				
***** BASE24 *****					
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:			
	F12-HELP				

Figure 3–15 KEY6 Screen 2

PIN KEY INFORMATION

The following fields contain working keys for PINs.

OUTBOUND KEYS

The following fields contain the PIN keys utilised for transactions sent from the interface to the interchange.

If the external PIN management is security module PIN management, the keys should be security module encrypted.

PIN KEY1 **2 fields of 16 hexadecimal characters M**

A double-length key used for the encryption or translation of PINs in transactions sent to the interchange. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 1. The valid values for each character in these fields are 0–9 and A–F. These fields must contain 32 valid characters, with 16 characters per field.

This value is known as the Zone Master Key (ZMK) and is manually exchanged between the member and MasterCard Europe.

For users of Thales e-security (formerly Racal) HSMs this key is encrypted under Local Master Key (LMK) pair 06-07 variant A6/5A.

For users of Atalla HSMs this key is encrypted under variant 1 of the Master File Key (MFK).

Default Value: 0000000000000000 per field

CHECK DIGITS1 **4 hexadecimal characters M**

The EPS-Net Interchange Interface process does not use this field.

Default Value: 0000

CURRENT INDEX **1 (N) M**

Indicates which outbound PIN encryption key is currently being used by the interface. Valid values are:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

Default Value: 1

PIN KEY2 **2 fields of 16 hexadecimal characters M**

A double-length key used for the encryption or translation of PINs in transactions sent to the interchange. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. The valid values for each character in these fields are 0–9 and A–F. These fields must contain 32 valid characters, with 16 characters per field.

This value is known as the Zone Master Key (ZMK) and is manually exchanged between the member and MasterCard Europe.

For users of Thales e-security (formerly Racal) HSMs this key is encrypted under Local Master Key (LMK) pair 06-07 variant A6/5A.

For users of Atalla HSMs this key is encrypted under variant 1 of the Master File Key (MFK).

Default Value: 0000000000000000

CHECK DIGITS2

4 hexadecimal characters M

The EPS-Net Interchange Interface process does not use this field.

Default Value: 0000

KEY COUNTER

6 hexadecimal characters M

The EPS-Net Interchange Interface process does not use this field.

Default Value: 000000

INBOUND KEYS

The following fields contain the PIN keys used in the transactions sent by the interchange to the interface.

If the external PIN management is security module PIN management, the keys should be security module encrypted.

PIN KEY1

2 fields of 16 hexadecimal characters M

A double-length key used for the decryption or translation of PINs in transactions sent from the interchange to the BASE24 transaction processing system. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 1. The valid values for each character in these fields are 0-9 and A-F. This field must contain 32 valid characters, with 16 characters per field.

This value is known as the Zone Master Key (ZMK) and is manually exchanged between the member and MasterCard Europe.

For users of Thales e-security (formerly Racal) HSMs this key is encrypted under Local Master Key (LMK) pair 06-07 variant A6/5A.

For users of Atalla HSMs this key is encrypted under variant 1 of the Master File Key (MFK).

Default Value: 0000000000000000 per field

CHECK DIGITS1 **4 hexadecimal characters M**

The EPS-Net Interchange Interface process does not use this field.

Default Value: 0000

CURRENT INDEX **1 (N) M**

Indicates which inbound PIN encryption key is currently being used by the interface. Valid values are:

1 = Value in the PIN KEY1 field

2 = Value in the PIN KEY2 field

Default Value: 1

PIN KEY2 **2 fields of 16 hexadecimal characters M**

A double-length key used for the decryption or translation of PINs in transactions sent from the interchange to the BASE24 transaction processing system. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 2. The valid values for each character in this field are 0-9 and A-F. These fields must contain 32 valid characters, with 16 characters per field.

This value is known as the Zone Master Key (ZMK) and is manually exchanged between the member and MasterCard Europe.

For users of Thales e-security (formerly Racal) HSMs this key is encrypted under Local Master Key (LMK) pair 06-07 variant A6/5A.

For users of Atalla HSMs this key is encrypted under variant 1 of the Master File Key (MFK).

Default Value: 0000000000000000

CHECK DIGITS2 **4 hexadecimal characters M**

The EPS-Net Interchange Interface process does not use this field.

Default Value: 0000

KEY COUNTER **6 hexadecimal characters M**

The EPS-Net Interchange Interface process does not use this field.

Default Value: 000000

3.6 SWITCH TERMINAL FILE

The Switch Terminal File (STF) may contain one record for each terminal that supports MasterCard transactions. The STF allows the EPS-Net Interchange Interface to identify whether a specific terminal has been certified for EMV processing by MasterCard.

For customers with some terminals certified for EMV processing by MasterCard and others not certified for EMV processing by MasterCard, STF records will be required.

The “default” certification status will be specified by the EMV CERTIFIED FLAG field value in a “wild card” STF record (i.e. where the TERMINAL ID is set to asterisks). Individual terminals with the opposite certification status must be configured individually in the STF (i.e. where the TERMINAL ID is not set to asterisks), with the EMV CERTIFIED FLAG field set to the opposite value to that in the “wild card” record.

Acquirers running both a BASE24-atm system and a BASE24-pos system must configure two “wild card” STF records, one containing a PROD ID of ‘01’ (ATM), and the other containing a PROD ID of ‘02’ (POS).

For acquirers running a BASE24-atm system, the MERCHANT ID must be set to asterisks in both the “wild card” STF record and individual terminal STF records. For acquirers running a BASE24-pos system, the MERCHANT ID must be set to asterisks in the “wild card” STF record, and may be set to asterisks in individual terminal STF records.

One screen is used to access records in the STF. The fields on this screen used by the EPS-Net Interchange Interface are described in the section below.

Operator access to the STF is permitted in accordance with the security access levels assigned in the BASE24 CRT access system. For more information on the BASE24 CRT access system, refer to the ***BASE24 User Security Manual***.

For more details on the STF, refer to the ***BASE24 Base Files Maintenance Manual***.

3.6.1 STF Screen 1

The STF screen 1 (Figure 3–16) contains switch terminal and merchant information.

BASE24-SWI		SWITCH TERMINAL FILE		TES9	07/05/05	05:06	01 OF 01
SWITCH FIID: _____		MERCHANT ID: _____					
TERMINAL FIID: _____		TERMINAL ID: _____		PROD ID: ____			
SWITCH MERCHANT INFORMATION				SWITCH TERMINAL INFORMATION			
ID: _____				ID: _____			
OFFSET: _____				OFFSET: _____			
LENGTH: _____				LENGTH: _____			
ON PREMISE FLAG: _____ (Y/N)							
EMV CERTIFIED FLAG: _____ (Y/N)							
CARDHOLDER ACTIVATED TERMINAL: _____ (NOT USED)							
TERMINAL INPUT CAPABILITIES: _____ (NOT USED)							
***** BASE24 *****							
NEW PAGE:		FILE DESTINATION:		NEW LOGICAL NETWORK ID:			
		F12-HELP					

Figure 3–16 STF Screen 1

SWITCH FIID

4 (A)

This field is an identifier for the financial institution of the interchange. The value entered in this field must be identical to a value entered in the INTERCHANGE FIID field on ICF screen 1.

MERCHANT ID

19 (A)

This field is an identifier for the merchant. When the BASE24-pos product is being used, this field is required. When the BASE24-pos product is not being used, this field should contain asterisks.

TERMINAL FIID

4 (A)

This field is the FIID associated with the terminal.

TERMINAL ID 16 (A)

This field is the terminal identifier. The value entered in this field should be left-justified.

PROD ID 2 (N) M

This field is an identifier for the product with which the terminal is associated. Valid values are:

01 = ATM

02 = POS

SWITCH MERCHANT INFORMATION

The following fields contain information about the interchange merchant.

ID 19 (A)

The EPS-Net Interchange Interface process does not use this field.

OFFSET 2(N) M

The EPS-Net Interchange Interface process does not use this field.

Default Value: 00

LENGTH 2(N) M

The EPS-Net Interchange Interface process does not use this field.

Default Value: 00

SWITCH TERMINAL INFORMATION

The following fields contain information about the interchange terminal.

ID 19 (A)

The EPS-Net Interchange Interface process does not use this field.

OFFSET **2(N) M**

The EPS-Net Interchange Interface process does not use this field.

Default Value: 00

LENGTH **2(N) M**

The EPS-Net Interchange Interface process does not use this field.

Default Value: 00

ON PREMISE FLAG **1 (A)**

The EPS-Net Interchange Interface process does not use this field.

EMV CERTIFIED FLAG **1 (A)**

This field specifies whether or not the terminal is certified for EMV processing by the interchange. Valid values are:

Y = Certified

N = Not certified

Default Value: Y

CARDHOLDER ACTIVATED TERMINAL **1(N)**

The EPS-Net Interchange Interface process uses this field to set data element 061 (POS Data) sub-field 10 (CAT Level Indicator) in messages sent to EPS-Net, if the information is not provided in the BASE24-pos internal message.

TERMINAL INPUT CAPABILITIES **1 (N)**

The EPS-Net Interchange Interface process does not use this field.

3.7 STATION NAME FILE

The Station Name File (SNF) may contain one record for each EPS-Net Interchange Interface process. Each record contains up to 16 station pairs on which messages are sent to and from EPS-Net. The key to the SNF record is the EPS-Net Interchange Interface process name.

The SNF is read at initialisation and on warmboots to obtain the names of stations to which the EPS-Net Interchange Interface is connected. If an SNF record does not exist for an interface process, then the process uses the stations configured on ICF screen 1 instead.

The SNF screen 1 is accessed from the Interchange Configuration File (ICF) Menu screen by placing the cursor to the left of the SNF menu entry and pressing function key F1. The ICF Menu screen itself is accessed from ICF screen 12 by pressing function key F8 (GOTO EPS-NET MENU).

3.7.1 SNF Screen 1

The SNF screen 1 (Figure 3–17) provides information associated with up to 16 stations used by BASE24 to communicate with EPS-Net.

BASE24-SNF SNF FILE MAINTENANCE EME3 99/08/18 13:40 01 OF 01

INTERCHANGE FIID: _____ PROCESS: _____

SEND STATION	RECEIVE STATION	TYPE	SWITCH DATA	SWITCH INFO
_____	_____	-	_____	_____
_____	_____	-	_____	_____
_____	_____	-	_____	_____
_____	_____	-	_____	_____
_____	_____	-	_____	_____
_____	_____	-	_____	_____
_____	_____	-	_____	_____
_____	_____	-	_____	_____
_____	_____	-	_____	_____
_____	_____	-	_____	_____
_____	_____	-	_____	_____
_____	_____	-	_____	_____
_____	_____	-	_____	_____
_____	_____	-	_____	_____
_____	_____	-	_____	_____
_____	_____	-	_____	_____

***** BASE24 *****

NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:

F12-HELP

Figure 3–17 SNF Screen 1

INTERCHANGE FIID

4 (A) M

This field is system-protected, and displays the INTERCHANGE FIID from ICF Screen 1.

PROCESS

16 (A) M

This field is system-protected, and displays the PROCESS from ICF Screen 1.

SEND STATION

16 (A)

This field specifies the symbolic name of one of the stations that will be used to send messages to the interchange. This name must match the name of a station in the NEF for the EPS-Net Interchange Interface, and must be paired with a receive station.

RECEIVE STATION

16 (A)

This field specifies the symbolic name of one of the stations that will be used to receive messages from the interchange. This name must match the name of a station in the NEF for the EPS-Net Interchange Interface, and must be paired with a send station. The name may be identical to the send station on the same line but must be unique between lines.

TYPE

1 (A)

This field specifies the type of station pair. Valid values are:

A = Acquirer

I = Issuer

B = Both (acquirer and issuer)

Default Value: B

SWITCH DATA

16 (A)

The EPS-Net Interchange Interface process does not use this field.

SWITCH INFO

16 (A)

The EPS-Net Interchange Interface process does not use this field.

3.8 EUROCARD INSTITUTION IDENTIFICATION FILE

The Eurocard Institution Identification File (ERIIF) may hold either the MasterCard Member Id or an internal reference number that uniquely identifies the acquiring institution. The key to the ERIIF record is the BASE24-assigned FIID (Terminal Owner FIID).

The EPS-Net Interchange Interface process accesses data maintained in the records to build an internal memory table. When a transaction is acquired by BASE24 and passed to the EPS-Net Interchange Interface, the terminal FIID in the transaction request is matched with the FIIDs found in the internal table, and the processing described in Section 3.7.2 is performed.

The ERIIF screen 1 is accessed from the Interchange Configuration File (ICF) Menu screen by placing the cursor to the left of the ERIIF menu entry and pressing F1. The ICF Menu screen itself is accessed from ICF screen 12 by pressing function key F8 (GOTO EPS-NET MENU).

3.8.1 ERIIF Screen 1

The ERIIF screen 1 (Figure 3–18) identifies the record to be added, updated, or deleted.

BASE24-BASE EURO INST. FILE EME3 99/08/18 13:40 01 OF 01

FINANCIAL ID: _____

EUROCARD ACQUIRER INSTITUTION ID: _____

CURRENCY INDICATOR: _

***** BASE24 *****

NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:

F12-HELP

Figure 3–18 ERIIF Screen 1

FINANCIAL ID 4 (A) M

The BASE24 financial institution identifier (FIID) of the institution that owns the terminals that can acquire MasterCard Europe transactions.

EUROCARD ACQUIRING INSTITUTION ID 11 (N) M

This field is either the MasterCard Member Id or an internal reference number that uniquely identifies the acquiring institution. This identification number is sent in authorisation requests to the EM.

CURRENCY INDICATOR 1 (A)

This field allows acquirers to configure different MasterCard Member Ids when multiple transaction currencies are supported.

3.8.2 Configuring MasterCard Member Ids

When setting DE 032 in authorisation request messages sent to EPS-Net, the EPS-Net Interchange Interface first checks the INSTITUTION ID field on Screen 1 of the ICF:

- If the INSTITUTION ID field on Screen 1 of the ICF contains a value other than spaces, then the value is used.
- If the INSTITUTION ID field on Screen 1 of the ICF contains spaces, then the ERIIF is accessed.

If an ERIIF record cannot be located (or the file does not exist), then the value received in the ACQ-INST-ID-NUM field in the (P)STM is used. If an ERIIF record can be located, then the EPS-Net Interchange Interface performs processing based on the value of the CURRENCY INDICATOR field on the ERIIF Screen:

- If the CURRENCY INDICATOR does not contain an alphabetic value, then the processing described in Section 3.7.2.1 is performed.
- If the CURRENCY INDICATOR does contain an alphabetic value, then the processing described in Section 3.7.2.2 is performed.

Examples of how acquirers can use the ERIIF to configure MasterCard Member Ids are given in Section 3.7.2.3.

3.8.2.1 Currency Indicator Not Configured

If the CURRENCY INDICATOR does not contain an alphabetic value, then the EUROCARD ACQUIRER INSTITUTION ID field on the ERIIF screen is retrieved from the ERIIF record. The value is put into either DE 032 (Acquiring Institution Identification Code) or DE 127 (Private Data) in the external message to EPS-Net, based on the following logic:

- If the INSTITUTION ID field on Screen 1 of the ICF contains a value other than spaces, then the ERIIF value will be sent in DE 127.
- If the INSTITUTION ID field on Screen 1 of the ICF contains spaces, then the value retrieved from the ERIIF will be sent in DE 032.

If the value retrieved from the ERIIF is sent in DE 032, or an ERIIF record cannot be located (or the file does not exist), then DE 127 is not included in the message.

3.8.2.2 Currency Indicator Configured

If the CURRENCY INDICATOR does contain an alphabetic value, then the EPS-Net Interchange Interface will attempt to reread the ERIIF using a key of the 1-byte CURRENCY INDICATOR and the 3-byte numeric transaction currency code.

- If a record can be located, then the value in the EUROCARD ACQUIRER INSTITUTION ID field on the screen of the second ERIIF record will be used.
- If a record cannot be located, then the value in the EUROCARD ACQUIRER INSTITUTION ID field on the screen of the first ERIIF record will be used.

If the EPS-Net Interchange Interface attempts to access a second ERIIF record when processing an authorisation request, then DE 127 is not included in the message.

The EPS-Net Interchange Interface never attempts to access more than two ERIIF records when processing a single authorisation request.

3.8.2.3 Configuration and Processing Example

A terminal owning institution is configured in a BASE24 system with an FIID of 'BNK1'. It supports some terminals operating in Pounds Sterling, some terminals operating in Euros, and a few terminals which can operate in a number of other currencies, e.g. U.S. dollars. A second terminal owning institution is configured in the BASE24 system with an FIID of 'BNK2'. All of its terminals operate in Euros only. A single EPS-Net Interchange Interface can process transactions received from terminals belonging to both BNK1 and BNK2.

For BNK1, the following rules apply when an authorisation request is sent to EPS-Net:

- If the transaction is in Pounds Sterling, then DE 032 must contain a Member ID of '111111';
- If the transaction is in Euros, then DE 032 must contain a Member ID of '222222'.
- If the transaction is in any other currency, then DE 032 must contain a Member ID of '333333'.

For BNK2, DE 032 in an authorisation request sent to EPS-Net must always contain a Member ID of '444444'.

To support the requirement, the INSTITUTION ID field on Screen 1 of the ICF must contain spaces, and ERIIF records must be configured as shown in the following table:

Financial Id	Eurocard Acquirer Institution Id	Currency Indicator
BNK1	333333	Alphabetic value, e.g. 'A'
A826	111111	Space
A978	222222	Space
BNK2	444444	Space

Note that '826' is the numeric currency code for Pounds Sterling (GBP), '978' is the numeric currency code for Euros (EUR), and '840' is the numeric currency code for Dollars (USD).

When the EPS-Net Interchange Interface receives an authorisation request from a terminal with an FIID of 'BNK1' and a currency code of '826', the interface reads the ERIIF, using a key of 'BNK1'. Finding an alphabetic value in the CURRENCY INDICATOR field, the interface reads the ERIIF again, using a key of the currency indicator and the currency code, i.e. 'A826'. Having located the second record, the interface sets DE 032 to '111111' in the authorisation request message sent to EPS-Net.

When the EPS-Net Interchange Interface receives an authorisation request from a terminal with an FIID of 'BNK1' and a currency code of '978', the interface reads the ERIIF, using a key of 'BNK1'. Finding an alphabetic value in the CURRENCY INDICATOR field, the interface reads the ERIIF again, using a key of the currency indicator and the currency code, i.e. 'A978'. Having located the second record, the interface sets DE 032 to '222222' in the authorisation request message sent to EPS-Net.

When the EPS-Net Interchange Interface receives an authorisation request from a terminal with an FIID of 'BNK1' and a currency code of '840', the interface reads the ERIIF, using a key of 'BNK1'. Finding an alphabetic value in the CURRENCY INDICATOR field, the interface reads the ERIIF again, using a key of the currency indicator and the currency code, i.e. 'A840'. Having failed to locate the second record, the interface sets DE 032 to '333333' in the authorisation request message sent to EPS-Net.

When the EPS-Net Interchange Interface receives an authorisation request from a terminal with an FIID of 'BNK2' and a currency code of '978', the interface reads the ERIIF, using a key of 'BNK2'. Finding a non-alphabetic value in the CURRENCY INDICATOR field, the interface sets DE 032 to '444444' in the authorisation request message sent to EPS-Net.

3.9 TOKEN FILE

To log or extract tokens, the Token File (TKN) must be configured appropriately. The token parameters can be set on the TKN file maintenance screen.

The following tokens used by the EPS-Net Interchange Interface can be logged or extracted:

- 03 - ATM 5.0 token
- 04 - POS 5.0 token
- 06 - PIN Change token
- 16 - Alternate Merchant ID token
- 20 - Interchange Compliance token
- 23 - Track-1 token
- 24 - ATM Flag1 token
- A5 - Non-Currency Dispense token
- A6 - ATM Interchange Compliance token
- B0 - Acquirer Switch token
- B1 - Issuer Switch token
- BE - Original Currency token
- BM - Transaction Subtype token
- C0 - POS 5.1 token
- C2 - Purchase token
- C4 - Point of Service Data token
- CE - Authentication Data token
- CF - IAVS Data token
- CH - POS Data1 token
- CI - POS Merchant token

For more details, refer to the *BASE24 Tokens Manual*.

3.10 OTHER BASE24 FILES

MasterCard Europe has a requirement for acquirers to provide a non-zero Merchant Category Code (MCC) in an authorisation request message, because the acquirer must identify the merchant type by providing a valid MCC.

To ensure compliance, the SIC Code fields in the POS Terminal Data files (PTD/PTDF) and the POS Retailer Definition File (PRDF) must be set to non-zero values by MasterCard Europe acquirers.

When processing POS Advice messages (acquired by BASE24), some acquirers may have a 'bilateral' agreement with MasterCard Europe issuers to send 0120 advice messages. To support the sending of 0120 advice messages, the configuration requirements depend on the release level of the BASE24 system:

- For Release 6.0 systems, ensure that 'Y' is configured in the appropriate Issuer Processing Code File (IPCF) Completion Required to Host field(s).
- For systems prior to Release 6.0, ensure that 'Y' is configured in the appropriate Institution Definition File (IDF) Completion Required field.

To allow ATM balance inquiry transactions to be supported nationally, issuers must ensure that the COUNTRY field on IDF Screen 1 is set to the correct value, so the ATM Authorisation process can compare the value against the country code received in the request message from EPS-Net.

3.11 EPS-NET IPF REFRESH CONFIGURATION

The EPS-Net Refresh process uses the following TACL assigns and parameters. These assigns and parameters are valid for the current TACL session, and can be entered at a TACL prompt or placed in an obey file.

3.11.1 EPS-Net IPF Refresh Assigns

These assigns are set using the standard TACL ASSIGN command.

Assign	Description
IPFX	This assign identifies the location of the Optimised Current Parameters File used as input for the IPF Refresh Process. Valid values are a HP NonStop (formerly Tandem) file name.
IPF	This assign indicates the location of the disk file used as output from the IPF Refresh Process. Valid values are a HP NonStop (formerly Tandem) file name.
TPLT	This assign indicates the template file that is used as template for the IPF. Valid values are a HP NonStop (formerly Tandem) file name.

3.11.2 EPS-Net IPF Refresh Parameters

These parameters are set using the standard TACL PARAM command.

Parameter	Description
NUM-OF-PROGRAMS	This param identifies the number of PROGRAM-XXX parameters that should follow this param. Valid values are 1–65535. Zero is not valid because at least one acceptance program is required. See PROGRAM-XXX param.

Parameter	Description
PROGRAM-XXX	Each of these params represents an acceptance program that is required for routing purposes. Replace XXX with the three-character acceptance program. The XXX value must be repeated as the first three characters of the param value. The last two characters of each param (YY) represent the card type that will be associated in the IPF with all prefixes under that acceptance program. Should a card type not be required for a program, these two characters should be kept blank. The separator ":" should remain after XXX. Valid values are 3 ASCII alphanumeric characters (XXX), ':' as a separator, and 2 ASCII alphanumeric characters (YY), i.e. XXX:YY. The number of acceptance programs required is dependent on the value of the NUM-OF-PROGRAMS param. In the RUN command, the NUM-OF-PROGRAMS param must precede all PROGRAM-XXX params.
SUPPRESS-DUP-MESS	This param controls whether duplicate key error messages are included in the error report. As POS and ATM records can both have the same key, duplicate records can be encountered if the TERMINAL-CATEGORY param is set to "BOTH". Valid values are: Y = Suppress error messages N = Do not suppress error messages
TERMINAL-CATEGORY	This param identifies the terminal category of BINs that should be compared to the range of acceptance programs required. Valid values are: ATM = ATM only POS = POS only BOTH = ATM and POS

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4 STARTUP AND CONTROL

This section provides information on the configuration of the EPS-Net Interchange Interface process. It also describes the processing involved for initialisation, network control commands, timers and the network management message flow between the EPS-Net Interchange Interface process and the EM.

4.1 PROCESS CONFIGURATION

The following paragraphs contain an explanation on how the EPS-Net Interchange Interface process is configured. They provide information about the startup logic of the EPS-Net Interchange Interface process and the method in which the EPS-Net Interchange Interface process logs messages.

4.1.1 Startup

In the Network Environment File (NEF), the STARTUP attribute for the PROCESS object type specifies the startup logic for an application process. Because of the length of time required for initialisation, the EPS-Net Interchange Interface process should be assigned a startup logic of AUTOMATIC in its process definition. AUTOMATIC means the EPS-Net Interchange Interface process is started automatically when the XPNET logical network is initiated. This ensures that the EPS-Net Interchange Interface process is ready when it receives its first transaction.

4.1.2 Log Messages

Log messages are routed to certain destinations, based on the routing code that appears in the log messages. The EPS-Net Interchange Interface process routes all of its log messages with a standard routing code of zero.

The EPS-Net Interchange Interface process generates log messages to assist network operators in performing their daily tasks. Log messages are unsolicited messages generated by BASE24 processes that provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Log messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

For a complete description of log messages output by the EPS-Net Interchange Interface, refer to Appendix C.

4.1.3 Transactional Interface Monitoring (T.I.M.)

The EPS-Net Interchange Interface supports the Transactional Interface Monitoring (T.I.M.) enhancement. T.I.M. offers the ability to monitor, in real-time, the actual transaction health between BASE24 and EPS-Net. This is achieved by keeping track of a number of transaction conditions in the interchange, for both acquirer and issuer processing. The T.I.M. monitoring facility can be started, stopped and have its counters reset using NCPCOM commands.

This enhancement works in conjunction with products such as NET24-OM, ENGUARD and PROGNOSIS. The critical information is held in appropriate Measure counters, enabling products that interface to the Measure utility available with HP NonStop (formally Tandem) hardware to provide display and reporting data.

4.1.4 Tokens

Additional disk space may be required on the HP NonStop (formerly Tandem) for the 5.0 or above BASE24 database depending on the number of tokens being logged to the files.

The Release 5.0 or above BASE24 user can indicate what tokens should be logged by the EPS-Net Interchange Interface to its Interface Log File (ILF). The user can also indicate to the Release 5.0 or above BASE24-atm Authorisation process the tokens that should be logged to the Transaction Log File (TLF), and to the Release 5.0 or above BASE24-pos Router/Authorisation module which tokens should be logged to the POS Transaction Log File (PTLF).

When deciding whether or not to log a token to any Release 5.0 or above log file, the user should consider what the length of the log record will be with the new token added. If logging a new token to a log file increases the length of a log file record so that fewer records can fit in the maximum HP NonStop (formerly Tandem) data block, more data blocks (i.e., more disk space) will be required to hold the same number of records. This will increase the user's disk space requirements.

The amount of token data logged to the ILF must not result in the record length exceeding 2034 bytes (Format-1 ILF) or 2022 bytes (Format-2 ILF), if it is required to write two ILF records to each disk block.

4.2 INITIALISATION

The EPS-Net Interchange Interface takes the following steps when it is first started. To initialise, the EPS-Net Interchange Interface process:

- Reads the name of its LCONF. The startup message from the XPNET process contains the name of the LCONF. This name allows the EPS-Net Interchange Interface process to access the LCONF for the assigns and parameters it needs for processing.
- Opens the LCONF, reads the assigns and parameters specified in section 3.2. If an error occurs, the EPS-Net Interchange Interface process logs an error message and abends.
- Locates the LMCF for its modified log messages, using the LCONF assign LMCF. The LMCF contains a record for each modified log message in the system. The EPS-Net Interchange Interface process opens the LMCF, retrieves its modified log messages, and closes the LMCF. Whenever the EPS-Net Interchange Interface process generates a log message, it checks the LMCF records in memory to determine whether the message requires modification.
- Initialises its timer table.
- Locates the ICF, using the LCONF assign ICF. The EPS-Net Interchange Interface process opens the Interchange Configuration File (ICF). If an error occurs, the EPS-Net Interchange Interface process logs an error message and abends. Using its process name, the EPS-Net Interchange Interface process reads its own ICF record.
- Builds the Process Control Table (PCT). The PCT contains configuration and control information for the process. This information is read from the ICF, and includes all of the controls needed to process transactions.
- Determines the posting dates for BASE24-atm and BASE24-pos by comparing the current time to the BASE24-atm cutover time and BASE24-pos cutover time parameters taken from the LCONF (ATM-LN-CUTOVER and POS-LN-CUTOVER).
- Locates the SNF, using the LCONF assign SNF. The EPS-Net Interchange Interface process opens the Station Name File (SNF). Using its process name, the EPS-Net Interchange Interface process reads its own SNF record.
- Locates the SAF, using the LCONF assign SAF. The EPS-Net Interchange Interface process opens the Store-and-Forward File (SAF). If an error occurs, the EPS-Net Interchange Interface process logs an error message and abends.
- Locates the ILF, using the LCONF assign ILF. The EPS-Net Interchange Interface process opens the Interchange Log Files (ILFs) for the previous day and the current day. If an error occurs, the EPS-Net Interchange Interface process logs an error message and abends.

- Locates the ENRLF, using the LCONF assign ENRLF. The EPS-Net Interchange Interface process opens the EPS-Net Request Log File (ENRLF).
- Initialises the extended memory required for this process.
- Locates the STF, using the LCONF assign STF. The EPS-Net Interchange Interface process opens the Switch Terminal File (STF) and loads STF records into extended memory, up to a maximum specified by the LCONF parameter MAX-STF-RECORDS.
- Locates the ERIIF, using the LCONF assign ERIIF. The EPS-Net Interchange Interface process opens the Eurocard Institution Identification File (ERIIF).
- Locates either the KEYF or KEY6, dependent on the contents of the DES-TRIPLE-SINGLE parameter, using the LCONF assign KEYF or KEY6. The EPS-Net Interchange Interface process opens the Key File (KEYF or KEY6). If an error occurs, the EPS-Net Interchange Interface process logs an error message and abends. Using its process name, the EPS-Net Interchange Interface process reads its own KEYF or KEY6 record.
- Initialises a security device if applicable.
- Updates the ICF with the current posting dates and then closes the ICF.
- Locates the TKN file, using the LCONF assign TKN. The EPS-Net Interchange Interface process opens the Token File (TKN). If an error occurs, the EPS-Net Interchange Interface process logs an error message and abends.
- Determines whether reports are required.

4.3 NETWORK CONTROL COMMANDS

This section contains the description and entry syntax of all network control text commands that the EPS-Net Interchange Interface process supports. The EPS-Net Interchange Interface process supports the following text commands:

- DEBUG
- DUMP
- LOGON
- LOGOFF
- PERFORMANCE
- PMONON
- PMONOFF
- PMONRESET
- STATUS
- SWL
- TRACE ON
- TRACE OFF
- WARMBOOT

The above commands are issued to the EPS-Net Interchange Interface by means of the DELIVER PROCESS command.

For more details, refer to the *NET24-XPNET Network Control Operations Guide* and the *BASE24 Text Command Reference Manual*.

More details of the above commands are given on the following pages.

4.3.1 **DEBUG Command**

A restricted use operator command that places the EPS-Net Interchange Interface process into the Inspect programming tool provided with HP NonStop (formerly Tandem) hardware, and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process).

Syntax

DELIVER PROCESS *object-name*, "DEBUG"

object-name — The symbolic name assigned to the EPS-Net Interchange Interface process in the XPNET node.

Processing

When the BASE24 process receives the DEBUG command, it performs the following processing steps:

1. Reads the message.
2. Places the EPS-Net Interchange Interface process under control of the Inspect programming tool and returns an Inspect prompt to the home terminal.

For more information on Inspect, refer to the appropriate HP NonStop (formerly Tandem) documentation.

4.3.2 **DUMP Command**

The DUMP command directs the EPS-Net Interchange Interface process to request a programmatic dump.

Syntax

DELIVER PROCESS *object-name*, "DUMP"

object-name — The symbolic name assigned to the EPS-Net Interchange Interface process in the XPNET node.

Processing

A programmatic dump allows a programmer to automatically capture an unusual situation for evaluation. The EPS-Net Interchange Interface process can continue running after the dump is generated. The information resulting from the dump can be read using RDEBUG. RDEBUG allows an image of the process environment (known as a dump) to be saved in an unstructured disk file called the dump file. The RDEBUG program allows interactive analysis of the dump file at a later time. For more information on RDEBUG, refer to the ***BASE24 RDEBUG Manual***.

The EPS-Net Interchange Interface process performs the following processing steps when it receives a DUMP command from an XPNET network control facility:

1. Creates a dump file containing an image of the process' environment.
2. Logs messages to indicate the status of the programmatic dump.

At this point, an operator can read and analyse the dump file. The dump file contains results of the programmatic dump. If necessary, analysis can be performed interactively through a terminal using the RDEBUG program at a later time. Information from the dump file can also be printed for further analysis.

4.3.3 LOGON Command

A general use command that initiates a logon sequence for one or more stations. It causes the EPS-Net Interchange Interface process to send a logon message to each station indicated in the message.

Syntax

```
DELIVER PROCESS object-name, "LOGON station-name"
```

object-name — The symbolic name assigned to the EPS-Net Interchange Interface process in the XPNET node.

station-name — The symbolic station name defined for the EPS-Net Interchange Interface process in the Interchange Configuration File (ICF) or the Station Name File (SNF).

Processing

If ALL is specified in this variable instead of a specific symbolic station name, a logon sequence is initiated for all stations defined to the process in the ICF or SNF. If an individual station is specified, a logon sequence is initiated for that station only.

4.3.4 LOGOFF Command

A general use command that initiates a logoff sequence for one or more stations. It causes the EPS-Net Interchange Interface process to send a logoff message to each station indicated in the message.

Syntax

```
DELIVER PROCESS object-name, "LOGOFF station-name"
```

object-name — The symbolic name assigned to the EPS-Net Interchange Interface process in the XPNET node.

station-name — The symbolic station name defined for the EPS-Net Interchange Interface process in the Interchange Configuration File (ICF) or the Station Name File (SNF).

Processing

If ALL is specified in this variable instead of a specific symbolic station name, a logoff sequence is initiated for all stations defined to the process in the ICF or SNF. If an individual station is specified, the logoff sequence is initiated for that station only.

4.3.5 PERFORMANCE Command

A general use command that causes the performance statistics kept in memory by the EPS-Net Interchange Interface process be sent to the EMS collector process. The EPS-Net Interchange Interface process tracks performance statistics for each station assigned to it. These statistics are gathered on an interval basis - the interval length being established in the Interchange Configuration File (ICF) - and statistics are available for the last five intervals.

Syntax

DELIVER PROCESS *object-name*, "PERFORMANCE"

object-name — The symbolic name assigned to the EPS-Net Interchange Interface process in the XPNET node.

Processing

This command requests the performance statistics kept in memory by the EPS-Net Interchange Interface process. The performance period is defined in the ICF record (screen 3) and is specified in minutes. The following is an example of a performance display consisting of a series of log messages:

```
*** ATM Performance Statistics Summary ***
Mins in period: <#> <Outbd/Inbd> rqst limit: <#>

      REQUESTS - TIMEOUTS - PERCENT - AVERAGE
PERIOD <#> <#####>      <#####>      <###>%      <##.##>
PERIOD <#> <#####>      <#####>      <###>%      <##.##>
PERIOD <#> <#####>      <#####>      <###>%      <##.##>
PERIOD <#> <#####>      <#####>      <###>%      <##.##>
PERIOD <#> <#####>      <#####>      <###>%      <##.##>

*** POS Performance Statistics Summary ***
Mins in period: <#> <Outbd/Inbd> rqst limit: <#>

      REQUESTS - TIMEOUTS - PERCENT - AVERAGE
PERIOD <#> <#####>      <#####>      <###>%      <##.##>
PERIOD <#> <#####>      <#####>      <###>%      <##.##>
PERIOD <#> <#####>      <#####>      <###>%      <##.##>
PERIOD <#> <#####>      <#####>      <###>%      <##.##>
PERIOD <#> <#####>      <#####>      <###>%      <##.##>
```

These log messages are described in Appendix C of this manual.

4.3.6 PMONON Command

A general use command that will mark the Performance Monitoring System as active.

Syntax

```
DELIVER PROCESS object-name, "PMONON"
```

object-name — The symbolic name assigned to the EPS-Net Interchange Interface process in the XPNET node.

Processing

This command will cause the EPS-Net Interchange Interface process to start Performance Monitoring and initialise the Performance Monitoring counters. When successfully completed, a message "Performance Monitoring is now on." is logged.

4.3.7 PMONOFF Command

A general use command that will mark the Performance Monitoring System as inactive.

Syntax

```
DELIVER PROCESS object-name, "PMONOFF"
```

object-name — The symbolic name assigned to the EPS-Net Interchange Interface process in the XPNET node.

Processing

This command will cause the EPS-Net Interchange Interface process to stop Performance Monitoring. When successfully completed, a message "Performance Monitoring is now off." is logged.

4.3.8 PMONRESET Command

A general use command that will initialise all the Performance Monitoring variables to zero.

Syntax

```
DELIVER PROCESS object-name, "PMONRESET"
```

object-name — The symbolic name assigned to the EPS-Net Interchange Interface process in the XPNET node.

Processing

This command will cause the EPS-Net Interchange Interface process to reset Performance Monitoring counters. When successfully completed, a message "Performance Monitoring counters have been reset" is logged.

4.3.9 STATUS Command

A general use operator command that causes the EPS-Net Interchange Interface process to send miscellaneous information on its current processing parameters to the EMS collector process.

Syntax

DELIVER PROCESS *object-name*, "STATUS"

object-name — The symbolic name assigned to the EPS-Net Interchange Interface process in the XPNET node.

Processing

This command requests the status of the EPS-Net Interchange Interface process. The EPS-Net Interchange Interface can log the following information:

```
BASE24-atm DATE:  yymmdd
BASE24-pos DATE:  yymmdd
Eurocard Date:   yymmdd
# Outbound Requests Pending of # Maximum
# Inbound Requests Pending of # Maximum
# Timers Active of # Maximum
SAF Cycle in progress
File:  SAF-name, # Percent Full
File:  ILF-name, # Records, # Percent Full
xxx Station is up/down and Logged on/off with # timeouts of # max
whether or not Performance is running
```

These log messages are described in Appendix C of this manual.

4.3.10 SWL Command

A general use operator command that changes the event message logging facility from the current facility to the indicated facility.

Syntax

DELIVER PROCESS *object-name*, "SWL *logging-facility*"

object-name — The symbolic name assigned to the EPS-Net Interchange Interface process in the XPNET node.

logging-facility — The facility to which logging is being changed.

Processing

The EPS-Net Interchange Interface begins sending event messages to the logging facility specified in the SWL command.

Note that this command is not the same as the command used to switch log files for the XPNET process. For more on the command used to switch log files for the XPNET process, refer to the **NET24-XPNET Network Control Command Reference Manual**.

4.3.11 TRACE ON Command

A restricted use operator command that activates a trace function within an EPS-Net Interchange Interface process. The EPS-Net Interchange Interface process contains trace log messages that are output to the BASE24 log. Trace log messages are requested only in a test environment. A trace log message is output when specified points are reached in the EPS-Net Interchange Interface software.

Syntax

DELIVER PROCESS *object-name*, "TRACE ON"

object-name — The symbolic name assigned to the process in the XPNET node.

Processing

The EPS-Net Interchange Interface process contains trace log messages that are output to the BASE24 log. Trace log messages are requested only in a test environment. A trace log message is output when specified points are reached in the EPS-Net Interchange Interface software.

The EPS-Net Interchange Interface process performs the following processing steps when it receives a TRACE ON command:

1. Reads the command.
2. Activates the trace function within the process.

4.3.12 TRACE OFF Command

A restricted use operator command that turns off the trace function within an EPS-Net Interchange Interface process. The EPS-Net Interchange Interface process contains trace log messages that are output to the BASE24 log. Trace log messages are requested only in a test environment. A trace log message is output when specified points are reached in the EPS-Net Interchange Interface software.

Syntax

DELIVER PROCESS *object-name*, "TRACE OFF"

object-name — The symbolic name assigned to the process in the XPNET node.

Processing

The EPS-Net Interchange Interface process performs the following processing steps when it receives a TRACE OFF command:

1. Reads the command.
2. Deactivates the trace function within the process.

4.3.13 WARMBOOT Command

A general use operator command that causes the EPS-Net Interchange Interface process to reinitialize.

Syntax

```
DELIVER PROCESS object-name, "9501"
```

```
DELIVER PROCESS object-name, "WARMBOOT"
```

object-name — The symbolic name assigned to the EPS-Net Interchange Interface process in the XPNET node.

Processing

This command warm boots the EPS-Net Interchange Interface process. Warmboot processing is similar to process initialization, except that a warmboot only includes those steps required to update the file information carried by the EPS-Net Interchange Interface process in memory. A message is logged to the BASE24 log indicating that the warmboot is completed and the new status implemented, or the warmboot was aborted and previous status retained.

4.4 NETWORK MANAGEMENT MESSAGE FLOW

Network management messages (NMMs) are used to control the message processing within the network.

Stations to the EM are defined in either the ICF or the SNF. These stations must be defined as being either issuer-only, acquirer-only, or both.

4.4.1 Signon Request from BASE24

The network control LOGON request is issued either for a specific station or for all stations defined for the process. For more information on the syntax of the network control command, refer to section 4.3.

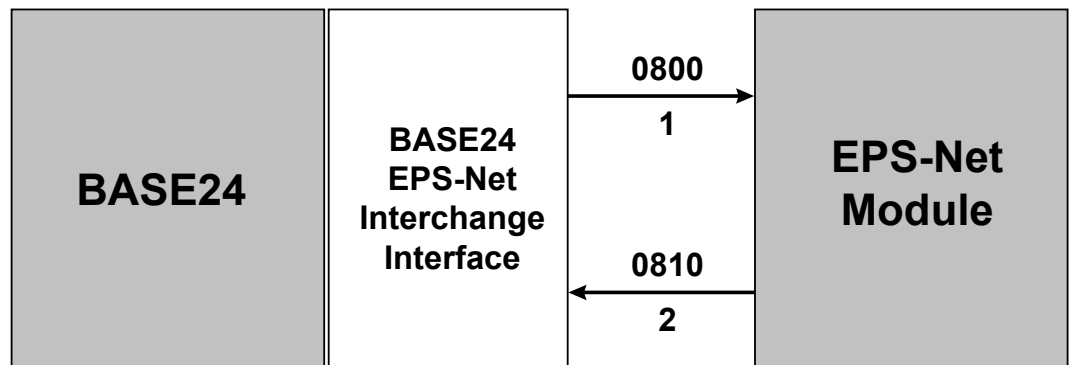


Figure 4–1 Signon Request from BASE24

The following explains the flow of the signon message, as illustrated in Figure 4–1:

1. 0800 NMM request is formatted with an NMM information code (ISO data element 070) of 001; additional data is formatted to include version number, traffic flags, and pin block format. The system trace number is incremented. The 0800 request is forwarded to the defined station. An NMM timer is started to wait for the logon response.
2. 0810 NMM response received on the defined station. The EPS-Net Interchange Interface locates the previously set NMM timer and deletes it. The response code (ISO data element 039) in the 0810 is verified to ensure the logon request was successful (response code of 00). The stations are marked up and logged on. A wait for traffic timer is started. This will contain the time specified between successive signons (refer to the *V5 Interface Specifications* for more details of Relaxed Session Management, plus section 3.3 in this manual). A log message is then output indicating the stations are logged on.

Notes:

- If the response code returned from the EM is not 00 (successful) then the stations will return to an idle state (logged off) and a logon can be attempted again after the time specified in the extended network timer (to comply with the Relaxed Session Management mentioned above, this value should be set to 30 seconds).
- If the EM detects one or more of the fields in the 0800 request are invalid, it will respond to the logon request with a response code 06 (error).
- If the EM encounters too many errors, it will send a logoff request to BASE24. BASE24 replies with an approved logoff response.

4.4.2 Signoff Request from BASE24

The LOGOFF request is either for a specific station or for all stations defined for the process. For more information on the syntax of the network control command, refer to section 4.3.

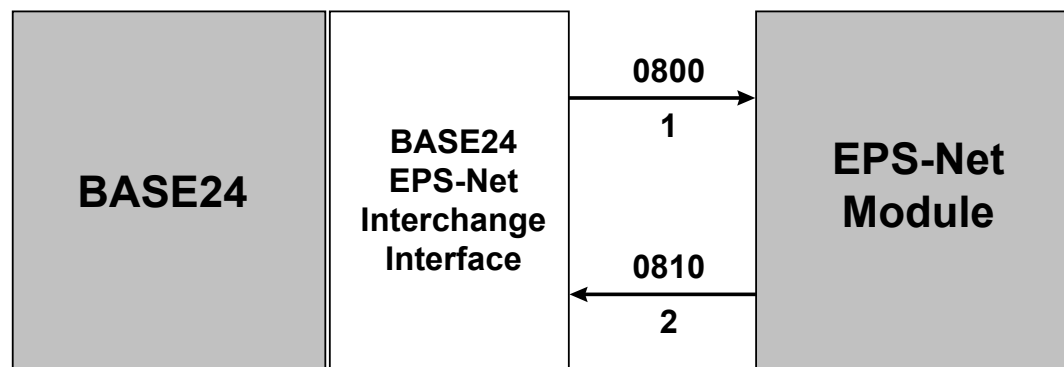


Figure 4–2 Signoff Request from BASE24

The following explains the flow of the signoff message, as illustrated in Figure 4–2:

1. 0800 NMM request is formatted with an NMM information code (ISO data element 70) of 002 and a valid logoff reason code. The system trace number is incremented. The 0800 request is forwarded to the defined station. An NMM timer is started to wait for the logoff response from the EM.
2. 0810 NMM response received on the defined station. The EPS-Net Interchange Interface locates the previously set NMM timer and deletes it. The response code (ISO data element 039) in the 0810 is verified to ensure the signoff request was successful (response code 00). The EPS-Net stations are marked down and logged off. A log message is then output indicating that the stations are now logged off.

Note that BASE24 will send a logoff request with a logoff reason code 03 (0810 timed out) if the EM does not reply to an 0800 signoff request.

4.4.3 Network Management—Reject Request/Confirm Messages

Reject request/confirm network management messages are supported by the BASE24 EPS-Net Interchange Interface process. The EM (or the EPS-Net Interchange Interface process) can generate a reject request at any time to reject an invalid message (0100, 0120, 0130, 0420, or 0430).

4.5 STATION MANAGEMENT

The *V5 Interface Specifications* includes some specific requirements for the management of communications sessions between an AHS or IHS and the EM.

4.5.1 Disconnect/Reconnect Station Control

The *V5 Interface Specifications* state that the Host must “drop” the communications line with the EM, and then attempt to re-establish data communications with the EM (i.e. disconnect and reconnect the station) under the following circumstances:

- When a logoff network management request message sent to the EM times out.
- When a “forced” sign-off (a logoff network management request message) is received from the EM.

The EPS-Net Interchange Interface achieves this by invoking a utility procedure that sends a disconnect control message to XPNET. This causes the XPNET process to disconnect the station in error, and the station configuration data in the NEF then results in the station being reconnected.

The control message is created by setting the Control Flag in the XPNET Message Header to “1” (to indicate a control message), and setting the first two bytes of the Message Text to a control code of “01” (to indicate a disconnect).

4.5.2 Detection of a Broken TCP/IP Socket

The *V5 Interface Specifications* describe the use of “Zero Length Probe” messages to detect broken TCP/IP sockets. The zero length probe functionality allows the network or the member to verify the session status and to close inactive sessions.

XPNET can receive zero length probe messages from the EM. XPNET discards the zero length probe message, and does not attempt to detect the “broken socket” condition by utilising an idle timer on the socket.

XPNET can be configured to send zero length probe messages to the EM by including a parameter in the STATION USERDATA within the declaration of a TCP/IP station in the NEF:

- A setting of –B+ will result in XPNET sending a zero length probe message when there has been no traffic detected on the station for the default time interval of 10 seconds.
- A setting of –BTnn (where nn is a time interval specified in seconds) will result in XPNET sending a zero length probe message when there has been no traffic detected on the station for a time interval of nn seconds.

XPNET does not attempt to perform any error recovery processing if it does not receive an acknowledgement to the zero length probe message. Instead, the standard TCP retry mechanism will ensure that any session problem is detected in a timely manner.

4.6 TIMERS

This section describes each timer used by the EPS-Net Interchange Interface. Each timer description includes the timer's function, and how and when the timer is set.

4.6.1 Outbound Requests Timer

An outbound request timer is set each time a 0100 message is sent to EPS-Net. The number of seconds used for this timer is taken from the OUTBOUND field on ICF screen 8 for BASE24-atm transactions and from the OUTBOUND field on ICF screen 10 for BASE24-pos transactions. The outbound request timer is used to indicate whether a transaction sent to EPS-Net has timed out.

If the timer expires before a response is received from EPS-Net for the 0100 message, the request is denied and the appropriate performance and timeout counters are incremented. If the expiration of this timer causes the maximum allowed timeouts to be reached, the link to EPS-Net is marked down.

4.6.2 Inbound Requests Timer

An inbound request timer is set each time a 0200 message is sent to a BASE24 process for authorisation. The number of seconds used for this timer is taken from the INBOUND field on ICF screen 8 for BASE24-atm transactions and from the INBOUND field on ICF screen 10 for BASE24-pos transactions. The inbound request timer is used to determine that an external transaction has timed out.

If the timer expires before a response is received from the authorizing process, the request is denied. The appropriate performance and timeout counters are incremented.

4.6.3 Store-And-Forward Timer

A store-and-forward timer is set each time a store-and-forward message is sent to EPS-Net. The number of seconds used for this timer is taken from the STORE AND FORWARD field on ICF screen 8 for BASE24-atm transactions and from the STORE AND FORWARD field on ICF screen 10 for BASE24-pos transactions. The store and forward timer is used to determine that a Store-and-Forward File (SAF) record sent to EPS-Net has timed out.

If the timer expires before an acknowledgment is received from EPS-Net, the SAF timeout counter is incremented. If the SAF timeout counter is equal to the maximum number of retries set in the MAX SAF RETRY field on ICF screen 3, the record is written to the log and deleted from the SAF. If the MAX SAF RETRY field is set to 0, the SAF message will be sent indefinitely until it is successfully processed.

4.6.4 Network Management Timer

A network management timer is set each time an 0800 message is sent to a station. The number of seconds used for this timer is taken from the TIMER LIMITS NETWORK MANAGEMENT field on ICF screen 3. The network management timer is used to monitor the communications between BASE24 and EPS-Net.

If the network management timer expires before a response is received from EPS-Net, an extended network management timer is set.

4.6.5 Extended Network Management Timer

An extended network management timer is set each time a network management timer expires without a response being received from EPS-Net. The number of seconds used for this timer is taken from the TIMER LIMITS EXTENDED NETWORK field on ICF screen 3.

The network management message is repeated at the interval specified by the extended network management timer until an acknowledgment is received from EPS-Net or the communications line is marked up.

4.6.6 Wait For Traffic Timer

A wait for traffic timer is set each time a station is marked up and each time a previously set wait for traffic timer expires. The number of seconds used for this timer is taken from the TIMER LIMITS WAIT FOR TRAFFIC field on ICF screen 3.

If the timer expires and message traffic has occurred on the line, a new wait for traffic timer is started.

If the timer expires and no message traffic has occurred on the line, a network management logon message is sent. The line is marked up or down, depending on the results of the logon.

4.6.7 Report Startup Timer

The report startup timer is used to start the SETL01 and STAT09 report programs. It is set at initialisation and at interchange cutover. The value for this timer is taken from the SW-RPT-STARTUP param in the LCONF.

When the report startup timer expires, the obey file containing the commands to run the report programs are executed.

4.6.8 Performance Timer

A performance timer is set at the start of each new performance interval. The number of minutes for this timer is taken from the TIMER LIMITS PERFORMANCE PERIOD field on ICF screen 3. The performance timer triggers the end of one performance period and the beginning of the next.

When the performance timer expires, the EPS-Net Interchange Interface process:

- Moves the interval four performance statistics to interval five.
- Moves the interval three performance statistics to interval four.
- Moves the interval two performance statistics to interval three.
- Moves the interval one performance statistics to interval two.
- Initialises the interval one performance statistics to zero.
- Starts a new performance timer.

4.6.9 Cutover Timer

A cutover timer is used to cut over the interface to the next business day. It is set at initialisation and at interchange cutover. The timer is set based on the SETTLEMENT HOUR and SETTLEMENT MINUTE fields on ICF screen 2.

When the cutover timer expires, the SWITCH POSTING DATE on ICF screen 2 is updated to the next business date. A new ILF is created (if required) and opened, the old ILF is closed, and the LCONF parameter SW-ILF-FILE-RETENTION is checked to determine whether any ILFs should be deleted.

5 TRANSACTION HANDLING

The following functions related to transaction processing are described in this section:

- BASE24 Acquirer Message Flows
- BASE24 Issuer Message Flows
- Transaction Matching
- Transaction Security
- Transaction Types
- Cash Withdrawals From Bank Branch Terminals
- Card Type Validation
- Merchant Category Code Validation
- Interchange Log File (ILF) Processing
- Store-and-Forward (SAF) Processing
- Card Details Processing
- Token Usage
- Response Message Processing
- Advice Message Processing
- EMV Support
- UK Domestic Maestro Support

5.1 BASE24 ACQUIRER MESSAGE FLOWS

This section describes transaction message flows where BASE24 is acting as an Acquirer Host System (AHS).

In this section the following conventions have been used:

- A 0110 is an approved external 0110 response
- A 0210 is an approved internal 0210 response
- D 0110 is a declined external 0110 response
- D 0210 is a declined internal 0210 response

5.1.1 Normal Customer Transaction (AHS)

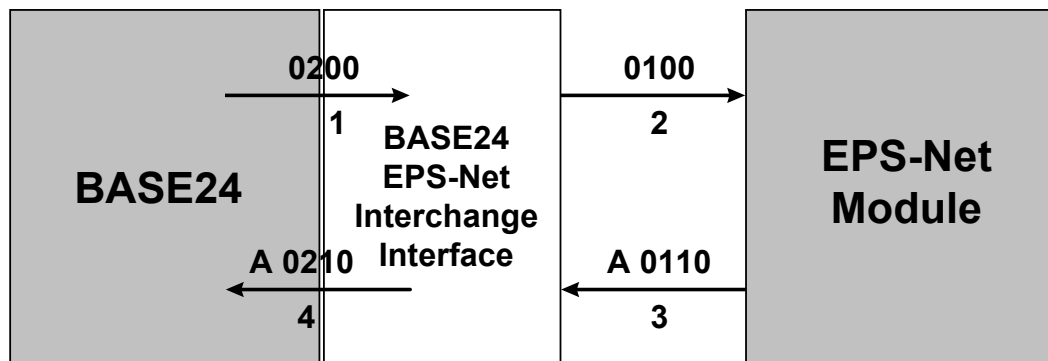


Figure 5–1 Normal Customer Transaction (AHS)

The following explains the flow of the message illustrated in Figure 5–1:

1. For a normal request message flow, the 0200 STM/PSTM authorisation request is received by the BASE24 EPS-Net Interchange Interface.
2. The 0200 STM/PSTM is converted into an 0100 Standard External Message (SEM), a copy of the 0100 is placed in extended memory, an *outbound* timer is set for a response to be received and the 0100 authorisation request is sent to the EM on an available station.
3. For a normal response message flow, the 0110 SEM is received from the EM.
4. The outbound timer is located and deleted by the EPS-Net Interchange Interface, the original 0100 request is located in extended memory and a 0210 STM/PSTM authorisation response is formatted and passed to the BASE24 Authorisation process for subsequent processing.

5.1.2 Late 0210 Response from EPS-Net (Terminal Timeout)

This situation may occur when the transaction timers between the terminal and BASE24 device handler (or acquiring host/interchange process) and the EPS-Net Interchange Interface and the EM are set-up incorrectly.

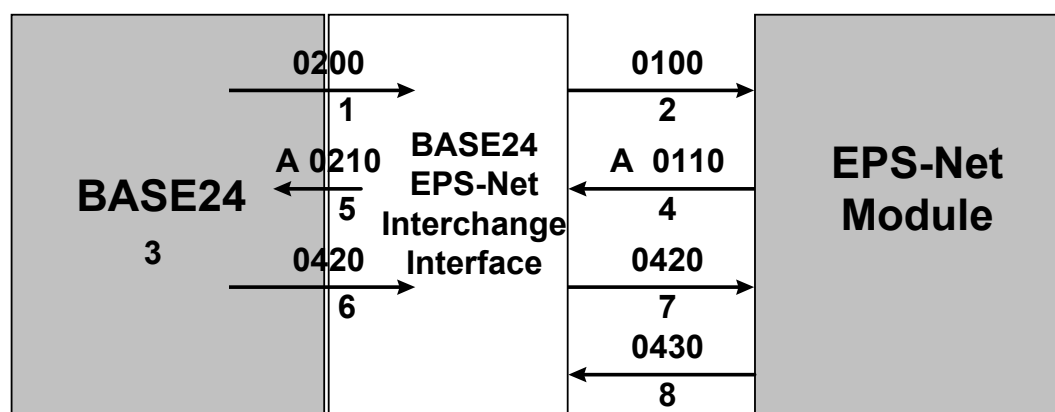


Figure 5–2 Late 0210 Response from EPS-Net (Terminal Timeout)

The following explains the flow of the message illustrated in Figure 5–2:

1. The 0200 transaction request is acquired by the BASE24 EPS-Net Interchange Interface.
2. Having been acquired by the BASE24 EPS-Net Interchange Interface, the 0200 transaction request is formatted into an 0100 SEM request, a copy held in extended memory, an *outbound* timer started and the request forwarded to the EM.
3. A timer expires at the terminal or at a different Interchange Interface (wherever the transaction was introduced into BASE24). The transaction is declined at the terminal (or Interchange Interface).
4. A late approved response is received at the BASE24 EPS-Net Interchange Interface from the EM.
5. The EPS-Net Interchange Interface deletes the outbound timer, locates the original 0200 in extended memory, and formats a 0210 authorisation response. The 0210 response is then forwarded to the BASE24 Authorisation process.
6. A 0420 reversal advice is generated by the device handler (or acquiring host/interchange process). This is then passed to the EPS-Net Interchange Interface.

7. The EPS-Net Interchange Interface then places the 0420 reversal advice on the SAF. When the SAF advice is sent to the EM a *SAF* timer is started to wait for a 0430 reversal advice acknowledgment from the EM.
8. A 0430 reversal advice acknowledgment is received from the EM, the SAF timer is located and deleted and the 0420 reversal advice is also deleted from the SAF.

5.1.3 BASE24 EPS-Net Interchange Interface Timeout

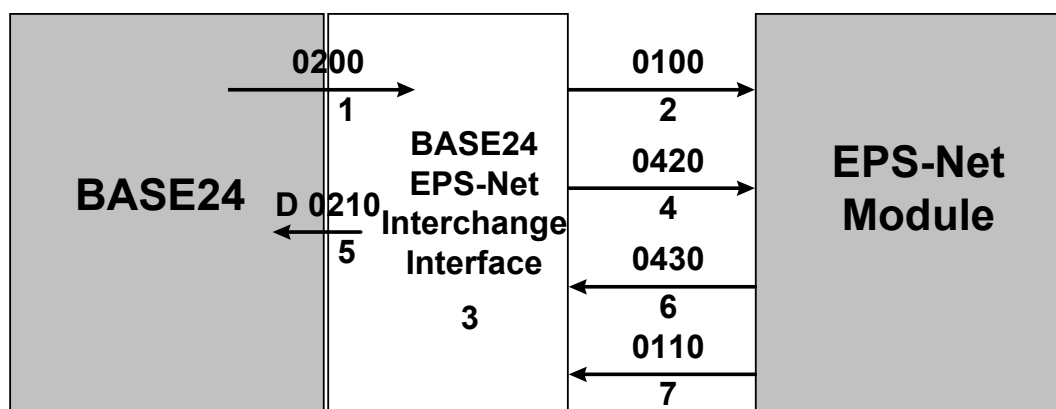


Figure 5–3 BASE24 EPS-Net Interchange Interface Timeout

The following explains the flow of the message illustrated in Figure 5–3:

1. The BASE24 EPS-Net Interchange Interface acquires the 0200 transaction request.
2. Having been acquired by the BASE24 EPS-Net Interchange Interface, the 0200 transaction request is formatted into an 0100 SEM request, a copy held in extended memory, an *outbound* timer started and the request forwarded to the EM.
3. The ICF outbound timer limit expires before a response has been received from EPS-Net.
4. The EPS-Net Interchange Interface formats a 0420 reversal advice and adds it to the SAF to be sent as soon as possible. When it is sent, a *SAF* timer is started to wait for the 0430 acknowledgment advice from the EM.
5. The EPS-Net Interchange Interface deletes the 0200 request from extended memory and formats a 0210 response; this is then forwarded to the specified BASE24 Authorisation process. The EPS-Net Interchange Interface will insert a response code of 068 (decline) for ATM transactions and 113 (timeout) for POS transactions.
6. The EM formats a 0430 reversal advice acknowledgment and forwards it to the BASE24 EPS-Net Interchange Interface. The original 0420 SAF timer is located, deleted and the SAF record is then deleted.
7. The late 0110 is received from the EM. The BASE24 EPS-Net Interchange Interface sends a reject (0800) request with a reject reason code of 02 (late response).

5.1.4 BASE24 Unable to Deliver a Request to the EM

This could be due to the Interchange line going down, a message failure on the Interchange line or the Interchange Interface not logged on.

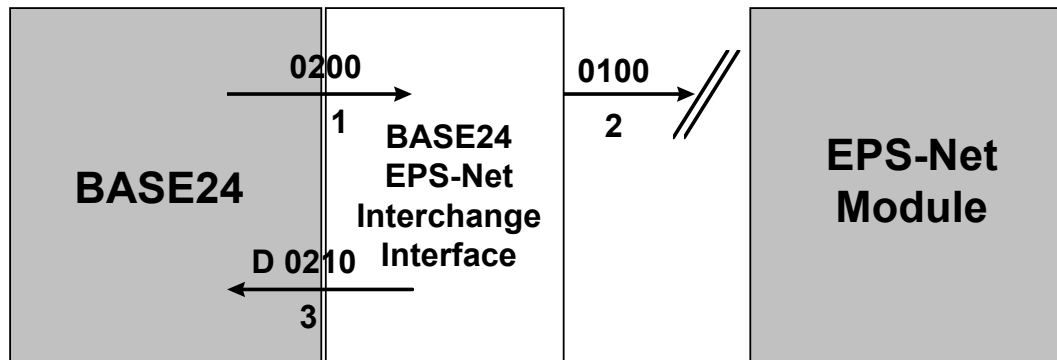


Figure 5–4 BASE24 Unable to Deliver a Request to the EM

The following explains the flow of the message illustrated in Figure 5–4:

1. A normal request message flow.
2. Depending on the configuration, the transaction will be denied, re-routed, or default authorised (POS only) for any of the following conditions:
 - ⇒ the line is down between BASE24 and the EM
 - ⇒ there is a line failure when the transaction is sent to the EM
 - ⇒ the BASE24 EPS-Net Interchange Interface is logged off
3. The EPS-Net Interchange deletes the transaction from extended memory and returns a 0210 response (response message flow). For an ATM transaction, the EPS-Net Interchange Interface returns a response code of 068 (decline); for a POS transaction, a response code of 100 (unable to process transaction) is returned, enabling the Device Handler/Router/Authorisation process to re-route the 0200 message.

5.2 BASE24 ISSUER MESSAGE FLOWS

This section describes transaction message flows where BASE24 is acting as an Issuer Host System (IHS).

In this section the following conventions have been used:

- A 0110 is an approved external 0110 response
- A 0210 is an approved internal 0210 response
- D 0110 is a declined external 0110 response
- D 0210 is a declined internal 0210 response

5.2.1 Normal Customer Transaction (IHS)

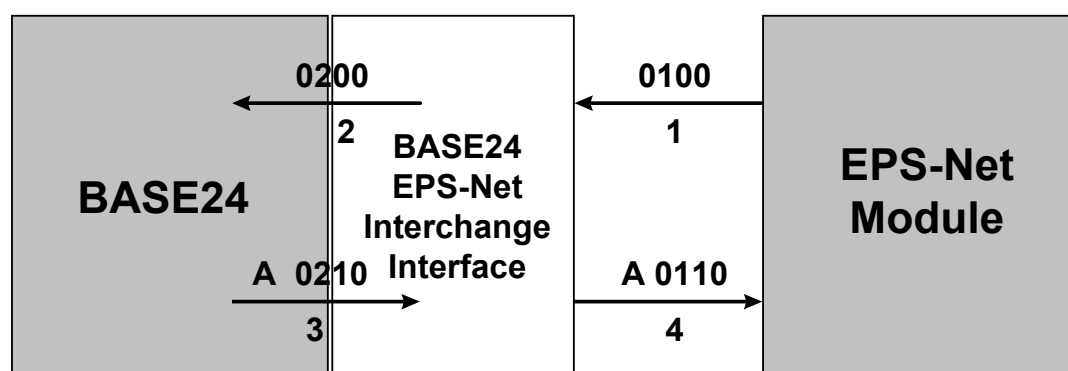


Figure 5–5 Normal Customer Transaction (IHS)

The following explains the flow of the message illustrated in Figure 5–5:

1. For a normal request message flow, the 0100 authorisation request is received by the BASE24 EPS-Net Interchange Interface.
2. The 0100 SEM is converted into an 0200 Standard Internal Message (PSTM/SEM), a copy of the 0100 is held in extended memory, an *inbound* timer is set for the response to be received for the request and the 0200 STM/PSTM authorisation request is then passed to the BASE24 Authorisation process specified in the Auth Process field of the ICF.
3. For a normal response message flow, the 0210 PSTM/SEM is received from the BASE24 Authorisation process.
4. The inbound timer is located and deleted by the EPS-Net Interchange Interface, the original 0100 SEM request is located in extended memory, and a 0110 SEM authorisation response is formatted and then sent to the EM.

5.2.2 EPS-Net Timeout of Response from the Authoriser

This occurs if BASE24 or a back-end host replies with a late 0210 response.

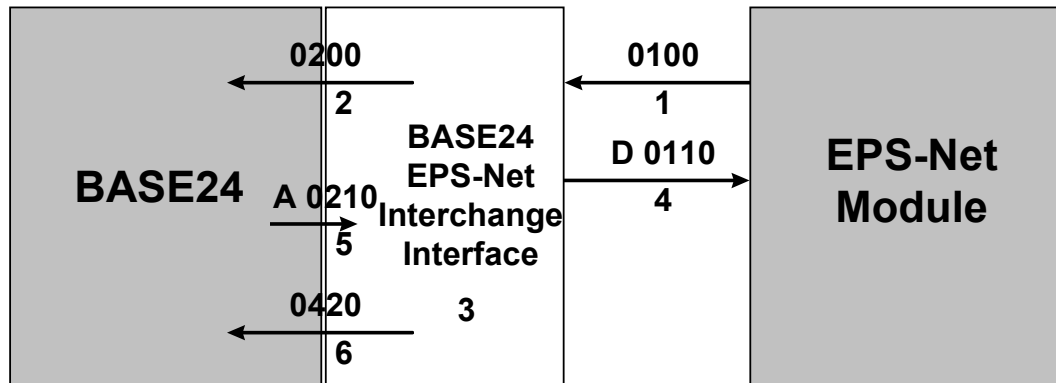


Figure 5–6 EPS-Net Timeout of Response from the Authoriser

The following explains the flow of the message illustrated in Figure 5–6:

1. The 0100 SEM authorisation request is received by the BASE24 EPS-Net Interchange Interface process.
2. The EPS-Net Interchange Interface formats an internal 0200 STM/PSTM from the request, places a copy of the 0100 in extended memory, starts an *inbound* timer for the transaction and passes it to the specified BASE24 Authorisation process.
3. The inbound timer expires before a response is received from the authorising entity.
4. The EPS-Net Interchange Interface deletes the transaction request from extended memory and formats and sends a declined 0110 transaction response to the EM. The EPS-Net Interchange Interface returns an ATM transaction with a response code of 072 (destination unavailable), and returns a POS transaction with a response code of 100 (unable to process).
5. The late 0210 response is received at the EPS-Net Interchange Interface. The Interface will not be able to locate the original request in extended memory.
6. If the response code indicates that the transaction is approved, the EPS-Net Interchange Interface formats a 0420 reversal advice and sends it to the specified BASE24 Authorisation process. Otherwise, the transaction will be ignored.

5.2.3 Unable to Deliver Response to EPS-Net

This could be due to a line failure between the EPS-Net Interchange Interface module and the EM after the 0100 authorisation request has been delivered to BASE24.

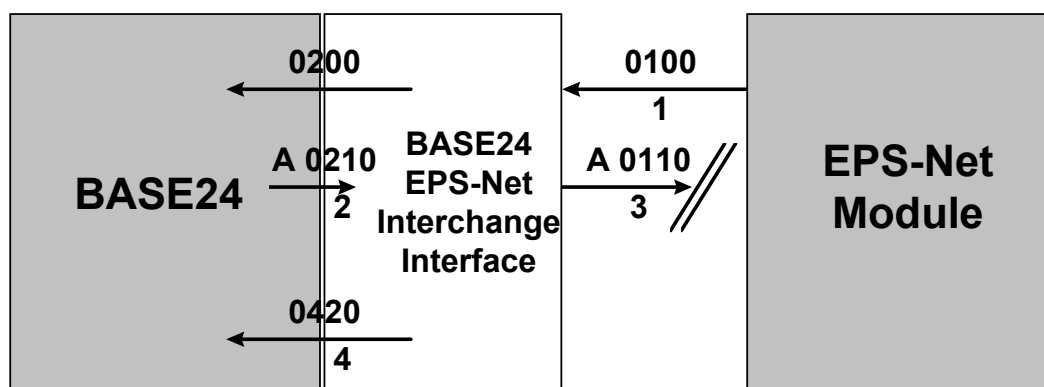


Figure 5–7 Unable to Deliver Response to EPS-Net

The following explains the flow of the message illustrated in Figure 5–7:

1. Normal transaction request flow.
2. Transaction is authorised by BASE24 (or a back-end host).
3. The communications line between BASE24 and the EM is down. The BASE24 EPS-Net Interchange Interface is unable to forward the approved response to the EM.
4. The BASE24 EPS-Net Interchange Interface generates a 0420 reversal advice and passes it to the originating Authorisation process.

5.2.4 Reversal Advice Received from EM (Acquirer Timeout)

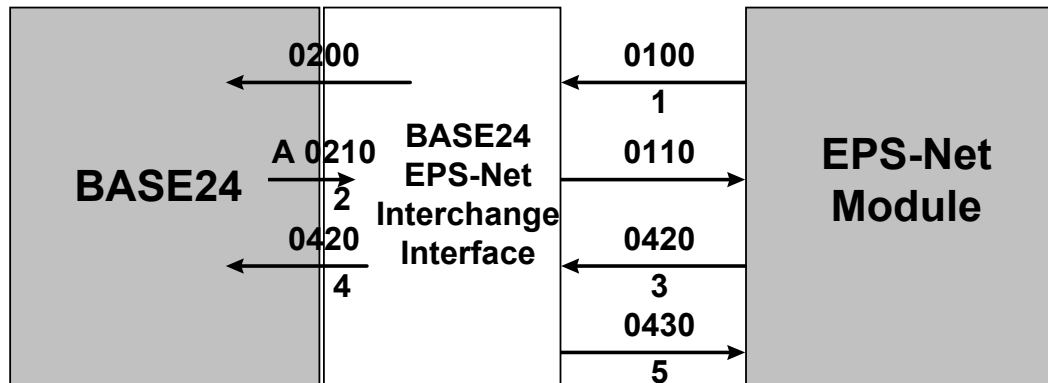


Figure 5–8 Reversal Advice Received from the EM (Acquirer Timeout)

The following explains the flow of the message illustrated in Figure 5–8:

1. Normal transaction request message flow.
2. Normal transaction response message flow.
3. A 0420 reversal advice is generated by the EM and forwarded to the BASE24 EPS-Net Interchange Interface.
4. The EPS-Net Interchange Interface receives the 0420 advice and sends it to the BASE24 Authorisation process.
5. The EPS-Net Interchange Interface formats a 0430 reversal advice acknowledgment and forwards it to the EM.

5.2.5 Reversal Advice Received from EM (Timeout at EM)

This occurs in the situation where the transaction is timed out by the local EM.

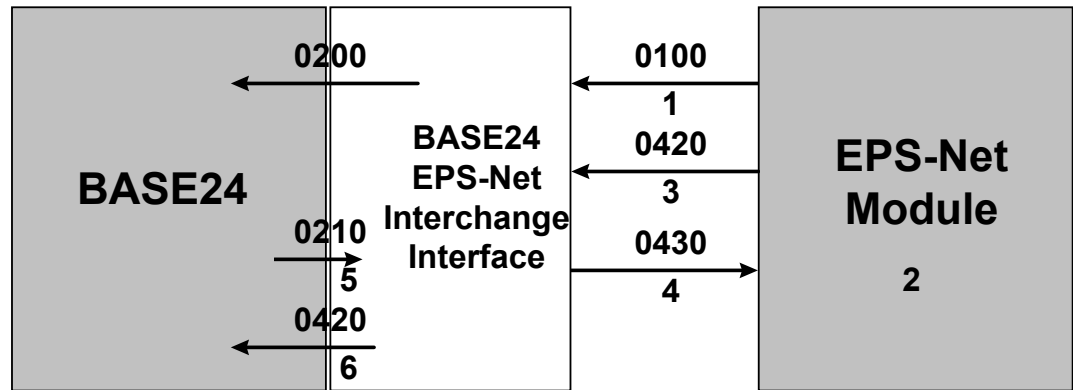


Figure 5–9 Reversal Advice Received from EM (Timeout at EM)

The following explains the flow of the message illustrated in Figure 5–9:

1. Normal transaction request message flow.
2. A timeout occurs at the EM.
3. The EM formats a 0420 reversal advice and sends it to the Issuer (BASE24 EPS-Net Interchange Interface). When the EPS-Net Interchange Interface receives the 0420 reversal advice it deletes the original 0100 transaction request from extended memory and also deletes the *inbound* timer. The 0420 is added to the ILF with a sub-state of timeout.
4. The BASE24 EPS-Net Interchange Interface formats a 0430 reversal advice acknowledgment and sends it to the EM.
5. The late 0210 approved response is received at the BASE24 EPS-Net Interchange Interface from the authorising entity. The original 0200 request cannot be located in extended memory.
6. If the response code indicates that it is an approved response, a reversal advice is formatted and forwarded to the originating Authorisation process.

5.2.6 EM Performs Stand-in Authorisation

This occurs if the Interchange line was down and the BASE24 EPS-Net Interchange Interface was unable to receive authorisation requests from the EM.

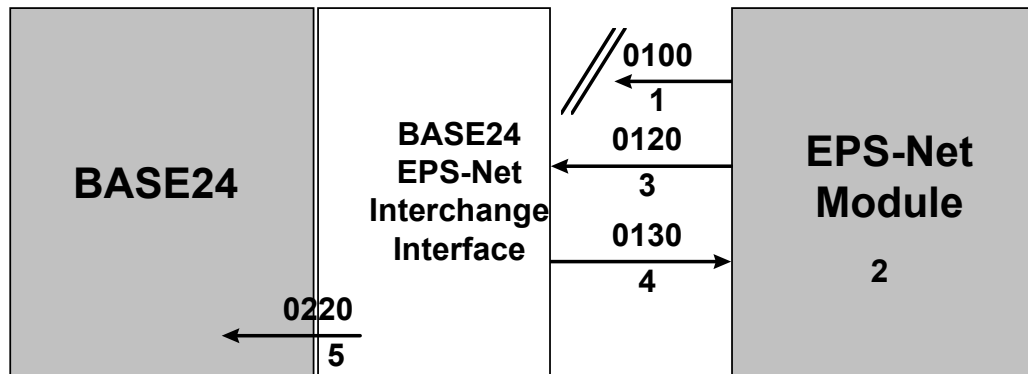


Figure 5–10 EM Performs Stand-in Authorisation

The following explains the flow of the message illustrated in Figure 5–10:

1. The EM cannot route the 0100 request to the Issuer system, for example, due to communications line failure.
2. The EM stands-in and authorises the transaction on behalf of the Issuer.
3. The communications line is re-established and the EM will forward the 0120 authorisation advice transaction to the BASE24 EPS-Net Interchange Interface on the Issuer system.
4. The BASE24 EPS-Net Interchange Interface formats a 0130 authorisation advice acknowledgment and sends it back to the EM.
5. The EPS-Net Interchange Interface formats the 0120 authorisation advice into a 0220 force post advice and sends it to the specified BASE24 Authorisation process for further processing.

5.3 TRANSACTION MATCHING

In order to trace transactions, the EPS-Net Interchange Interface uses a combination of fields from the external message, together with a flag indicating whether BASE24 is acting as acquirer or issuer for the transaction. The data elements from the external message that are used to trace transactions are listed below.

- Data element 032 is the Acquiring Institution Identification Code
- Data element 002 is the PAN
- Data element 011 is the Systems Trace Audit Number (STAN)
- Data element 004 is the Transaction Amount (BASE24-originated transactions only)
- Data element 033 is the Forwarding Institution Identification Code (EPS-Net-originated transactions only)

Data element 033 is used for tracing purposes in BASE24 Release 6.0 systems (BASE24-REL = '60') only.

5.3.1 Message Matching

In order to match a response message to the corresponding request message, the EPS-Net Interchange Interface uses the data elements listed above.

In order to match a reversal message to the corresponding request/response message, the EPS-Net Interchange Interface retrieves the STAN and the Forwarding Institution Identification Code from the Original Data Elements (data element 090) in the reversal message.

5.3.2 Definition of a Unique STAN

To ensure a unique Systems Trace Audit Number (STAN) is used when multiple EPS-Net Interchange Interface processes exist in a single BASE24 system, two LCONF parameters need to be configured. These are FIRST-EURO-TRACE-NUM and INCR-EURO-TRACE NUM-BY. These parameters are optional, as they have no value for a BASE24 system that operates with a single EPS-Net Interchange Interface process. If these parameters are not present, then the EPS-Net Interchange Interface process will assume that they have a value of 1.

The FIRST-EURO-TRACE-NUM needs to be defined per EPS-Net Interchange Interface process. It is used to define the starting Trace Number for each process; therefore this parameter should be set to a different value for each EPS-Net Interchange Interface process operating within a BASE24 system. The INCR-EURO-TRACE NUM-BY is the value by which the Trace Number is to be incremented, and should be set to the number of EPS-Net Interchange Interface processes that are configured.

When an EPS-Net Interchange Interface process is started, the Trace Number is set to the value contained within the FIRST-EURO-TRACE-NUM parameter.

When the Trace Number is incremented in the processing, it is incremented by the value contained within the INCR-EURO-TRACE NUM-BY parameter.

When incrementing the Trace Number, if the value exceeds 999999 then the number is reset to the value in the FIRST-EURO-TRACE-NUM parameter.

5.4 TRANSACTION SECURITY

Certain transactions are valid only if the cardholder has entered a PIN. When sending authorisation request messages to MasterCard Europe, the EPS-Net Interchange Interface determines whether a PIN is required for the transaction type. ATM transactions must contain a PIN block, and both cash withdrawals from a bank branch terminal and CAT Level 1 transactions must be PIN-based.

PIN blocks are encrypted using Triple DES in the interface with the EM.

The processing of PIN blocks depends on whether the BASE24 system is supporting the Transaction Security Services (TSS).

5.4.1 TSS Not Supported

This processing is applicable to BASE24 Release 5.x and Release 6.0 systems.

The Inbound and Outbound double-length ZMKs are manually exchanged between the EPS-Net Interchange Interface and the interchange. These values are stored on screen 2 of the KEY6 (see section 3.5). A single-length intermediate PIN block encryption key (KPE) is stored on screen 1 of the KEY6 file.

A PIN-based transaction acquired by BASE24 and forwarded to the BASE24 EPS-Net Interchange Interface contains a PIN block encrypted under a single-length key held in the internal STM/PSTM message. At the EPS-Net Interchange Interface, the PIN block is translated from being encrypted under this key to being encrypted under the double-length Outbound ZMK from the KEY6. The transaction is then passed to the EM.

For a PIN-based transaction request received at the BASE24 EPS-Net Interchange Interface, the PIN block in the external message is encrypted under the double-length Inbound ZMK. At the EPS-Net Interchange Interface, the PIN block is translated from being encrypted under this key to being encrypted under the single-length intermediate PIN block encryption key (KPE) from the KEY6. The PIN block and the intermediate PIN block encryption key (KPE) are moved to the BASE24 Standard Internal Message (STM) or BASE24-pos Standard Internal Message (PSTM).

5.4.2 TSS Supported

This processing is applicable to BASE24 Release 6.0 systems only. Further details of the TSS are given in the *BASE24 Transaction Security Services Processing Guide*.

The Inbound and Outbound double-length ZMKs are manually exchanged between the EPS-Net Interchange Interface and the interchange. These values are stored in the TSS database, and locators to these keys are entered on screen 2 of the KEYF (see section 3.4). The Inbound ZMK is used to decrypt all PIN blocks in the inbound requests from the MasterCard Europe network. The Outbound ZMK is used to encrypt all PIN blocks in the outbound requests to the MasterCard Europe network.

A PIN-based transaction acquired by BASE24 and forwarded to the BASE24 EPS-Net Interchange Interface contains a PIN block encrypted under a single-length or double-length key identified by a key locator in the internal STM/PSTM message. At the EPS-Net Interchange Interface, the PIN block is translated from being encrypted under this key to being encrypted under the double-length Outbound ZMK, identified by the key locator on the KEYF. The transaction is then passed to the EM.

For a PIN-based transaction request received at the BASE24 EPS-Net Interchange Interface, the PIN block in the external message is encrypted under the double-length Inbound ZMK. At the EPS-Net Interchange Interface, the PIN block and the key locator to the Inbound ZMK from the KEYF are moved to the BASE24 Standard Internal Message (STM) or BASE24-pos Standard Internal Message (PSTM).

5.5 TRANSACTION TYPES

When BASE24 acts as an acquirer host system (AHS), the EPS-Net Interchange Interface provides support for both BASE24-atm and BASE24-pos transactions.

When BASE24 acts as an issuer host system (IHS), the EPS-Net Interchange Interface provides support for all financial transaction types defined by MasterCard Europe.

5.5.1 BASE24-atm Processing

This section describes specific processing performed on ATM transactions by the EPS-Net Interchange Interface.

5.5.1.1 Processing/Transaction Code Mappings

When a transaction message is received from EPS-Net, the MasterCard Europe processing code is mapped to a BASE24-atm transaction, as shown below.

EPS-Net Processing Code		BASE24-atm Transaction Code	
01	ATM Withdrawal	10	Withdrawal
30	Balance Inquiry	30	Balance Inquiry
91	PIN Unblock	82	EMV PIN Unblock
92	PIN Change	81	PIN Change

Note that an EPS-Net processing code of '00' (purchase) will be mapped to a BASE24-atm transaction code of '10' (withdrawal), if DE 018 (Merchant Category Code) contains '6012' (indicating a non-currency dispense transaction), and the NCD segment is present in the ICF record.

When a transaction message is sent to EPS-Net, the BASE24-atm transaction code is mapped to a MasterCard Europe processing code, as shown below.

BASE24-atm Transaction Code		EPS-Net Processing Code	
10	Withdrawal	01	ATM Withdrawal
30	Balance Inquiry	30	Balance Inquiry
81	PIN Change	92	PIN Change
82	EMV PIN Unblock	91	PIN Unblock

Note that a BASE24-atm transaction code of '10' (withdrawal) will be mapped to an EPS-Net processing code of '00' (purchase), if the internal message contains the NCD token (indicating a non-currency dispense transaction), and the NCD segment is present in the ICF record.

Other BASE24-atm transaction types may be routed to an alternate destination for authorisation, (e.g., the ATM Authorisation process).

5.5.2 BASE24-pos Processing

This section describes specific processing performed on POS transactions by the EPS-Net Interchange Interface.

5.5.2.1 Electronic Commerce (e-commerce) Transactions

Electronic commerce transactions are characterised by the use of an open network to transmit the PAN of a card between a cardholder network access device and a merchant server to initiate a card payment. The BASE24 POS 5.1 token contains a field used to identify e-commerce transactions.

Authorisation messages for e-commerce transactions are allowed to contain chip data. The EPS-Net Interchange Interface already supports these messages, but the setting of the Point of Service Entry Mode (data element 022) ensures that e-commerce takes precedence over chip information. New POS Data values (data element 061) have been defined to support e-commerce.

Security Level Indicator values have been defined to support e-commerce with chip data.

The Universal Cardholder Authentication Field (UCAF) is a universal, multipurpose data transport infrastructure that will be used to communicate authentication information between the cardholder, the merchant, the issuer and the acquirer.

The Security Level Indicator and UCAF data are each included in a Primary Data Sub-element (PDS) within data element 048. These PDSs may be present for telephone order transactions in addition to e-commerce transactions. Merchants may also provide UCAF data in telephone order transactions.

UCAF is a variable length (maximum 32-byte) sub-element with a flexible data structure. A variety of issuer security schemes, including Secure Payment Architecture (SPA), 3D-Secure and other future security solutions can use it.

The data structure of UCAF includes a control byte, assigned to MasterCard Europe, to represent a particular use of UCAF, such as SPA. The control byte is followed by issuer discretionary data that may be standardised for specific applications i.e. SPA.

SPA is an issuer security scheme designed to authenticate cardholders when paying online. It is a design specification based on MasterCard intellectual property. It will be licensed to technology vendors and end users for integration into current security solutions such as server-based wallets and pseudo account number systems.

Note that data element 055 (ICC System Related Data) will not be processed for e-commerce transactions.

5.5.2.1.1 Acquirer Processing

E-commerce transactions are those where the PS51 Token is present and E-COM-FLG contains 5, 6, 7, 8 or 9. The first two bytes of POS-ENTRY-MDE must be set to '81' for e-commerce transactions.

If the Point of Service Data Token is not present, data element 061 (POS Data), sub-field 10 (CAT Level Indicator) must be set to '6' for e-commerce transactions.

PDS 42, which contains SEC-LEVEL-IND, should only be added to the Standard External Message (SEM) for an e-commerce, telephone order or recurring payment transaction. The first two bytes of SEC-LEVEL-IND depend on the value of E-COM-FLG and the presence or absence of an EMV Application Cryptogram (the ARQC field in the EMV Request Token), as follows:

E-COM-FLG	ARQC	SEC-LEVEL-IND (first 2 bytes)
5	No	21
5	Yes	23
6	No	21
6	Yes	23
7	No	21
7	Yes	23
8	N/A	91
9	N/A	91

The third byte of SEC-LEVEL-IND indicates the use of UCAF for a transaction. The third byte of SEC-LEVEL-IND is set from AUTHN-COLL-IND of the PS51 token, as follows:

SEC-LEVEL-IND (third byte)	Description
0	UCAF not supported by merchant
1	UCAF supported by merchant but not provided by issuer
2	UCAF is present

PDS 43, which contains UCAF-DATA, should only be added to the SEM if the third byte of SEC-LEVEL-IND is set to 2. The AUTHN-DATA token is used to transport UCAF data within BASE24.

5.5.2.1.2 Issuer Processing

The first two bytes of SEC-LEVEL-IND are mapped into E-COM-FLG as follows:

SEC-LEVEL-IND (first 2 bytes)	E-COM-FLG
21	7
23	7
91	8

The third byte of SEC-LEVEL-IND is moved to AUTHN-COLL-IND of the PS51 token, as follows:

SEC-LEVEL-IND (third byte)	Description
0	UCAF not supported by merchant
1	UCAF supported by merchant but not provided by issuer
2	UCAF is present

If the third byte of SEC-LEVEL-IND is set to 2, the AUTHN-DATA token is created with AUTHN-IND-FLG set to “01” and AUTHN-DATA populated from PDS 43. If PDS 43 is not present in the message from MasterCard Europe, the EPS-Net Interchange Interface will decline the transaction, because the data is inconsistent.

An “on-behalf” service provides issuers with the option of requesting MasterCard Europe to validate the value of the UCAF-DATA, before an authorisation request is forwarded to the issuer. If MasterCard Europe validates the UCAF-DATA on behalf of the issuer, then the authorisation request message will contain data element 048, PDS 71. This comprises 2 sub-fields: sub-field 1 indicates the type of on-behalf service (‘05’), and sub-field 2 indicates the result of the verification. The EPS-Net Interchange Interface maps sub-field 2 to the CAVV-AAV-RSLT-CDE field in the PS51 Token, as follows:

PDS 71 Sub-field 2		CAVV-AAV-RSLT-CDE	
Value	Description	Value	Description
V	Validation successful	2	Passed validation
I	Invalid	1	Failed validation
U	Unable to validate	0	Not validated

5.5.2.2 Address Verification Service

The Address Verification Service (AVS) is a service for online members that verifies the cardholder's billing address. Using the Address Verification Service is one way to combat fraud associated with Card Not Present (CNP) transactions.

Note that the AVS data in the external message (DE 120) is not logged to the ILF.

5.5.2.2.1 Acquirer Processing

The presence of AVS data in a request message is identified by a value of '98' in the PSTM field ADDR-TYP. If AVS data is present, then the EPS-Net Interchange Interface checks the contents of the AVS data for non-numeric characters.

If non-numeric characters are found, then the AVS data must be in the uncondensed format (format 1), as required for non-UK acquirers. The EPS-Net Interchange Interface formats the external request message as follows:

- If the BASE24-pos transaction code is '16' (card verification), then set DE 048 PDS 82 (AVS Request) to '51' (AVS only). Otherwise, set DE 048 PDS 82 (AVS Request) to '52' (AVS and authorisation request transaction).
- In DE 120, set the address data sub-element tag to '01' (format 1). If the International Address Verification Service (IAVS) Data Token is present, then use the contents to format the remainder of DE 120. Otherwise, format DE 120 from the PSTM ADDR-FLDS and ZIP-CDE fields.

If non-numeric characters are not found, then the AVS data must be in the condensed format (format 2), as supported by UK acquirers. The EPS-Net Interchange Interface formats the external request message as follows:

- In DE 120, set the address data sub-element tag to '02' (format 2). Format DE 120 from the PSTM ADDR-FLDS and ZIP-CDE fields.

The result of the AVS data validation is returned to the acquirer in DE 048 PDS 83 (AVS Response) in an external response message.

5.5.2.2.2 Issuer Processing

In DE 094 in network management request messages, the issuer specifies the preferred format of AVS data received in authorisation request messages. Issuers can specify the preferred format by configuring the LCONF parameter AVS-IND. Note that the AVS data may always be in condensed format (format 2) when a UK acquirer sends the request message.

The presence of AVS data in an authorisation request message is identified by the presence of DE 120 (Record Data). If AVS data is present, then the EPS-Net Interchange Interface checks for the presence of DE 048 PDS 82 (AVS Request).

If DE 048 PDS 82 (AVS Request) is present in the message, then the AVS data must have originated from a non-UK acquirer. The EPS-Net Interchange Interface formats the internal request message as follows:

- If DE 048 PDS 82 (AVS Request) is '51' (AVS only) and DE 004 (Transaction Amount) contains zero, then set the BASE24-pos transaction code to '16' (card verification).
- If the address data sub-element tag in DE 120 is set to '01' (format 1) and the address data is more than 20 characters long, then format and add the International Address Verification Service (IAVS) Data Token. Format the PSTM ADDR-FLDS (using the first 20 characters of the address data) and ZIP-CDE fields from DE 120.
- Set PSTM.ADDR-TYP to '98' (AVS processing required).

If DE 048 PDS 82 (AVS Request) is not present in the message, then the AVS data must have originated from a UK acquirer. The EPS-Net Interchange Interface formats the internal request message as follows:

- Format the PSTM ADDR-FLDS and ZIP-CDE fields from DE 120.
- Set PSTM.ADDR-TYP to '98' (AVS processing required).

The result of the AVS data validation is returned to the acquirer in DE 048 PDS 83 (AVS Response) in an external response message.

5.5.2.3 Processing/Transaction Code Mappings

When a transaction message is received from EPS-Net, the MasterCard Europe processing code is mapped to a BASE24-pos transaction, as shown below.

EPS-Net Processing Code		BASE24-pos Transaction Code	
00	Purchase	10	Normal Purchase
01	Cash Withdrawal	15	Cash Advance
09	Purchase with Cash Back	18	Purchase with Cash Back
17	Cash Advance	15	Cash Advance
28	Payment Transaction	14	Merchandise Return

Note that an EPS-Net processing code of '00' (purchase) will be mapped to the following BASE24-pos transaction codes, if the specified conditions are satisfied:

- Mail/Phone Order (13), if DE 061 (POS Data) sub-field 4 (POS Cardholder Presence Indicator) contains a non-zero value (cardholder not present).
- Card Verification (16), if DE 004 (Transaction Amount) contains zero, and DE 048 PDS 82 (AVS Request) contains '51' (AVS only).

When a transaction message is sent to EPS-Net, the BASE24-pos transaction code is mapped to a MasterCard Europe processing code, as shown below.

BASE24-pos Transaction Code		EPS-Net Processing Code	
10	Normal Purchase	00	Purchase
11	Pre-Authorisation Purchase	00	Purchase
13	Mail/Phone Order	00	Purchase
15	Cash Advance	17	Cash Advance
16	Card Verification	00	Purchase
18	Purchase with Cash Back	09	Purchase with Cash Back
14	Merchandise Return	28	Payment Transaction

Note that a BASE24-pos card verification transaction is supported only if the request message contains Address Verification Service (AVS) data, in which case the transaction is sent to EPS-Net as an "AVS only" request.

Note that a BASE24-pos merchandise return transaction is mapped to a payment transaction only if at least one or both of the following conditions are satisfied:

- The SIC Code (MCC) in the PSTM is set to either '6532' (Member Financial Institution) or '6533' (Merchant).
- The Transaction Subtype Token (token ID = 'BM') is present, and the transaction subtype is set to 'B000' (payment from third party).

5.5.2.4 Adjustment to Reversal Translation

The Host-EM Interface supports only a subset of the POS transaction types supported by BASE24.

However, adjustment transactions may be mapped to reversals (if the transaction being adjusted can be located in the ILF).

The ILF is checked for an entry for the adjusted transaction (see section 5.9.3), if the corresponding request message was sent to EPS-Net. If a record is located, then the message in the ILF record is converted to a reversal for transmission to EPS-Net. If a record is not located, then the transaction is authorised blind (see below).

5.5.2.5 Blind Authorisation

The Host-EM Interface supports only a subset of the POS transaction types supported by BASE24.

However, some other POS transaction types are authorised blind, (i.e., the transaction is approved by the interface without attempting to forward the message to EPS-Net).

The EPS-Net Interchange Interface supports the blind authorisation of the following POS transaction types:

- Preauthorisation completions
- Refunds (not Payment Deposits)
- Refund adjustments (not Payment Deposits)
- Adjustments which could not be mapped to reversals (i.e., a corresponding ILF record could not be located)

Other BASE24-pos transaction types may be routed to an alternate destination for authorisation (e.g., the POS Router/Authorisation process).

5.5.2.6 Alternate Routing

If an authorisation request cannot be sent to EPS-Net, e.g. because of a communications error, then it is possible for “alternate” routing to occur. The transaction message is routed either to another interface process or back to the POS Router/Authorisation process in an attempt to obtain authorisation for the transaction.

It is also possible to invoke alternate routing when the EPS-Net Interchange Interface receives a response message containing a response code indicating some type of system malfunction, e.g. 31, 91 or 96. This processing may be configured to apply to any response code returned by EPS-Net, by specifying the relevant response code value(s) in the LCONF parameter EURO-STAND-IN-RCS.

5.5.2.7 Proximity Chip Payment Transactions

Proximity Chip payment transactions are identified by specific values in sub-field 1 ‘Card Data Entry Mode’ of DE 022 ‘Point of Service Entry Mode’, and sub-field 11 ‘POS Card Data Terminal Input Capability Indicator’ of DE 061 ‘POS Data’.

For DE 022, sub-field 1, the values are as follows:

- 07 (PAN auto-entry via contactless M/Chip)
- 91 (PAN auto-entry via contactless magnetic stripe)

For DE 061, sub-field 11, the values are as follows:

- 3 (PAN auto-entry via contactless M/Chip)
- 4 (PAN auto-entry via contactless magnetic stripe)

Acquirers of Proximity Chip cards may optionally provide the values for sub-field 1 of DE 022 and for sub-field 11 of DE 061. These values are set if the Cardholder Activated Terminal (CAT) Indicator in the Point of Service Data Token is set to ‘7’ (radio frequency device). Alternatively, if the first two digits of the Point of Service Entry Mode in the PSTM are set to ‘07’ or ‘91’, then the value is moved directly to DE 022 sub-field 1. Similarly, if the Terminal Input Capability Indicator in the Point of Service Data Token is set to ‘3’ or ‘4’, then the value is moved directly to DE 061 sub-field 11.

Issuers of Proximity Chip cards must be prepared to receive the new values for sub-field 1 of DE 022 and the modified values for sub-field 11 of DE 061. The new values are moved directly to the Point of Service Entry Mode in the PSTM and the Terminal Input Capability Indicator in the Point of Service Data Token.

5.5.2.8 MasterCard Electronic Program

MasterCard Electronic is a MasterCard brand designed to meet the needs of issuers targeting higher-risk cardholders. MasterCard Electronic requires 100% online authorization to allow close risk management.

The value of DE 048 PDS 76 (MasterCard Electronic Acceptance Indicator) in 0100 authorisation request messages indicates whether the acquirer participates in the MasterCard Electronic program. The LCONF parameter DENY-MC-E-COMM-TXN allows the issuer to decline MasterCard Electronic transactions if the acquirer (or the merchant) does not participate in the program.

If DE048 PDS 76 (MasterCard Electronic Acceptance Indicator) contains “C” or “M” and the LCONF parameter DENY-MC-E-COMM-TXN is not set to “N”, then the transaction will be declined with a response code of ‘57’ (transaction not supported). The message will then be dropped, unless the interface has been configured to forward declined advices to the POS Router/Authorisation process.

The EPS-Net Interchange Interface will not populate DE 048 PDS 76 in 0100 authorisation request messages sent to EPS-Net.

5.5.3 Cash Withdrawals from Bank Branch Terminals

It is possible to authorise cash withdrawals from terminals in bank branches through BASE24-atm or through BASE24-pos.

Where BASE24 is acting as issuer, the EPS-Net Interchange Interface uses an LCONF parameter to allow customers to configure whether cash withdrawals from terminals in bank branches should be routed to BASE24-atm or BASE24-pos. The LCONF parameter is called CASH-WDL-AUTH-PROD, and valid values are 'ATM' or 'POS', with 'ATM' being the default value.

MasterCard Europe requires PIN entry, and prohibits manually entered card details. Where BASE24 is acting as acquirer, the EPS-Net Interchange Interface ensures that debit card transactions from terminals in bank branches are rejected if either of these requirements is not met.

A merchant category code of '6010' (manual cash disbursement) is used to identify bank branches, and an LCONF parameter called DEBIT-CARD-TYPES is used to identify the valid card types, (i.e., debit cards).

If the above criteria are satisfied, the EPS-Net Interchange Interface formats an appropriate external message:

- The processing code (data element 003) is set to '01' instead of '17'
- The merchant category code (data element 018) is set to '6010'
- The point of service entry mode (data element 022) indicates PIN entry capability
- The point of service data (data element 061) is set appropriately

Note that this processing will result in transactions performed on ATM-only cards (e.g., CIRRUS) being logged to the PTLF.

To acquire cash withdrawals from terminals in bank branches through BASE24-pos, the following system changes are required:

- Configure a PRDF record for the bank branch with a SIC Code of '6010'
- Communicate with bank branch terminals using a POS Device Handler (e.g., SPDH)
- Ensure transactions are routed to the EPS-Net Interchange Interface

If there is a requirement to return an account balance in a cash withdrawal transaction initiated at a bank branch terminal, then the transaction must be routed through BASE24-atm (this applies to both acquirer and issuer processing).

5.6 CARD DETAILS PROCESSING

This section describes the processing performed by the EPS-Net Interchange Interface on the card details included in a transaction message.

5.6.1 Card Entry Options

BASE24 requires Track-2 data to perform routing and authorisation processing.

For acquirer processing, the EPS-Net Interchange Interface will always receive Track-2 data from the ATM Authorisation or POS Router/Authorisation process. The Track-2 data could have been either read from the card by the terminal or manually entered.

If the Track-2 data has been manually entered (i.e., POS transactions only), then the EPS-Net Interchange Interface will ensure that manual entry is allowed for the card type. This is achieved by comparing the card type in the PSTM with the contents of the LCONF parameter NO-MANUAL-ENTRY-CARD-TYPES. If a match is found, then the transaction is declined. Note that this processing does not apply to e-commerce transactions.

Certain transactions are valid only if the card details have not been manually entered. When sending authorisation request messages to MasterCard Europe, the EPS-Net Interchange Interface determines whether the transaction type allows the card details to have been manually entered. In addition to ATM transactions, card details must not be manually entered in any CAT transactions (except for e-commerce) or in cash withdrawals from a bank branch terminal.

For issuer processing, the EPS-Net Interchange Interface supports the following modes of entry for card details:

- Track-2
- Track-1
- PAN/Expiration Date manual entry (POS transactions only)

If the Track-2 field (data element 035) is not received in a request message from EPS-Net, a “pseudo” Track-2 is built from the contents of either the Track-1 (data element 045) or the PAN (data element 002) and Expiration Date (data element 014). The ATM Authorisation and POS Router/Authorisation processes require this information.

5.6.2 CVC1 Support

As a security feature, issuers must encode the Card Validation Code (CVC1) on both Track-1 and Track-2. This provides a mechanism for validating the card, allowing issuers to respond with reasonable assurance that the track data is present and accurate.

In order to validate CVC1 successfully, acquirers must provide the complete and unaltered track data. This capability is terminal-specific, and therefore cannot be derived by the EPS-Net Interchange Interface.

- In Release 5.0 and above BASE24 systems, this information is passed in a token from the acquiring interface or device handler to the EPS-Net Interchange Interface.
- In Release 3.4/4.0 BASE24 systems, the EPS-Net Interchange Interface uses the LCONF parameter INCOMPLETE-TRACK2-TERM-TYPES to identify which transactions do not contain complete and unaltered track data. Similarly, the EPS-Net Interchange Interface uses the LCONF parameters HOST-COMplete-TRACK2 and SWITCH-COMplete-TRACK2 to identify which acquiring Hosts and Interchanges supply complete and unaltered track data.

Where BASE24 is acting as an issuer system, the ATM Authorisation process and POS Router/Authorisation process can be configured to verify the CVC1 data (on either Track-1 or Track-2).

5.6.3 CVC2 Support

Card Validation Code 2 (CVC2) support is designed to provide added security for the card authentication process for POS authorisation requests, primarily in non-face-to-face environments.

CVC2 data is printed on the signature panel of the card, and can therefore be provided by a merchant. BASE24-pos can then forward the data to EPS-Net in an authorisation request message.

Where BASE24 is acting as an issuer system, the POS Router/Authorisation process can be configured to verify the CVC2 data.

MasterCard Europe will allow CVC1 and CVC2 to be transmitted in the same transaction. The EPS-Net Interchange Interface can include the results of CVC1 validation and CVC2 validation in a single response message returned to MasterCard Europe.

5.6.4 CVC3 Support

In a contactless magnetic stripe transaction, the card creates a dynamic Card Validation Code 3 (CVC3), unique to the transaction. CVC3 is supported in both Track-1 and Track-2 transactions. The CVC3 is sent to the acquirer in the Track-1 or Track-2 discretionary data area, together with other dynamic data used in the generation of the CVC3. This provides a mechanism for validating the card, allowing issuers to respond with reasonable assurance that the track data is present and accurate.

Where BASE24 is acting as an issuer system, the POS Router/Authorisation process can be configured to verify the CVC3 data (on either Track-1 or Track-2).

CVC1 and CVC3 cannot be transmitted in the same transaction, but MasterCard Europe will allow CVC3 and CVC2 to be transmitted in the same transaction. The EPS-Net Interchange Interface can include the results of CVC3 validation and CVC2 validation in a single response message returned to MasterCard Europe.

5.7 MERCHANT CATEGORY CODE VALIDATION

Certain transactions are valid only if performed on particular Merchant Category Codes (MCCs).

5.7.1 Incoming Messages from EPS-Net

When receiving transaction messages from MasterCard Europe, the EPS-Net Interchange Interface validates the transaction type against the MCC.

- Where the Merchant Category Code is 6011, the EPS-Net processing code must not be 00 (Purchase), 09 (Purchase with Cash Back), 17 (Cash Advance) or 28 (Payment Deposit).
- Where the Merchant Category Code is 6010, the EPS-Net processing code must not be 00 (Purchase), 09 (Purchase with Cash Back) or 28 (Payment Deposit).
- Where the EPS-Net processing code is 01 (Cash Withdrawal) the Merchant Category Code must be 6011 (for ATM withdrawals) or 6010 (for Bank Branch withdrawals).

5.7.2 Outgoing Messages to EPS-Net

When sending transaction messages to MasterCard Europe, the EPS-Net Interchange Interface validates the transaction type against the MCC.

- Where the Merchant Category Code is 6011, the EPS-Net processing code must not be 00 (Purchase), 09 (Purchase with Cash Back), 17 (Cash Advance) or 28 (Payment Deposit).
- Where the Merchant Category Code is 6010, the EPS-Net processing code must be 01 (Cash Withdrawal) or 17 (Cash Advance).
- The Merchant Category Code must not be 0000.

5.8 CARD TYPE VALIDATION

Certain transactions are valid only if performed on particular card types. When sending transaction messages to MasterCard Europe, the EPS-Net Interchange Interface validates the transaction type against the card type.

The DEBIT-CARD-TYPES parameter is used to specify debit card types explicitly (except Debit MasterCard). The DEBIT-MC-CARD-TYPES parameter is used to specify Debit MasterCard cards explicitly.

5.8.1 Purchase with Cash Back

These transactions are only allowed on debit cards and Debit MasterCards. When the DEBIT-CARD-TYPES parameter is configured, the transaction is allowed if the card type is contained in the parameter list. If it is not contained in the list, then the transaction is rejected, unless the DEBIT-MC-CARD-TYPES parameter is also configured and the card type is contained in the parameter list. Where the DEBIT-CARD-TYPES parameter has not been configured, the transaction is allowed.

5.8.2 Cash Advance

These transactions are only allowed on credit cards and Debit MasterCards. Therefore, when the DEBIT-CARD-TYPES parameter is configured, the transaction is allowed only when the card type does not appear in the parameter list. If it is contained within the list, then the transaction is rejected. Where the DEBIT-CARD-TYPES parameter has not been configured, the transaction is allowed.

5.8.3 Cash Withdrawal From Bank Branch Terminal

These transactions are only allowed on debit cards, but not on Debit MasterCards. Therefore, when the DEBIT-CARD-TYPES parameter is configured, the transaction is allowed only when the card type is contained in the parameter list. If it is not contained in the list, then the transaction is rejected. Acquirers must configure the DEBIT-CARD-TYPES parameter if support for cash withdrawals from bank branch terminals is required.

5.8.4 Proximity Chip Payment Transactions

These transactions must be authorised offline (i.e. at the terminal) if a debit card is used. Proximity chip payment transactions are identified by specific values in sub-field 1 'Card Data Entry Mode' of DE 022 'Point of Service Entry Mode':

- 07 (PAN auto-entry via contactless M/Chip)
- 91 (PAN auto-entry via contactless magnetic stripe)

Therefore, when the DEBIT-CARD-TYPES parameter is configured, the transaction is sent to EPS-Net only when the card type does not appear in the parameter list. If it is contained within the list, then the transaction is rejected. Where the DEBIT-CARD-TYPES parameter has not been configured, the transaction is sent to EPS-Net.

5.8.5 CVC2 Data

According to the *V5 Interface Specifications*, CVC2 data applies only to non-face-to-face MasterCard (including Debit MasterCard) transactions. However, the EPS-Net Interchange Interface also includes the CVC2 data (if present) in non-face-to-face debit card transactions. Therefore when the DEBIT-CARD-TYPES parameter is configured, the CVC2 data is included in included in face-to-face transaction messages sent to EPS-Net only when the card type does not appear in the parameter list. If it is contained within the list, then the CVC2 data is not included in face-to-face transaction messages. Where the DEBIT-CARD-TYPES parameter has not been configured, the CVC2 data is included in the message.

5.8.6 AVS Data

According to the *V5 Interface Specifications*, AVS data applies only to MasterCard (including Debit MasterCard) transactions. Therefore when the DEBIT-CARD-TYPES parameter is configured, the AVS data is included in the message sent to EPS-Net only when the card type does not appear in the parameter list. If it is contained within the list, the AVS data is not included in the message. Where the DEBIT-CARD-TYPES parameter has not been configured, the AVS data is included in the message.

5.8.7 Gambling Transactions

For credit card (including Debit MasterCard) transactions, CVC2 data is mandatory in non-face-to-face gambling transactions (MCC = 7995), where the EPS-Net processing code is 00 (Purchase). Therefore, when the DEBIT-CARD-TYPES parameter is configured, this type of transaction is allowed only when the card type does not appear in the parameter list. If it is contained within the list, then the transaction is rejected. Where the DEBIT-CARD-TYPES parameter has not been configured, the transaction is allowed.

5.8.8 Telephone Orders

For debit card (excluding Debit MasterCard) transactions, telephone order transactions are allowed, but only if the merchant supports UCAF. Therefore, when the DEBIT-CARD-TYPES parameter is configured, and the card type appears in the parameter list, the transaction is allowed only if the merchant is UCAF-enabled.

5.9 INTERCHANGE LOG FILE (ILF) PROCESSING

Transaction messages handled by the EPS-Net Interchange Interface are logged to the ILF.

5.9.1 ILF Formats

The EPS-Net Interchange Interface has traditionally supported Format-1 ILFs.

The EPS-Net Interchange Interface now supports Format-2 ILFs for D46 and higher HP NonStop operating systems. Format-2 file support allows BASE24 customers the ability to format ENSCRIBE disk file (ILF) partitions to accommodate Format-2 files. Files with format 2 partitions can contain much more data than the existing Format-1 files. Files, whether Format-1 or Format-2, can have up to 16 partitions. Format-1 files are limited to 4 gigabytes (each partition can have up to 2 gigabytes). Format-2 files have new block and label formats and can hold as much as 1024 gigabytes of data per partition, and are currently limited only by disk volume size. Large Format-1 files, limited by the capacity of a single disk volume, have used partitioning as a means to increase file capacity.

Customers wishing to implement Format-2 ILFs should start at cutover, by following the procedures described:

- Rename the existing ILF template and alternate key file(s), as the new files created should use the same locations.
- Create a new ILF Format-2 template file and its alternate key file(s).
- The Format-2 ILF template file in the create command must be set to the value specified in the LCONF assign for the ILF for the interface process.

When the interface process is cutover, it will create and begin using the new Format-2 ILF and new Format-2 alternate key file(s). Previous ILFs can be Format-1 and current ILFs can be Format-2.

5.9.2 ILF Record Structure

The transactions logged to the ILF by the EPS-Net Interchange Interface follow a standard structure that contains a fixed length portion (818 bytes), followed by the expanded external message structure, followed by any tokens configured for logging to the ILF.

Each ILF record has the following format:

Bytes 1–55	Primary Key
Bytes 56–98	ICF Record Information
Bytes 99–150	External Message Information
Bytes 151–818	STM or PSTM Message Data
Bytes 819–x	External Message and token data (variable length depending on the token data)

(x = 4072 for Format-1 ILFs and x = 4048 for Format-2 ILFs.)

The primary key for the EPS-Net ILF contains the following information:

- Flag indicating whether BASE24 is acting as acquirer or issuer for this transaction
- Data element 002 is the PAN
- Data element 004 is the Transaction Amount (BASE24-originated transactions only)
- Data element 032 is the Acquiring Institution Identification Code
- Data element 033 is the Forwarding Institution Identification Code (EPS-Net-originated transactions only)
- Data element 037 is the Retrieval Reference Number

The EPS-Net Interchange Interface provides users with the capability of identifying when the ILF is becoming full. The EPS-Net Interchange Interface maintains a counter of records written to the ILF. Each time the counter exceeds the value of the LCONF parameter ILF-NOTIFY-INTERVAL, the interface resets the counter and calculates how full the file has become. If the percentage of the file used exceeds the value of the LCONF parameter ILF-NOTIFY-PERCENTAGE, a warning message is logged.

5.9.3 Enhanced Adjustment Matching

The EPS-Net Interchange Interface attempts two additional matches if the above one is unsuccessful for BASE24-originated adjustment transactions where the corresponding request message was sent to EPS-Net.

The data elements from the external message and original external message that are used in the second attempt to trace these adjustment transactions are listed below:

- Data element 032 is the Acquiring Institution Identification Code
- Data element 002 is the PAN
- Data element 037 is the Retrieval Reference Number from the original transaction
- Data element 004 is the Transaction Amount

If a match is not successful, a reduced primary key is used. The data elements from the external message that are used in the third attempt to trace these adjustments are listed below:

- Data element 032 is the Acquiring Institution Identification Code
- Data element 002 is the PAN

If a match is successful, further checks on the following fields are made to ensure that the correct record has been located:

- Data element 004 is the Transaction Amount
- Data element 041 is the Card Acceptor Terminal Identification (Terminal ID)
- Data element 042 is the Card Acceptor Identification Code (Retailer ID).

5.9.4 Data Suppression

MasterCard does not allow members to store sensitive transaction data on their systems. This includes the discretionary data from the card's magnetic stripe and/or chip (Track-1 and Track-2), manual card verification digits (CVC2), and AVS data.

Acquirers can configure the EPS-Net Interchange to ensure that this data is suppressed, i.e. not logged to the ILF, using the following LCONF parameters:

- SWI-ACQ-DISCR-DATA-SUPPRS
- SWI-ACQ-CVD2-DATA-SUPPRS
- SWI-ACQ-AVS-DATA-SUPPRS

Issuers can configure the EPS-Net Interchange to ensure that this data is suppressed, i.e. not logged to the ILF, using the following LCONF parameters:

- SWI-ISS-DISCR-DATA-SUPPRS
- SWI-ISS-CVD2-DATA-SUPPRS
- SWI-ISS-AVS-DATA-SUPPRS

Note that issuers should suppress the Track-1/Track-2 discretionary data only if the EPS-Net Interchange Interface is running in a BASE24 Release 6.0 system. Earlier releases of BASE24-atm and BASE24-pos use the discretionary data in reversal message processing, and the track data is retrieved from the ILF when the EPS-Net Interchange Interface receives an external reversal advice.

5.10 STORE-AND-FORWARD (SAF) PROCESSING

The following is a description of the Store-and-Forward (SAF) processing performed by the EPS-Net Interchange Interface. Only reversal advices and authorisation advices are placed on the SAF.

1. If a reversal or force post transaction (advice) is received from BASE24, an external message will be formatted and placed on the EPS-Net Interchange Interface SAF for processing. In addition, if a late authorisation response is received from the EM, an external reversal advice will also be formatted and placed on the EPS-Net SAF for processing.
2. When the EPS-Net Interchange Interface determines that there are transactions on the SAF (the SAF counter is greater than zero), the EPS-Net Interchange Interface will send an external reversal advice or force post (advice) transaction to MasterCard Europe. The ICF SAF timer is set in the EPS-Net Interchange Interface for an acknowledgment to the transaction.
3. The EM will process the reversal advice or force post (advice) transaction and send an acknowledgment message to the BASE24 EPS-Net Interchange Interface. The EPS-Net Interchange Interface process will delete the transaction from the SAF.
4. If an acknowledgment is not received before the SAF acknowledgment timer expires at the BASE24 EPS-Net Interchange Interface, the Interface will attempt to send the transaction again as a repeat advice (message type 0421) until an acknowledgment is received or until the maximum number of SAF retries has been reached (specified in the ICF).

The EPS-Net Interchange Interface provides users with the capability of identifying when the SAF is becoming full. The EPS-Net Interchange Interface maintains a counter of records written to the SAF. Each time the counter exceeds the value of the LCONF parameter SAF-NOTIFY-INTERVAL, the interface resets the counter and calculates how full the file has become. If the percentage of the file used exceeds the value of the LCONF parameter SAF-NOTIFY-PERCENTAGE, a warning message is logged.

5.11 TOKEN USAGE

The EPS-Net Interchange Interface supports the following BASE24 tokens:

- Generic BASE24 Switch token (Token Ids = B0 and B1)
- Original Currency token (Token Id = BE)
- Transaction Subtype token (Token Id = BM)
- Track-1 token (Token Id = 23)
- ATM 5.0 token (Token Id = 03)
- PIN Change token (Token Id = 06)
- ATM Flag1 token (Token Id = 24)
- Non-Currency Dispense token (Token Id = A5)
- ATM Interchange Compliance token (Token Id = A6)
- POS 5.0 token (Token Id = 04)
- Alternate Merchant ID token (Token Id = 16)
- Interchange Compliance token (Token Id = 20)
- POS 5.1 token (Token Id = C0)
- Purchase token (Token Id = C2)
- Point of Service Data token (Token Id = C4)
- Authentication Data token (Token Id = CE)
- IAVS Data token (Token Id = CF)
- POS Data1 token (Token Id = CH)
- POS Merchant token (Token Id = CI)

The EPS-Net Interchange Interface also supports the BASE24 EMV tokens for customers who have licensed the EMV extension to the EPS-Net Interchange Interface (see section **Error! Reference source not found.**).

Tokens are supported in BASE24 Release 5.x systems (BASE24-REL = '50') and BASE24 Release 6.0 systems (BASE24-REL = '60') only.

5.11.1 Generic BASE24 Switch Token

The generic switch token is a variable-length token definition that can be used by any BASE24 interchange interface. The EPS-Net Interchange Interface supports the generic switch token.

Two token IDs are available for the switch token:

- B0 (Acquirer Switch token)
- B1 (Issuer Switch token)

The generic switch token includes data taken from the external message. For information on the EPS-Net definition of the generic switch token, refer to Appendix B.

For requests, advices and advice repeats received from the MasterCard Europe network, the interface adds the Acquirer Switch token (B0).

For reversal advices and reversal advice repeats received from EPS-Net, the interface checks for the presence of the Acquirer Switch token (B0) in the ILF. If the token exists, the interface updates the token; otherwise the interface will add the token. The EPS-Net Interchange Interface checks the version of the generic switch token during update processing. If the token version is not the current version, it is ignored.

For responses received from the MasterCard Europe network, the interface adds the Issuer Switch token (B1).

5.11.2 Original Currency Token

Card management systems such as ACI CMS require certain information that is not available in standard BASE24 transaction log file records. In particular, for transactions acquired through an international interchange, details of the currency in which the transaction was acquired are not written to either the Transaction Log File (TLF) or the POS Transaction Log File (PTLF).

The EPS-Net Interchange Interface includes the required information in the Original Currency token, and creates the token on all transaction messages received from EPS-Net, where the transaction currency differs from the cardholder billing currency.

5.11.3 Transaction Subtype Token

This token is used to carry data to identify payment transactions.

On payment transaction requests received from EPS-Net, the TXN-SUBTYP field in this token is set to indicate payment from third party “B000”, and field ISS-PROC-CDE field in this token is populated with the value in data element 003 (Processing Code).

On merchandise return requests received from BASE24, the EPS-Net Interchange Interface checks for the presence of this token. If the token is present, and the TXN-SUBTYP field in this token is set to indicate payment from third party “B000”, then data element 003 (Processing Code) is set to “28” (payment transaction) and the authorisation request message is sent to EPS-Net.

5.11.4 Track-1 Token

This token is used to hold Track-1 data when a transaction containing the Track-1 field (data element 045) is received from EPS-Net.

5.11.5 ATM 5.0 Token

This token is used to pass assorted data items specific to ATM processing. The data items used by the EPS-Net Interchange Interface are listed below.

- Error Flag
- Card Verify Flag
- Complete Track Data Flag

5.11.6 PIN Change Token

This token is used to carry information about the new PIN requested by a customer through a BASE24 system.

On requests sent to EPS-Net, the NEW-PIN-1 field in this token (if present) is used to populate data element 125 (New PIN Data).

On requests received from EPS-Net, the NEW-PIN-1 field in this token is populated from data element 125 (New PIN Data).

5.11.7 ATM Flag1 Token

This token is used to pass assorted data items specific to ATM processing. The data items used by the EPS-Net Interchange Interface are listed below.

- Card Status

When formatting a response message to EPS-Net, the EPS-Net Interchange Interface uses this data item to differentiate between a lost card (value '2') and a stolen card (value '3'). This allows the response code (DE 039) to be set to the appropriate value ('41' for a lost card, '43' for a stolen card).

5.11.8 Non-Currency Dispense Token

This token is used to identify non-currency dispense transactions, and to carry data related to the non-currency dispense transaction.

On requests sent to EPS-Net, the presence of this token is used to populate data element 003 (Processing Code) with a value of '00' (purchase).

On requests received from EPS-Net, this token is added if data element 003 (Processing Code) contains '00' (purchase) and data element 018 (Merchant Category Code) is set to '6012'. The HOPR-CONTENTS field is set to the value of the LCONF parameter NCD-DFLT-HOPR-CONTENTS.

5.11.9 ATM Interchange Compliance Token

This token is used to carry reference data from other MasterCard networks.

On ATM responses received from EPS-Net, the following fields are populated:

- TRACE-ID field in this token is populated with the first 9 bytes of the reference number from data element 063 (Banknet Reference Data) and the value in DE-15 (Settlement Date). This field forms the compliance reference number for the INET Settlement Record.
- VALID-CDE field is not populated by the interface.
- MONITORING-STAT field in this token is populated with the value in data element 048 (Additional Data) PDS 088.
- ERR-IND field in this token is populated with the value in data element 048 (Additional Data) PDS 089.

5.11.10 POS 5.0 Token

This token is used to pass assorted data items specific to POS processing. The data items used by the EPS-Net Interchange Interface are listed below.

- Error Flag
- Card Verify Flag
- Complete Track Data Flag

5.11.11 Alternate Merchant ID Token

In the POS Retailer Definition File (PRDF), it is possible to configure an alternate merchant identifier by card type. Therefore, for card types routed to an interchange for authorisation, the identifier of the merchant as recognised by the network can be specified. This value is passed through the BASE24 system in the Alternate Merchant ID token.

The value in the token is used (instead of the BASE24 Retailer ID) to populate data element 042 (Card Acceptor ID) in requests sent to EPS-Net.

5.11.12 Interchange Compliance Token

This token is used to carry reference data from other MasterCard networks.

On POS responses received from EPS-Net, the following fields are populated:

- TRACE-ID field in this token is populated with the first 9 bytes of the reference number from data element 063 (Banknet Reference Data) and the value in DE-15 (Settlement Date). This field forms the compliance reference number for the INET Settlement Record.
- VALID-CDE field is not populated by the interface.
- MONITORING-STAT field in this token is populated with the value in data element 048 (Additional Data) PDS 088.
- ERR-IND field in this token is populated with the value in data element 048 (Additional Data) PDS 089.

5.11.13 POS 5.1 Token

This token is used to pass assorted data items specific to POS processing. The data items used by the EPS-Net Interchange Interface are listed below.

- CVC2 Data
- MOTO/e-commerce Flag
- Authentication Collection Indicator

- CAVV/AAV Result Code

5.11.14 Purchase Token

This token is used to carry data for Fleet Card transactions.

On requests sent to EPS-Net, the ID-NUM and VEHICLE-NUM fields in this token (if present) is used to populate data element 048 (Additional Data) PDS 098 and 099 (respectively) if the card used is a MasterCard Fleet card.

On invalid transaction responses (Response Code '12') sent to EPS-Net, the DCLN-RSN-CDE field in this token (if present) is used to populate data element 044 (Additional Response Data).

On requests received from EPS-Net, the ID-NUM and VEHICLE-NUM fields in this token is populated from data element 048 (Additional Data) PDS 098 and PDS 099 (respectively).

On invalid transaction responses (Response Code '12') received from EPS-Net, the DCLN-RSN-CDE field in this token is populated from data element 044 (Additional Response Data).

5.11.15 Point of Service Data Token

This token is used to indicate the local conditions at the point-of-service device, at the time a card payment transaction was initiated.

On requests sent to EPS-Net, this token (if present) is used to populate data element 061 (POS Data).

On requests received from EPS-Net, this token is populated from data element 061 (POS Data).

5.11.16 Authentication Data Token

This token is used to carry authentication data for e-commerce transactions. At present, it is only used to carry the Universal Cardholder Authentication Field (UCAF). UCAF data is indicated by setting the AUTHN-IND-FLG to "01".

5.11.17 IAVS Data Token

This token is used to carry AVS data for POS transactions, when the address data is too big for the appropriate PSTM field (more than 20 characters).

5.11.18 POS Data1 Token

This token is used to carry data identifying the authoriser of the transaction.

On responses received from MasterCard Europe, the following fields are populated:

- RESP-SRC-RSN-CDE field in this token is populated based on the value in data element 100 (Receiving Institution Identification Code).

If the field is not present, then it will be assumed that the issuer authorised the transaction, and the RESP-SRC-RSN-CDE field is set to '5' (response provided by issuer).

If the field is present, and the sixth digit (application number) of the code is set to '3' (alternate processing application), then the RESP-SRC-RSN-CDE field is set to 'A' (third party agent).

If the field is present, and the sixth digit (application number) of the code is not set to '3', then the RESP-SRC-RSN-CDE field is set to '4' (issuer is not available for processing).

- CRD-VRFY-FLG2 field in this token is populated with the value in data element 048 (Additional Data) PDS 087.

5.11.19 POS Merchant Token

This token is used to hold the MasterCard Electronic Acceptance Indicator when an authorisation request containing data element 048 (Additional Data) PDS 076 is received from EPS-Net.

5.12 RESPONSE MESSAGE PROCESSING

This section describes the processing performed by the EPS-Net Interchange Interface on response messages.

5.12.1 Response Code Translation

When sending a response message to EPS-Net, BASE24 maps the BASE24 internal response code to an ISO response code supported by the MasterCard Europe network. When receiving a response message from EPS-Net, BASE24 maps the ISO response code used by the MasterCard Europe network to a BASE24 internal response code.

For a complete description of response code mappings performed by the EPS-Net Interchange Interface, refer to Appendix A.

5.12.2 Card Verification Results

When sending a response message to EPS-Net, BASE24 includes the result of CVC1, CVC2 or CVC3 verification in DE 048 PDS 87. For card-read transactions, BASE24 maps the CVC1 or CVC3 result code from the CRD-VRFY-FLG field in the PS51 token, if CVC1 or CVC3 verification was not successful. If CVC1 or CVC3 verification was successful, and CVC2 was present in the request message, then BASE24 maps the CVC2 result code from the CRD-VRFY-FLG2 field in the POS Data1 token. The CVC1 or CVC3 result code is included only on MasterCard (including Debit MasterCard) transactions, as specified in the *V5 Interface Specifications*. For manually-entered transactions, BASE24 maps the CVC2 result code from the CRD-VRFY-FLG field in the PS51 token.

When receiving a response message from EPS-Net, BASE24 maps the result of CVC1, CVC3 and/or CVC2 verification from DE 048 PDS 87 to the CRD-VRFY-FLG field in the PS51 token. If CVC2 was present in the request message, then BASE24 also maps the CVC2 result code to the CRD-VRFY-FLG2 field in the POS Data1 token.

5.12.3 Address Verification Results

The result of the AVS data validation is returned to the acquirer in DE 048 PDS 83 (AVS Response) in an external response message.

When sending a response message to EPS-Net, BASE24 maps DE 048 PDS 83 from the AVS result code field in the PSTM. When receiving a response message from EPS-Net, BASE24 maps DE 048 PDS 83 to the AVS result code field in the PSTM.

5.12.4 Recurring Payments

A card recurring payment arises following an arrangement between a cardholder and a merchant or service provider in which the cardholder pre-authorises billing against his/her account at predetermined or variable intervals. The billing amount may be the same value each time, or it may fluctuate.

As part of the process of improving the acceptance of card recurring payments for members, MasterCard Europe allows issuers to provide more detailed information to acquirers and merchants about the reasons for declining a recurring payment authorisation request. The objective is to help direct merchants as to the action to take following a declined recurring payment request.

If a recurring payment is rejected, the issuer may return a Merchant Advice Code in DE 048 PDS 84.

5.12.4.1 Acquirer Processing

The Merchant Advice Code is used to set the ERR-FLG field in the PS50 token as detailed below:

Merchant Advice Code (PDS 84)	ERR-FLG
01	1
02	2
03	3

5.12.4.2 Issuer Processing

The Merchant Advice Code is included only on MasterCard (including Debit MasterCard) transactions, as specified in the *V5 Interface Specifications*. The Merchant Advice Code is set from the ERR-FLG field in the PS50 token as detailed below:

ERR-FLG	Merchant Advice Code (PDS 84)
1	01
2	02
3	03

5.12.5 Account Balances

ATM issuer processing in the EPS-Net Interchange Interface allows an available balance and, optionally, a ledger balance to be returned to MasterCard acquirers in DE 054 in responses messages.

The EPS-Net Interchange Interface returns one or more balance under the following conditions:

- The LCONF parameter BAL-ENQ-SUPPORT contains a non-zero value;
- The transaction is an ATM cash withdrawal or an ATM balance inquiry;
- The transaction is approved;
- An available balance is included in the response message received from the ATM Authorisation process.

If a ledger balance is returned, then it will always be the second balance included in DE 054.

ATM acquirer processing in the EPS-Net Interchange Interface allows balances to be received from MasterCard issuers in DE 054 in responses messages.

5.13 ADVICE MESSAGE PROCESSING

This section describes the processing performed by the EPS-Net Interchange Interface on advice messages.

5.13.1 Acquirer Support

The Host-EM Interface supports acquirer-generated POS Advice messages sent to issuers, with whom the acquirer has concluded a 'bilateral' agreement.

To support the sending of 0120 advice messages, the configuration requirements depend on the release level of the BASE24 system:

- For Release 6.0 systems, ensure that 'Y' is configured in the appropriate Issuer Processing Code File (IPCF) Completion Required to Host field(s).
- For systems prior to Release 6.0, ensure that 'Y' is configured in the appropriate Institution Definition File (IDF) Completion Required field.

5.13.2 Issuer Support

The Host-EM Interface supports acquirer-generated POS Advice messages sent to issuers, with whom the acquirer has concluded a 'bilateral' agreement, and also advices generated by EPS-Net.

An EPS-Net Interchange Interface on an issuer system will accept 0120 advice messages from EPS-Net, and convert them to the BASE24 internal message format before forwarding them to the ATM Authorisation process or the POS Router/Authorisation process.

When processing an advice message received from EPS-Net the interface may forward the advice to the BASE24-atm Authorisation process or the BASE24-pos Router/Authorisation process, even if the transaction is not approved. The EPS-Net Interchange Interface can also forward an advice to BASE24 if a request message is declined by the interface itself, e.g. if the acquirer (or the merchant) does not participate in the MasterCard Electronic program. The LCONF parameter FRWD-DCLN-ADVC allows the issuer to configure whether declined advices should be forwarded to BASE24 or logged to the ILF and dropped. Transaction messages rejected because of formatting errors will not be forwarded to the BASE24-atm Authorisation process or the BASE24-pos Router/Authorisation process, regardless of the setting of the LCONF parameter.

5.14 REVERSAL MESSAGE PROCESSING

This section describes the processing performed by the EPS-Net Interchange Interface on reversal messages.

5.14.1 Acquirer Support

The Host-EM Interface supports acquirer-generated reversal advice messages sent to issuers.

The EPS-Net Interchange Interface sends 0420 reversal advice messages to MasterCard Europe under the following circumstances:

- An internal reversal message is received from BASE24-atm or BASE24-pos.
- No response to a 0100 request message is received from EPS-Net before the outbound requests timer expires.

5.14.2 Issuer Support

The Host-EM Interface supports acquirer-generated reversal messages sent to issuers, and also reversals generated by EPS-Net.

An EPS-Net Interchange Interface on an issuer system accepts 0420 reversal advice messages from EPS-Net, and locates the original transaction message in the ILF. The EPS-Net Interchange Interface converts the reversal advice to the BASE24 internal message format before forwarding the message to the ATM Authorisation process or the POS Router/Authorisation process.

When processing a reversal message received from EPS-Net, the interface may forward the reversal to the BASE24-atm Authorisation process or the BASE24-pos Router/Authorisation process, even if a matching request/response message cannot be found. The LCONF parameter FRWD-UNMATCHED-RVSLs allows the issuer to configure whether unmatched reversals should be forwarded to BASE24 or logged to the ILF and dropped. This functionality is not available for PIN Change or PIN Unblock reversal messages or for UKDM adjustment transactions.

5.14.3 Reversal Code Translation

When sending a reversal message to EPS-Net, BASE24 maps the BASE24 reversal code to an ISO response code supported by the MasterCard Europe network in a reversal message. When receiving a reversal message from EPS-Net, BASE24 maps the ISO response code used by the MasterCard Europe network in the reversal message to a BASE24 internal reversal code.

For a complete description of reversal code mappings performed by the EPS-Net Interchange Interface, refer to Appendix A.

5.15 UK DOMESTIC MAESTRO (UKDM) SUPPORT

The BASE24 EPS-Net UKDM module is an additional licensable module, built on top of the Switch 6.0 version of the EPS-Net Interchange Interface.

The source files of the standard EPS-Net Interchange Interface include a number of 'hooks' or user exits to the UKDM module. These provide an interface between the standard EPS-Net Interchange Interface and the EPS-Net UKDM module. The EPS-Net UKDM module is bound together with the standard EPS-Net Interchange Interface code to provide support for UK Domestic Maestro transactions.

The UK Domestic Maestro card scheme was formerly known as the SWITCH debit card scheme. The Release 6.0 EPS-Net Interchange Interface has replaced the SWITCH Authorisation Network Interface (SWANI) as the mechanism for routing POS transactions from the acquirer to the issuer. The enhanced processing in the EPS-Net Interchange Interface required to support UK Domestic Maestro transactions is summarised below.

5.15.1 Identification of UK Domestic Maestro Transactions

The PIN Block Format Code (DE 048 sub-field 08) is not present in UK Domestic Maestro logon messages. Also, these message types include the Forwarding Institution Identification Code (DE 033), and institutions have been assigned different values for UK Domestic Maestro support. Because different types of logon message cannot be supported within a single EPS-Net Interchange Interface process, this effectively means that UK Domestic Maestro transactions must be routed via a different interface process to all other types of EPS-Net transaction. The PROCESSING MODE field on Screen 3 of the Interchange Configuration File (ICF) is used to identify a process supporting UK Domestic Maestro transactions.

5.15.2 Backwards-Compatibility with Release 5.x

Some BASE24 customers who are also SWITCH members still run Release 5.x of BASE24-pos. This means that the changes applied to the EPS-Net Interchange Interface must be compatible with Release 5.x BASE24 systems, as well as Release 6.0 BASE24 systems.

In Release 5.1.2 and Release 5.3, the SWITCH Data Token is used to carry information within BASE24. This token has been retired in Release 6.0 (the corresponding data is carried in other tokens). To maintain backwards-compatibility, the token definition is included with the other data definitions built for the EPS-Net Interchange Interface, and the token identifier ('XH') is also defined locally.

Full support for EMV was not introduced into BASE24 until Release 5.3. Therefore, the EMV processing performed by the Release 6.0 EPS-Net Interchange Interface for UKDM transactions is not backwards-compatible with the ICC processing performed in Release 5.1 systems.

5.15.3 Message Format Differences

There are a number of differences in the message formats used for UK Domestic Maestro transactions and those used by all other transactions (known as the International Message Format). These differences are described fully in the *Host-EM Programmer Specification for UK Domestic Maestro*.

5.15.4 Secure Telephone Order Transactions

MasterCard Europe specifies that telephone order transactions performed on debit cards are allowed, but only if the merchant supports UCAF. However, this requirement does not apply to UK Domestic Maestro cards. Processing in the EPS-Net Interchange Interface ensures that that telephone order transactions can be performed on UKDM cards, regardless of whether UCAF is supported by the merchant.

5.15.5 Adjustment Transaction Processing

The SWITCH debit card scheme has always supported Adjustment transactions in authorisation messages. In the EPS-Net Interchange Interface, this type of transaction is not supported in the International Message Format. Instead, Adjustment transactions acquired by BASE24-pos are either converted to a reversal message or not sent to EPS-Net at all. Support for Adjustments has been added to the EPS-Net message format for UK Domestic Maestro cards (in advice messages only), ensuring that the EPS-Net Interchange Interface performs appropriate message processing for Adjustment transactions. Adjustment transactions received from BASE24-pos are converted to reversal messages before being sent to EPS-Net (as with other EPS-Net transactions), and Adjustment transactions received from EPS-Net are also converted to reversal messages before being forwarded to BASE24-pos.

5.15.6 Response/Reversal Code Values

The SWITCH debit card scheme supports a smaller set of ISO response codes than MasterCard Europe. However, the SWITCH debit card scheme supports an additional reversal code ('00') to those supported by MasterCard Europe. In the EPS-Net message format, the response code is passed in DE 039 in 0110 messages, and the reversal code is passed in DE 039 in 0420/0421 messages.

When a UK Domestic Maestro transaction is not approved, DE 039 returned to EPS-Net must contain one of the permitted response codes. Similarly, a value of '00' in DE 039 is valid when a UK Domestic Maestro transaction is reversed. Processing in the EPS-Net Interchange Interface ensures that valid response/reversals codes are used for UK Domestic Maestro transactions.

5.15.7 CVC2/AVS Support

The SWITCH debit card scheme supports the UK Card Not Present (CNP) initiative. However, MasterCard Europe specifies that CVC2/AVS data can be included in MasterCard transactions only. Processing in the EPS-Net Interchange Interface ensures that CVC2/AVS data is included in UK Domestic Maestro messages sent to EPS-Net.

5.15.8 Support for SWITCH-Specific Data

MasterCard Europe has defined DE 112 to carry data specific to UK Domestic Maestro transactions. In the SWITCH Inter-Member message, this data is carried in field P-17 and field P-48. Within BASE24-pos, this data is carried in token fields. Processing in the EPS-Net Interchange Interface ensures that the SWITCH-specific data is mapped from/to the appropriate token fields.

In addition, Track-2 Data (DE 035) must be included in authorisation request and authorisation advice messages, even when the card details have been manually-entered at the point-of-sale. This means that the Track-2 Data must be constructed for manually-entered transactions, as described in the *Host-EM Programmer Specification for UK Domestic Maestro*. The EPS-Net Interchange Interface ensures that DE 035 (Track-2 Data) is processed appropriately in UK Domestic Maestro transaction messages.

5.15.9 Support for 4-Digit Authorisation Codes

The rules of the SWITCH debit card scheme specify that the approval code for UK Domestic Maestro transactions must be four digits in length. The EPS-Net Interchange Interface ensures that DE 027 (Authorization Identification Response Length) and DE 038 (Authorization Identification Response) are processed appropriately in UK Domestic Maestro transaction messages.

5.15.10 Stand-In Authorisation Processing

The SWITCH debit card scheme has always specified detailed rules about whether/when an acquirer can “stand-in” to authorise transactions on behalf of an issuer, i.e. when an authorisation request cannot be sent to the issuer. No such rules exist for MasterCard Europe. Processing in the EPS-Net Interchange Interface ensures that a UK Domestic Maestro transaction can be authorised within BASE24-pos if the authorisation request cannot be sent to EPS-Net. If the BASE24-pos system approves the transaction using “stand-in” authorisation, then a subsequent advice message is sent to the issuer via the EPS-Net Interchange Interface.

The requirement to support “stand-in” authorisation and to generate an advice message if the transaction is approved, means that XPNET service-based routing cannot be configured for UKDM transactions sent to an EPS-Net Interchange Interface process. If a transaction approved using “stand-in” authorisation is subsequently reversed, then the reversal message must be sent to the same EPS-Net Interchange Interface process that received the advice message. Otherwise, the EPS-Net Interchange Interface process will not be able to locate a corresponding authorisation message in its ILF, and will therefore drop the reversal message as unmatched. Where service-based routing is configured for a set of EPS-Net Interchange Interface processes, it is not possible to ensure that a reversal message is sent to the same process as the corresponding advice message.

6 FILE ACTION MESSAGE HANDLING

The following functions related to file action message processing are described in this section:

- Automated Hot Card Updates
- File Action Message Flows
- File Action Transaction Types
- EPS-Net Request Log File (ENRLF) Processing

6.1 AUTOMATED HOT CARD UPDATES

The BASE24 Automated Hot Card Update function allows the BASE24 From Host Maintenance (FHM) ISO Host Interface (HISO) process to initiate a Card Status Update message to as many as three interchanges, with or without updating a cardholder file on BASE24.

When a Card Status Update Request message is received from the host, the From Host Maintenance ISO Host Interface performs either or both of the following functions:

- The FHM HISO module builds a request message and sends it to the BASE24 From Host Maintenance process, which updates the Cardholder Authorization File (CAF) or Negative Card File (NEG) on BASE24 with information from the Card Status Update Request message.
- The FHM HISO module builds a request message and sends it to the specified interchange interface process, so that the interchange databases can be updated. A single Card Status Update Request message can be forwarded to as many as three interchange interface processes simultaneously.

Issuers can utilise the Automated Hot Card Update facility in order to send File Action request messages to EPS-Net.

The ***BASE24 External Message Manual*** includes the format of FHM messages sent to the HISO Interface. It also specifies the ISO field format for File Action request data, which is forwarded to the EPS-Net Interchange Interface as a PATH Request message.

6.2 FILE ACTION MESSAGE FLOWS

The Automated Hot Card Update facility is utilised in order to allow the EPS-Net Interchange Interface to generate File Action request messages. The process is shown in Figure 6–1.

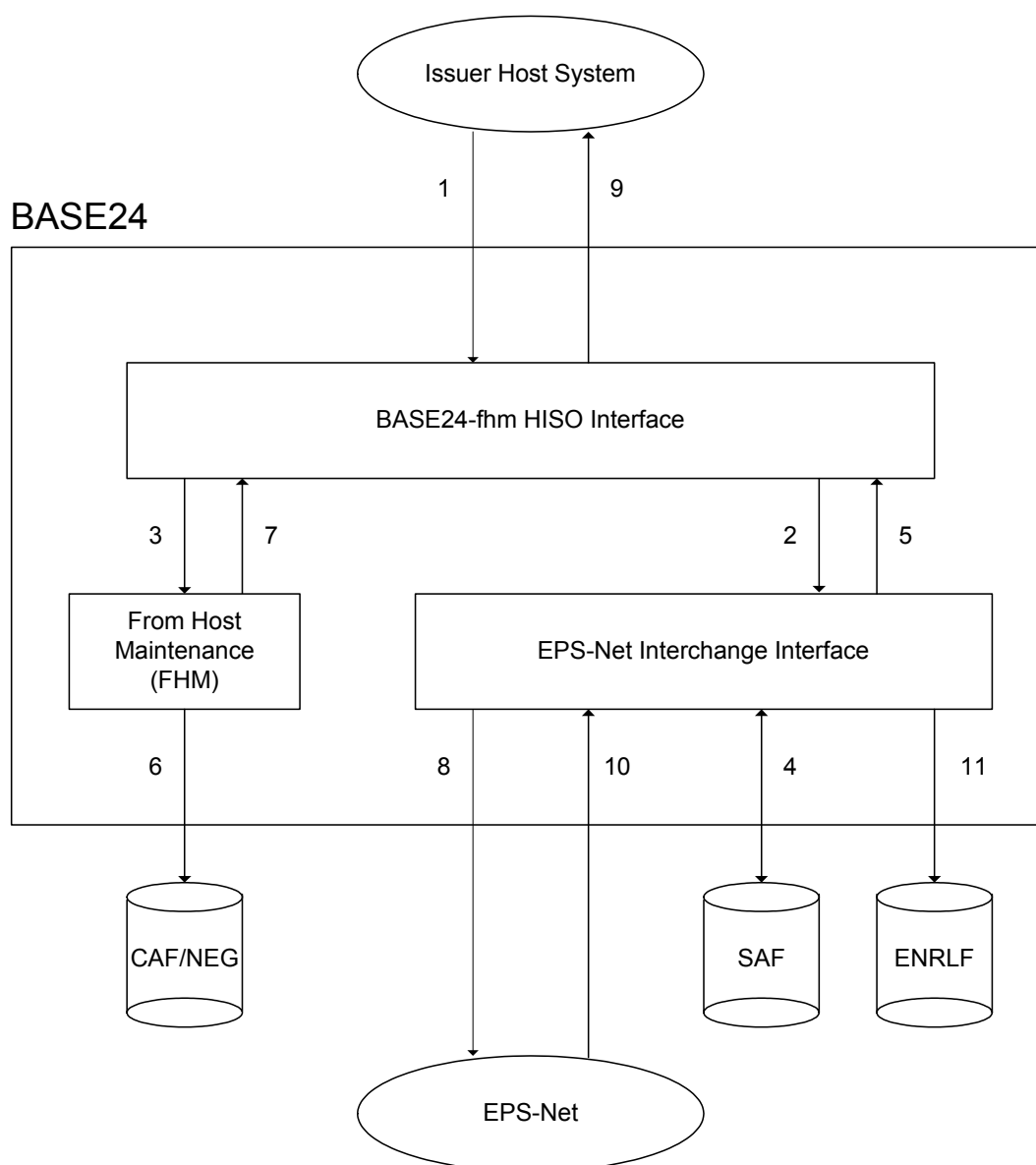


Figure 6–1 File Action Request Message Flow

1. The BASE24 HISO Interface receives an 0300 request message from the Issuer Host System, in the format specified in the BASE24 External Messages Manual.
2. The HISO Interface extracts the data required by the EPS-Net Interchange Interface, and sends a PATH request message to the EPS-Net Interchange Interface.
3. The HISO Interface forwards the 0300 request message to the FHM process, if the BASE24 database must also be updated.
4. The EPS-Net Interchange Interface processes the PATH request message, and adds a 1304 external message to the Store and Forward (SAF) File.
5. The EPS-Net Interchange Interface returns a PATH response message to the HISO Interface.
6. The FHM process updates the CAF or the NEG with the information in the 0300 request message.
7. The FHM process returns an 0310 response message to the HISO Interface.
8. The EPS-Net Interchange Interface retrieves the 1304 request message from the SAF, and sends the message to EPS-Net. Note that the station used to send the 1304 request message must be configured for issuer processing.
9. The HISO Interface returns an 0310 response message to the Issuer Host System, if the BASE24 database was updated, but drops the PATH response message received from the EPS-Net Interchange Interface.
10. The EPS-Net Interchange Interface receives a 1314 response message from EPS-Net.
11. The EPS-Net Interchange Interface logs a record to the EPS-Net File Request Log File (ENRLF).

The structure and content of the File Action Request/Response messages used by the EPS-Net Interchange Interface process to communicate with the MasterCard Europe network is specified in the ***V5 Interface Specifications***.

6.3 FILE ACTION TRANSACTION TYPES

When a File Action request message is sent to EPS-Net, the transaction code in the PATH Request message is mapped to a MasterCard Europe function code (DE 024), as shown below.

PATH Request Transaction Code		EPS-Net Function Code	
1	Add a new record	301	Add record
2	Update an existing record	302	Modify record
3	Delete an existing record	303	Delete record

In addition, the negative reason code in the PATH Request message is mapped to a MasterCard Europe message reason code (DE 025), as shown below.

PATH Request Negative Reason Code		EPS-Net Message Reason Code	
L	Lost card	3000	Lost card
S	Stolen card	3001	Stolen card
U	Undelivered card	3002	Undelivered card
X	Counterfeit card	3000	Counterfeit card
F	Fraudulent use of card	3701	Fraudulent use of card
O	Other reasons	3702	Other reasons

The EPS-Net Interchange Interface does not allow issuers to make inquiries of the AMS file.

6.4 EPS-NET REQUEST LOG FILE (ENRLF) PROCESSING

File Action request messages handled by the EPS-Net Interchange Interface are logged to the ENRLF.

The ENRLF screen 1 is used to peruse File Action messages processed by the EPS-Net Interchange Interface, and to delete ENRLF records after the update has been verified. It is not possible to enter data on the ENRLF screen for inclusion in File Action request messages.

A record is written to the Online Maintenance File (OMF) whenever an ENRLF record is read or deleted.

The ENRLF screen 1 is accessed from the Interchange Configuration File (ICF) Menu screen by placing the cursor to the left of the ENRLF menu entry and pressing function key F1. The ICF Menu screen itself is accessed from ICF screen 12 by pressing function key F8 (GOTO EPS-NET MENU).

6.4.1 ENRLF Screen 1

The ENRLF screen 1 (Figure 6–2) displays information from a maximum of 12 File Action messages.

```

BASE24-BASE  ENRLF PERUSAL SCREEN  LLLL      YY/MM/DD  HH:MM  01 OF 01
INTERCHANGE FIID: _____        PROCESS: _____

      FILE REQUEST DATE (YYMMDD): _____

DATE  TIME  TRAN  CARDHOLDER #      MBR  RSN  STAN  RESP
yyymmdd hh:mm xxx  xxxxxxxxxxxxxxxxxxxx  xxx  x  xxxxxxx  xxx
yyymmdd hh:mm xxx  xxxxxxxxxxxxxxxxxxxx  xxx  x  xxxxxxx  xxx
yyymmdd hh:mm xxx  xxxxxxxxxxxxxxxxxxxx  xxx  x  xxxxxxx  xxx
yyymmdd hh:mm xxx  xxxxxxxxxxxxxxxxxxxx  xxx  x  xxxxxxx  xxx
yyymmdd hh:mm xxx  xxxxxxxxxxxxxxxxxxxx  xxx  x  xxxxxxx  xxx
yyymmdd hh:mm xxx  xxxxxxxxxxxxxxxxxxxx  xxx  x  xxxxxxx  xxx
yyymmdd hh:mm xxx  xxxxxxxxxxxxxxxxxxxx  xxx  x  xxxxxxx  xxx
yyymmdd hh:mm xxx  xxxxxxxxxxxxxxxxxxxx  xxx  x  xxxxxxx  xxx
yyymmdd hh:mm xxx  xxxxxxxxxxxxxxxxxxxx  xxx  x  xxxxxxx  xxx
yyymmdd hh:mm xxx  xxxxxxxxxxxxxxxxxxxx  xxx  x  xxxxxxx  xxx
yyymmdd hh:mm xxx  xxxxxxxxxxxxxxxxxxxx  xxx  x  xxxxxxx  xxx
yyymmdd hh:mm xxx  xxxxxxxxxxxxxxxxxxxx  xxx  x  xxxxxxx  xxx

***** BASE24 *****
NEW PAGE:      FILE DESTINATION:      NEW LOGICAL NETWORK ID:
                F12-HELP

```

Figure 6–2 ENRLF Screen 1

INTERCHANGE FIID

4 (A) M

This field is system-protected, and displays the INTERCHANGE FIID from ICF Screen 1.

PROCESS

16 (A) M

This field is system-protected, and displays the PROCESS from ICF Screen 1.

FILE REQUEST DATE (YYMMDD)

6 (N) M

This field identifies the date that the record was added to the ENRLF.

The remaining fields are system-protected, and display the following information from up to 12 ENRLF records:

Screen Field Name	Description/Text Displayed
DATE	Date the record was added to the ENRLF
TIME	Time the record was added to the ENRLF
TRAN	ADD UPD DEL
CARDHOLDER #	Card number
MBR	Member number
RSN	L = Lost card S = Stolen card U = Undelivered card X = Counterfeit card F = Fraudulent use of card O = Other reasons
RESP	The action code (DE 039) returned from EPS-Net

To comply with the requirements of the PCI data security standards, some or all of the CARDHOLDER # may be displayed masked with asterisks (*). For Release 6.0 users, the setting of the MASKING FLAG in the AFT operator's Security File record determines whether the data in this field is masked. If masking is enabled for the AFT operator, then the LCONF parameter AFT-PAN-DIGITS determines which digits are masked.

6.4.2 ENRLF Maintenance

The EPS-Net Interchange Interface provides users with the capability of identifying when the ENRLF is becoming full. The EPS-Net Interchange Interface maintains a counter of records written to the ENRLF. Each time the counter exceeds the value of the LCONF parameter ENRLF-NOTIFY-INTERVAL, the interface resets the counter and calculates how full the file has become. If the percentage of the file used exceeds the value of the LCONF parameter ENRLF-NOTIFY-PERCENTAGE, a warning message is logged.

The EPS-Net Interchange Interface does not provide automated maintenance for the ENRLF, i.e. there is no periodic closing of an old file and opening of a new file, and no deleting of old files. It is the responsibility of the user to ensure that the file does not become full, which can be achieved by deleting ENRLF records via the AFT screen.

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7 CUTOVER AND REPORTING

Within the BASE24 system, interchange cutover processing is a transaction management and reporting function that produces reports for each business day. These reports can be used in conjunction with BASE24-atm and BASE24-pos reports to reconcile the differences between the BASE24 processing day and the EPS-Net processing day.

7.1 SETTLEMENT WITH MASTERCARD EUROPE

Settlement with MasterCard Europe is performed through the MasterCard Global Clearing Management System (GCMS). Settlement data is captured by the Member and transmitted between the MasterCard Europe Member and the local EM using communications lines or magnetic tape. The EM validates the settlement data and then forwards it to GCMS. All transactions received before 00:00 hours (GMT) are settled the next bank working day.

Settlement is performed offline from BASE24 using extracts from transaction files (e.g., the ILF). Settlement data is transmitted to MasterCard Europe in the Integrated Product Message (IPM) Format, which is supported by card management systems such as ACI CMS and is replacing the EPS-Net Common Clearing Format (ECCF).

7.2 INTERCHANGE REPORTS

The BASE24 EPS-Net Interchange Interface can be configured to produce the standard BASE24 Interchange reports.

The EPS-Net Interchange Interface produces two reports of interchange transaction activity:

- SETL01 Clearings Statement Report
- STAT09 Transaction Detail Report

For details of the standard BASE24 Interchange reports, refer to the *BASE24-atm Settlement and Reporting Manual* and the *BASE24-pos Settlement and Reporting Manual*.

7.2.1 SETL01 Report

The Clearings Statement Report for the EPS-Net Interchange Interface reflects the debit, credit, and net position of the MasterCard Europe network to BASE24.

This report contains the totals of all transactions logged to the ILF. The EPS-Net Interchange Interface calculates totals as an acquirer and as an issuer.

7.2.2 STAT09 Report

The Transaction Detail Report for the EPS-Net Interchange Interface provides a complete listing of all interchange activity. The report is based on the transactions logged to the current day's ILF.

This report contains one detail item for each transaction processed by the EPS-Net Interchange Interface during its processing day. The information provided for each transaction includes the cardholder number, transaction type, transaction time, posting date, and all transaction amounts. This report also includes a summary of the transaction detail information contained in the report.

To comply with the requirements of the PCI data security standards, some or all of the cardholder number in the report may be displayed masked with asterisks (*). The setting of the DATA MASK FLAG on Screen 2 of the ICF record determines whether the data in this field is masked. If masking is enabled for the interface process, then the values of the other three report data masking parameters on ICF Screen 2 determine which digits are masked.

7.2.3 Configuration

The support of Interchange reports for the EPS-Net Interchange Interface process is not mandatory. If it is required to produce the standard BASE24 Interchange reports, then they can be defined in the Logical Network Configuration File (LCONF) and produced for the EPS-Net Interchange Interface process.

The LCONF assigns required to specify the settlement report production are:

- RPT
- RPT-ERROR-FILE
- SW-RPT01-PROG-NAME
- SW-RPT09-PROG-NAME
- SW-RPT-OBEY-FILE

The LCONF parameter required for settlement report production is:

- SW-RPT-STARTUP

The LCONF parameters also used in settlement report production are:

- ACCT-NUM-TYP
- DATE-DISPLAY-TYPE

For a detailed explanation of the above assigns and parameters, refer to section 3.2.

7.2.4 Starting Reports

The EPS-Net Interchange Interface is responsible for starting the interface reports. It does so by creating an obey file to start the report programs and then running the obey file automatically at the appropriate time.

Using the time specified in the LCONF param SW-RPT-STARTUP, the EPS-Net Interchange Interface sets a timer to expire when the reports are to be run. When the timer expires, the EPS-Net Interchange Interface creates and runs an obey file containing the commands necessary to run the report programs using Tandem Advanced Command Language (TACL).

Under normal circumstances, the report programs start and stop without operator intervention. However, if the EPS-Net Interchange Interface does not start the programs, they can be started manually using TACL by running the report obey file created by the EPS-Net Interchange Interface.

If the report programsabend or fail, they can be restarted in the same way.

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8 EMV SUPPORT

The BASE24 EPS-Net EMV product module is an additional licensable module which, when bound with the BASE24 EPS-Net ISO Interchange Interface process (or EPS-Net Interchange Interface), complies with the message formats and rules developed in the *V5 Interface Specifications*.

8.1 ARCHITECTURE

The BASE24 EPS-Net EMV extension module is built on top of the Switch 6.0 version of the EPS-Net Interchange Interface.

The source files of the standard EPS-Net Interchange Interface include a number of EMV-specific ‘hooks’ or user exits. These provide an interface between the standard EPS-Net Interchange Interface and the BASE24-atm and BASE24-pos EPS-Net EMV modules. The EPS-Net EMV modules are bound together with the standard EPS-Net Interchange Interface code to provide support for EPS-Net EMV transactions, in addition to non-EMV transactions.

The main function of the BASE24 EPS-Net EMV module is the formatting of internal message token data to and from EMV-specific external message fields and subfields. It also handles the initialization, compression, and expansion of the EMV-related fields in the external message.

8.2 EPS-NET EMV EXTERNAL MESSAGE

The EPS-Net interchange uses data element P-55, Integrated Circuit Card System Related Data, to pass chip information from the acquirer to the issuer. This data element contains many subelements and is supported only for transactions where chip data is successfully read from the chip at a chip-capable terminal. This information is passed only in Authorization Request (0100), Authorization Response (0110), and Authorization Advice (0120) messages. Most of the information in data element P-55 is also contained in the EMV-specific tokens. Although data element P-55 may be present in the external message, it is not logged to the ILF.

8.3 EMV TOKENS

The format of EMV-specific tokens is described in the *BASE24 Tokens Manual*.

8.3.1 EMV Request Data Token

The following table shows the mapping between fields in the EMV Request Data token and the EMV tag values in data element P-55 of the request message. All data elements in this token are required.

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
CRYPTO-INFO-DATA	9F27
TVR	95
ARQC	9F26
AMT-AUTH	9F02
AMT-OTHER	9F03
AIP	82
ATC	9F36
TERM-CNTRY-CDE	9F1A
TRAN-CRNCY-CDE	5F2A
TRAN-DAT	9A
TRAN-TYPE	9C
UNPREDICT-NUM	9F37
ISS-APPL-DATA	9F10

8.3.2 EMV Discretionary Data Token

The following table shows the mapping between fields in the EMV Discretionary Data token and the EMV tag values in data element P-55 of the request message.

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
TERM-SERL-NUM	9F1E
EMV-TERM-CAP	9F33
EMV-TERM-TYPE	9F35
APPL-VER-NUM	9F09
CVM-RSLTS	9F34
DF-NAME	84

8.3.3 EMV Status Token

The following table shows the mapping between fields in the EMV Status token and the interchange message.

Token Field	Value
PT-SRV-ENTRY-MDE	Set to the value 01x if field 22 (Point-of-Service Entry Mode) equals the value 79x, set to the value 02x if field 22 equals the value 80x or 90x; otherwise set to the value of field 22.
TERM-ENTRY-CAP	Set to the value 5 (EMV-capable terminal) if field 61 subfield 11 (POS Card Data Terminal Input Capability Indicator) is equal to the value 5, 8, or 9.
LAST-EMV-STAT	Set to the value 1 if field 22 (Point-of-Service Entry Mode) equals the value 05x, 79x, or 80x; otherwise set to the value 0.
DATA-SUSPECT	Set to the value 0.
APPL-PAN-SEQ-NUM	Set to the value of field 23 (Card Sequence Number), if present; otherwise set to spaces.
DEV-INFO	Set to spaces.
RSN-ONL-CDE	Set to spaces.
ARQC-VRFY	Mapped from field 48 subfield 71 (On-Behalf Services), if present; otherwise set to the value 0.

8.3.4 EMV Response Data Token

The following table shows the mapping between fields in the EMV Response Data token and the EMV tag values in data element P-55 of the response message.

Token Field	EMV Tag Values in Data Element P-55 of the Response Message
ISS-AUTH-DATA	91

8.3.5 EMV Script Data Token

The following table shows the mapping between fields in the EMV Script Data token and the EMV tag values in data element P-55 of the response message.

Token Field	EMV Tag Values in Data Element P-55 of the Response Message
ISS-SCRIPT-DATA	71 or 72

8.4 EMV SUPPORT OPTIONS

Two levels of support for Chip traffic have been defined to ease member implementation: Chip Partial Grade and Chip Full Grade.

8.4.1 Chip Partial Grade

This mode indicates that a transaction was performed using a chip card at a chip-capable terminal. However, chip data is not included in authorisation messages. For partial grade transactions, the following considerations apply:

- Values ‘05’ (ICC), ‘79’ (manual card read, in fallback to Chip reading), and ‘80’ (magnetic stripe, unaltered and read, in fallback to Chip reading) must be supported for digits 1 and 2 of the Point of Service Entry Mode (data element 022).
- Card Sequence Number (data element 023) must be supported when the Application PAN Card Sequence Number data (EMV tag 5F34) is present on the ICC.
- Track-2 Data (data element 035) must be present for chip transactions. The current Track-2 structure defined for non-chip transactions also applies to chip transactions.
- Retrieval Reference Number (data element 037) may contain the Transaction Sequence Counter generated by the ICC terminal.
- Cardholder Verification Method (data element 048, sub-field 20) must contain ‘S’ for transaction messages where the PIN has been verified offline or has a signature or none. This field contains ‘P’ when the PIN has been verified online.
- For transaction requests where the PIN has been verified offline, the following data elements do not need to be present in the message:
 - ⇒ PIN Capture Code (data element 026)
 - ⇒ PIN Data (data element 052)
 - ⇒ Security Information (data element 053)

8.4.2 Chip Full Grade

This mode indicates both that a transaction was performed using a chip card, and that additional chip data is included in authorisation messages. For full grade transactions, the considerations described for Chip Partial Grade apply. In addition:

- Chip Data (data element 055) must be supported.

8.5 BASE24 AS ACQUIRER

Support for *partial grade* and *full grade* options, as described in the **V5 Interface Specifications**, is provided for acquirer processing within the standard EPS-Net Interchange Interface module.

For BASE24 as an acquirer, the Device Handler module formats the internal message. The EPS-Net Interchange Interface receives a PSTM/STM, which may have one or more EMV tokens attached. External message formatting is performed and the BASE24 EPS-Net EMV module completes the formatting of the external message. The extent of this formatting is dependent on the level of support, as determined by the contents of the transaction message. For partial grade messages, the EMV Status token is present in the internal message. For full grade messages, both the EMV Status token and the EMV Request Data token are present in the internal message.

On receiving a response from EPS-Net containing EMV data, the PSTM/STM is formatted, and the BASE24 EPS-Net EMV module completes the formatting of the PSTM/STM, which may include adding applicable tokens (i.e. the EMV Response Data token and possibly the EMV Script Data token).

8.5.1 Fallback Transactions

Chip fallback occurs when both the card and the terminal support EMV technology, and a failure of either the card or the terminal prevents the use of this technology. If appropriate, a secondary, or “fallback” technology, e.g. reading the data on the magnetic stripe, may then be used in order to complete the transaction. When fallback occurs, the first two digits of the POS Entry Mode (data element 022) are set to ‘80’ (fallback to magnetic stripe) or ‘79’ (fallback to PAN key entry).

8.5.1.1 Fallback Allowed

MasterCard Europe does not permit fallback for ATM transactions (both credit and debit), PIN based in-branch transactions (i.e. cash withdrawals from a bank branch terminal), and CAT Level 1 and CAT Level 2 transactions. If fallback occurs, the acquirer should decline these types of transaction.

However, MasterCard Europe has waived this rule for some countries/members. Therefore, the EPS-Net Interchange Interface supports an LCONF parameter that indicates whether fallback is permitted for ATM transactions (both credit and debit), PIN based in-branch transactions (i.e. cash withdrawals from a bank branch terminal), and CAT Level 1 and CAT Level 2 transactions. Acquirer processing checks the FALLBACK-ALWD parameter, and uses the value to determine whether the authorisation request should be declined or forwarded to the issuer.

8.5.1.2 EMV Certification

A MasterCard transaction can be identified as “fallback”, only if the terminal has been certified for EMV processing by MasterCard. The Switch Terminal File (STF) allows the EPS-Net Interchange Interface to identify whether a specific terminal has been certified for EMV processing by MasterCard.

When processing a BASE24-acquired non-EMV POS transaction containing the EMV Status Token (indicating that the terminal is EMV-capable), the EPS-Net Interchange Interface will use the STF memory table to determine whether the transaction is “fallback”. If an STF record is located, and the record contents indicate that the terminal has not been certified for EMV processing by MasterCard, then the transaction will not be identified as “fallback”, i.e. the POS Entry Mode (DE 022) will not be set to a value of ‘80x’ (card read) or ‘79x’ (key entered) in the message sent to EPS-Net.

When processing a BASE24-acquired non-EMV ATM transaction containing the EMV Status Token (indicating that the terminal is EMV-capable), the EPS-Net Interchange Interface will use the STF memory table to determine whether the transaction is “fallback”. If an STF record is located, and the record contents indicate that the terminal has not been certified for EMV processing by MasterCard, then the transaction will not be identified as “fallback”.

8.6 BASE24 AS ISSUER

For issuer processing, chip card issuers are mandated to upgrade their authorisation systems to support the receipt and generation of the new chip data elements included in authorisation messages.

Depending on the acquirer, chip issuers can receive requests formatted to Chip Partial Grade or Chip Full Grade. Response messages must be formatted accordingly (i.e., the response will include data element 055 only if it was included in the request). Also, the issuer must not provide a cryptogram in the response if there was no cryptogram in the request.

Where BASE24 is the issuer, the EPS-Net Interchange Interface process receives external messages and formats the PSTM/STM. The EPS-Net EMV module completes the formatting, dependent on the level of acquirer chip support, as determined by the contents of the transaction message.

If the transaction is partial grade, the EPS-Net EMV module appends the EMV Status token to the PSTM/STM for forwarding to the BASE24-atm Authorization process or the BASE24-pos Device Handler/Router/Authorization process.

If the transaction is full grade, the processing described above for Partial Grade messages occurs. In addition, data is added to the EMV Request Data token, which is appended to the PSTM/STM. The PSTM/STM, with the tokens, is sent to the BASE24-atm Authorization process or the BASE24-pos Device Handler/Router/Authorization process.

On receiving a PSTM/STM response containing EMV data, a response to the EPS-Net interchange is formatted. The BASE24 EPS-Net EMV module completes the formatting, which may include extracting data from the EMV Response Data token and (if present) the EMV Script Data token.

8.6.1 Issuer Support Options

An “on-behalf” service provides issuers with the option of requesting MasterCard Europe to validate the value of the Authorisation Request Cryptogram (ARQC), before an authorisation request is forwarded to the issuer. If MasterCard Europe validates the ARQC on behalf of the issuer, then the authorisation request message will contain data element 048, PDS 71. This comprises 2 sub-fields: sub-field 1 indicates the type of on-behalf service (‘02’), and sub-field 2 indicates the result of the verification. The EPS-Net Interchange Interface maps sub-field 2 to the ARQC-VERIFY field in the EMV Status Token, as follows:

PDS 71 Sub-field 2		ARQC-VERIFY	
Value	Description	Value	Description
V	Valid ARQC, valid TVR/CVR	2	ARQC was checked by acquiring system and passed verification
T	Valid ARQC, invalid TVR/CVR	2	ARQC was checked by acquiring system and passed verification
I	Invalid ARQC, status of TVR/CVR unknown	1	ARQC was checked by acquiring system but failed verification
U	Unable to validate	0	ARQC not verified
G	Valid cryptogram but not an ARQC, status of TVR/CVR unknown	0	ARQC not verified

Issuers may notify EPS-Net of any errors encountered with the validation of the Authorisation Request Cryptogram (ARQC). Under certain error conditions, BASE24 will include data element 048, PDS 74 in the authorisation response message. This comprises 2 sub-fields: sub-field 1 indicates the function performed, and sub-field 2 indicates the result of the processing. The EPS-Net Interchange Interface sets sub-field 1 to ‘50’ (issuer chip validation), and sets sub-field 2 (based on the contents the ARQC-VERIFY field in the EMV Status Token and the RESP-CDE field in the PSTM/STM), to one of the following values:

PDS 74 Sub-field 2	
Value	Description
I	Invalid ARQC, status of TVR/CVR unknown
T	Valid ARQC, invalid TVR/CVR
U	Unable to validate

9 EPS-Net IPF REFRESH

The EPS-Net Interchange Prefix File (IPF) Refresh Process constructs a full positive list of all BINs that must be sent through EPS-Net for authorisation. The External Interchange Prefix File (IPFX) is the disk file format in which BASE24 receives input records to the IPF. Tape processing is not supported.

The EPS-Net IPF Refresh module supports full refresh only (i.e. where a new IPF is created). Partial refresh (i.e. where an existing IPF is updated) is not supported.

9.1 PROCESSING OVERVIEW

MasterCard provides Acquirers with the Integrated Product Message (IPM) Member Parameter Extract (MPE) file. The IPM MPE file is used as input to MasterCard's IPM MPE Maintenance Process (Pre-Edit Utility), which runs on a PC and updates an existing database of BINs. This program creates optimised parameter files in a directory called "OneDayFile", including a file called CURRENTPARM.TXT. This file must be transferred as a structured (Type E) file to the HP NonStop (formerly Tandem) system, for the EPS-Net IPF Refresh process to use as input data (IPFX).

The IPFX is a structured, entry-sequenced file containing variable length records, with a maximum record length of 4072 bytes. The first record in the file is an IP0000T1 Keys Table record, and is used to identify a valid IPFX input file. Only IP0040T1 Account Range Table entries within the file will be processed as detail records. MasterCard IP0040T1 Account Range Table records provide the required BIN information.

The EPS-Net IPF Refresh module validates run parameters, renames the current IPF to "OLDIPF", converts the detail records from the Optimised Current Parameters File record format on disk to the BASE24 IPF record format, and writes the BASE24 IPF records to a new IPF. When the refresh is completed, the new IPF is moved into production.

9.2 CONFIGURATION

Configuration information for the EPS-Net IPF Refresh module must be set using standard TACL ASSIGN and TACL PARAM commands. These commands are valid for the current TACL session, and can be entered at a TACL prompt or placed in an obey file.

The assigns are:

- IPFX
- IPF
- TPLT

The params are:

- NUM-OF-PROGRAMS
- PROGRAM-XXX
- SUPPRESS-DUP-MESS
- TERMINAL-CATEGORY

When creating params, the NUM-OF-PROGRAMS param must precede all PROGRAM-XXX params, and must match the number of PROGRAM-XXX params.

For more details of assign and param settings, refer to section 3.11.

9.3 RUNNING IPF REFRESH

The EPS-Net IPF Refresh module is located on subvolume SW60IPEU.

The RUNIPEUR file provided with the EPS-Net IPF Refresh module is an OBEY file containing TACL ASSIGN and TACL PARAM commands. The file also contains a command to purge an obsolete IPF, plus a TACL RUN command to execute the EPS-Net Refresh program (IPEUREF). The RUNIPEUR file can be edited to set the required values of the assigns and params listed above.

The RUNIPEUR obey file contains the following RUN command:

```
RUN IPEUREF / OUT outfile /
```

where *outfile* specifies the device or spooler location used to display informational messages and errors.

9.4 IPF REFRESH REPORT

The EPS-Net IPF Refresh module provides a report that displays informational messages and error messages.

The report layout is shown below.

```
BASE24 EPS-Net IPF Refresh
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TOTAL RECORDS READ FROM IPFX DISK FILE:          aaaaaaaaaa
TOTAL IP0040T1 RECORDS READ FROM IPFX DISK FILE:  bbbbbbbbbb
TOTAL IP0040T1 RECORDS PROCESSED FROM IPFX DISK FILE: cccccccc
TOTAL DUPLICATE RECORDS NOT WRITTEN TO IPF DISK FILE: dddddddd
TOTAL PREFIX RECORDS WRITTEN TO IPF DISK FILE:    eeeeeeeee

IPF REFRESH IS COMPLETED
```

Where: aaaaaaaaaa is the number of records read;

bbbbbbbbb is the number of IP0040T1 records read;

ccccccc is the number of records processed (i.e. this value excludes those records that don't meet the TERMINAL-CATEGORY and/or PROGRAM-XXX requirements);

ddddddd is the number of duplicate records encountered;

eeeeeee is the number of records actually added to the IPF.

9.5 PROCESSING CONSIDERATIONS

Acquirers should configure the IPF Refresh parameters carefully, to ensure that the resulting IPF records contain accurate data.

Terminal Categories and Acceptance Programs, the ranges of which are configurable by the Acquirer, identify a subset of the input file's records. The Acquirer can use these parameters to limit the number of records added to the IPF. For example, a BASE24-pos acquirer may not want to include BINs for ATM-only (e.g. CIRRUS) cards.

The Acquirer can also configure Card Types associated with Acceptance Programs that will be included in the specific IPF records. These provide a means of alternate routing.

Acquirers running both a BASE24-atm and a BASE24-pos system may want to create a separate IPF for ATM and POS transactions. This approach can ensure that all transactions routed to EPS-Net are valid for a specific card program. For example, a BASE24 acquirer may want to reject POS transactions performed on ATM-only (e.g. CIRRUS) cards.

This solution may not be appropriate for institutions using BASE24-pos to acquire debit card cash withdrawals from bank branch terminals. These transactions are equivalent to ATM cash withdrawals, and are therefore valid when performed on ATM-only cards. For these transactions, a separate entry can be configured in the PREGEN file, e.g. by specifying a unique routing group, which would reference the IPF used for ATM transactions. To access the additional entry, the same routing group value must be configured on the PTD/PTDF record for the bank branch terminal.

A RESPONSE CODE TRANSLATION

This section gives a description of the response code and reversal code mappings performed by the EPS-Net Interchange Interface.

A.1 BASE24-pos TO EPS-Net RESPONSE CODE

BASE24-pos RESPONSE CODE		EPS-Net RESPONSE CODE	
000	Approved - with balances	00	Approved
001	Approved - no balances	00	Approved
002	Approved - country club	00	Approved
003	Approved - maybe	08	Honour with ID
004	Approved - with ID	08	Honour with ID
005	Approved - blind	00	Approved
006	Approved - VIP	00	Approved
007	Approved - admin	00	Approved
008	Approved - NNEG Hit OK	00	Approved
009	Approved - commercial	00	Approved
050	Declined	05	Do not honour
051	Expired card	54	Expired card
052	PIN tries exceeded	75	PIN tries exceeded
053	No sharing	05	Do not honour
054	No Atalla box	96	System problem
055	Invalid transaction	12	Invalid transaction
056	No support	57	Invalid tran for cardholder
057	Lost/Stolen card	41	Lost card - pick up
058	Declined card status	05	Do not honour
059	Restricted card	62	Restricted card
060	No accounts	05	Do not honour
061	No PBF	14	Invalid card number
062	PBF update error	96	System problem
063	Invalid Auth type	05	Do not honour
064	Bad Track2	14	Invalid card number
065	Adjustment not allowed	12	Invalid transaction
066	Invalid CCD advance incr	13	Invalid amount

BASE24-pos RESPONSE CODE		EPS-Net RESPONSE CODE	
067	Invalid transaction date	05	Do not honour
068	TLF error	96	System problem
069	Message edit error	96	System problem
070	No IDF	05	Do not honour
071	Invalid Auth destination	96	System problem
072	Card on Nat'l Neg	05	Do not honour
073	Invalid service destination	96	System problem
074	Unable to authorise	05	Do not honour
075	Invalid PAN length	14	Invalid card number
076	Low funds	51	Not sufficient funds
077	Pre-Auth limit reached	65	Over withdrawal freq limit
078	Duplicate transmission	94	Duplicate transmission
079	Refund max reached	05	Do not honour
080	Offline refund max reach	05	Do not honour
081	Max credit/refund reach	05	Do not honour
082	Usage exceeded	65	Over withdrawal freq limit
083	Max refund credit reach	05	Do not honour
084	Cust-selected Neg reason	05	Do not honour
085	Enquiry not allowed	57	Invalid transaction for card
086	Over floor limit	05	Do not honour
087	Max # of refunds reached	05	Do not honour
088	Place Call	01	Refer to Card Issuer
089	CAF status - "09"	05	Do not honour
090	Referral file full	05	Do not honour
091	Neg file problem	96	System problem
092	Advance less than minimum	13	Invalid amount
093	Delinquent	05	Do not honour
094	Over limit table	05	Do not honour
095	Amount over maximum	61	Amount > withdrawal limit
096	PIN required	05	Do not honour
097	Mod 10 check	14	Invalid card number
099	Bad PBF	05	Do not honour
100	Referral; Unable to process	01	Refer to Card Issuer
101	Referral; Issue call	01	Refer to Card Issuer
102	Referral; Call	01	Refer to Card Issuer
103	Referral; NEG file problem	01	Refer to Card Issuer

BASE24-pos RESPONSE CODE		EPS-Net RESPONSE CODE	
104	Referral; CAF file problem	01	Refer to Card Issuer
105	Referral; No support	14	Invalid Card Number
106	Referral; Amount over max	01	Refer to Card Issuer
107	Referral; Over daily limit	61	Amount > withdrawal limit
108	Referral; CAPF not found	01	Refer to Card Issuer
109	Referral; Advance < min	01	Refer to Card Issuer
110	Referral; Number times	65	Over withdrawal freq limit
111	Referral; Delinquent	01	Refer to Card Issuer
112	Referral; Over limit	01	Refer to Card Issuer
113	Referral; Timeout	01	Refer to Card Issuer
115	Referral; PTLF full	01	Refer to Card Issuer
120	UAF file problem	01	Refer to Card Issuer
121	Admin file problem	01	Refer to Card Issuer
122	Admin file problem	01	Refer to Card Issuer
130	ARQC verification failure	01	Refer to Card Issuer
131	CVR check failure	01	Refer to Card Issuer
132	TVR check failure	01	Refer to Card Issuer
133	Reason online code check failure	01	Refer to Card Issuer
134	Fallback transaction	01	Refer to Card Issuer
150	Invalid service code	03	Invalid merchant
200	Error; Account	14	Invalid card number
201	Error; Incorrect PIN	55	PIN incorrect
202	Error; Advance < minimum	13	Invalid amount
203	Error; Need admin card	05	Do not honour
204	Error; Enter less amount	61	Over w/drawal amt limits
205	Error; Invalid advance amt	13	Invalid amount
206	Error; CAF not found	14	Invalid card number
207	Error; Invalid tran date	05	Do not honour
208	Error; Invalid expiry date	05	Do not honour
209	Error; Invalid tran code	05	Do not honour
251	Purchase w/cash back-nsf	51	Not sufficient funds
400	ARQC verification failure	05	Do not honour
401	HSM parameter error	05	Do not honour
402	HSM failure	05	Do not honour
403	KEYI not found	05	Do not honour

BASE24-pos RESPONSE CODE		EPS-Net RESPONSE CODE	
404	ATC check failure	05	Do not honour
405	CVR check failure	05	Do not honour
406	TVR check failure	05	Do not honour
407	Reason online code check failure	05	Do not honour
408	Fallback transaction	05	Do not honour
900	Capture; PIN tries exceeded	38	PIN tries exceeded, pick up
901	Capture; Expired card	33	Expired card, pick up
902	Capture; NEG capture	43	Stolen card, pick up
903	Capture; CAF stolen	43	Stolen card, pick up
904	Capture; Advance < min	04	Pick up card
905	Capture; Num times used	04	Pick up card
906	Capture; Delinquent	04	Pick up card
907	Capture; Over limit tbl	04	Pick up card
908	Capture; Amount > max	04	Pick up card
909	Capture	04	Pick up card
910	Capture; ARQC check failure	04	Pick up card
911	Capture; CVR check failure	04	Pick up card
912	Capture; TVR check failure	04	Pick up card

Any BASE24 codes not listed are translated to 05 (do not honour).

If the ERR-FLG element of the standard POS token indicates security check failure (R or S), the external response code is set to 96.

A.2 EPS-Net TO BASE24-pos RESPONSE CODE

EPS-Net RESPONSE CODE		BASE24-pos RESPONSE CODE	
00	Approved	001	Approved – no balances
01	Refer to card issuer	101	Refer – Issue call
03	Invalid merchant	150	Merchant not on file
04	Pick up	909	Capture
05	Do not honour	050	Declined
08	Honour with identification	004	Approved – with ID
12	Invalid transaction	055	Invalid transaction
13	Invalid amount	205	Invalid advance amount
14	Invalid card number	200	Ineligible account
15	Unable to route at issuer EM	073	Invalid destination
30	Format error	069	Bad message edit
31	Issuer sign-off	053	Sharing not allowed
33	Expired card, pick up	901	Capture; expired card
36	Restricted card, pick up	909	Capture
37	Card acceptor call acquirer security, pick up	903	Capture; CAF stolen
38	Allowable PIN tries exceeded, pick up	900	Capture; PIN tries exceeded
41	Lost card, pick up	902	Capture; NEG Capture
43	Stolen card, pick up	903	Capture; CAF stolen
51	Not sufficient funds	076	Low funds
54	Expired card	051	Expired card
55	Incorrect PIN	201	Error, incorrect PIN
57	Transaction not permitted to cardholder	055	Invalid transaction
58	Transaction not permitted to terminal	055	Invalid transaction
61	Exceeds withdrawal amount limit	204	Over daily limit
62	Restricted card	059	Restricted
64	Original amount incorrect	205	Invalid advance amount
65	Exceeds withdrawal frequency limit	082	Number of times use
66	Card acceptor call acquirer's security department	088	Place call

EPS-Net RESPONSE CODE		BASE24-pos RESPONSE CODE	
67	Card to be picked up at ATM	909	Capture card
75	Allowable number of PIN tries exceeded	052	PIN tries exceeded
76	Wrong PIN, allowable number of PIN tries exceeded	052	PIN tries exceeded
80	Network error	074	Unable to process
81	Foreign network failure	074	Unable to process
82	Timeout at Issuer EM	073	Destination not available
83	Transaction failed	074	Unable to process
86	PIN validation not possible	054	No Atalla box
88	Cryptographic failure	054	No Atalla box
89	Authentication failure	054	No Atalla box
91	Issuer or switch is inoperative	074	Unable to process
92	Unable to route at acquirer EM	073	Invalid destination
94	Duplicate transmission	078	Duplicate transmission
96	System malfunction	074	Unable to process

Any codes not listed are translated to the BASE24 response code 050 (Declined).

A.3 BASE24-atm TO EPS-Net RESPONSE CODE

BASE24-atm RESPONSE CODE		EPS-Net RESPONSE CODE	
00	Approved	00	Approved
01	Approved – no balances	00	Approved
50	Unauthorised usage	62	Restricted card
51	Expired card	54	Expired card, card return
52	Invalid card	14	Invalid card number
53	Incorrect PIN	55	Invalid PIN
54	BASE24 database problem	96	System problem
55	Ineligible transaction	57	Transaction not permitted to cardholder
56	Ineligible account	62	Restricted card
57	Transaction not supported	57	Transaction not permitted to cardholder
58	NSF – no available balance	51	Not sufficient funds
59	NSF – w/ available balance	51	Not sufficient funds
60	No further withdrawals	65	Over withdrawal frequency limit
61	Enter lesser amount	61	Over withdrawal limits
62	PIN tries exceeded	75	PIN tries exceeded
63	Daily w/drawal lmt reached	61	Exceeds withdrawal limits
64	Inv amt – CCD request	13	Invalid amount
65	No statement print info	57	Transaction not permitted to cardholder
66	No statement print info	57	Transaction not permitted to cardholder
67	Invalid cash back amount	57	Transaction not permitted to cardholder
68	External decline	57	Transaction not permitted to cardholder
69	No sharing with issuer	58	Transaction not permitted to terminal
70	System error	96	System problem
71	Contact bank	57	Transaction not permitted to cardholder
72	Destination not available	57	Transaction not permitted to cardholder
73	Routing lookup problem	57	Transaction not permitted to cardholder

BASE24-atm RESPONSE CODE		EPS-Net RESPONSE CODE	
74	Message edit error	57	Transaction not permitted to cardholder
80	Signature check failed	62	Restricted card
81	HSM parameter error	62	Restricted card
82	HSM failure	62	Restricted card
83	KEYI not found	62	Restricted card
84	ATC check failure	62	Restricted card
85	CVR check failure	62	Restricted card
86	TVR check failure	62	Restricted card
87	ARQC verification failure	62	Restricted card
88	Fallback transaction	62	Restricted card

If the BASE24-atm response code is 50, and the ERR-FLG element of the AT-FLG1 token indicates a lost card (2), the external response code is set to 41.

If the BASE24-atm response code is 50, and the ERR-FLG element of the AT-FLG1 token indicates a stolen card (3), the external response code is set to 43.

If the first byte of the response code is 1 (Capture) and bytes 2 and 3 are 51 (Expired Card), the external response code is set to 33.

If the first byte of the response code is 1 (Capture) and bytes 2 and 3 are 62 (PIN Tries Exceeded), the external response code is set to 38.

If the first byte of the response code is 1 (Capture) and bytes 2 and 3 are **not** 51 or 62, the external response code is set to 67.

Any BASE24 codes not listed are translated to 05 (do not honour).

If a PIN Unblock transaction is declined, the external response code is set to 70 (contact card issuer).

If a PIN Change transaction is declined, the external response code is set to 71 (PIN not changed).

If a Balance Inquiry transaction is declined, the external response code is set to 78 (account balance unavailable).

A.4 EPS-Net TO BASE24-atm RESPONSE CODE

EPS-Net RESPONSE CODE		BASE24-atm RESPONSE CODE	
00	Approved	001	Approved w/o balance
01	Refer to card issuer	068	External decline
03	Invalid merchant	068	External decline
04	Pick up	150	Unauthorised usage
05	Do not honour	068	External decline
08	Honour with identification	068	External decline
12	Invalid transaction	055	Invalid transaction
13	Invalid amount	064	Invalid advance amount
14	Invalid card number	056	Ineligible account
15	Unable to route at issuer EM	052	Invalid card
30	Format error	074	Bad message edit
31	Issuer sign-off	073	Routing lookup problem
33	Expired card, pick up	151	Expired card
36	Restricted card, pick up	152	Invalid card
37	Card acceptor call acquirer security, pick up	150	Unauthorised usage
38	Allowable PIN tries exceeded, pick up	162	PIN tries exceeded
41	Lost card, pick up	150	Unauthorised usage
43	Stolen card, pick up	150	Unauthorised usage
51	Not sufficient funds	058	NSF No amount - 3
54	Expired card	051	Expired card
55	Incorrect PIN	053	Invalid PIN
57	Transaction not permitted to cardholder	055	Ineligible transaction
58	Transaction not permitted to terminal	055	Ineligible transaction
61	Exceeds withdrawal amount limit	061	Enter lesser amount
62	Restricted card	052	Invalid card
64	Original amount incorrect	068	External decline
65	Exceeds withdrawal frequency limit	060	No further withdrawal
66	Card acceptor call acquirer's security department	068	External decline

EPS-Net RESPONSE CODE		BASE24-atm RESPONSE CODE	
67	Card to be picked up at ATM	150	Unauthorised usage
70	Contact card issuer	071	Contact bank
71	PIN not changed	071	Contact bank
75	Allowable number of PIN tries exceeded	062	PIN tries exceeded
76	Wrong PIN, allowable number of PIN tries exceeded	062	PIN tries exceeded
77	Issuer does not participate in the service	057	Transaction not supported
78	Account balance unavailable	072	Destination not available
79	Unacceptable PIN – transaction declined. Retry	053	Invalid PIN
80	Network error	068	External decline
81	Foreign network failure	068	External decline
82	Timeout at Issuer EM	072	Destination not available
83	Transaction failed	068	External decline
86	PIN validation not possible	070	System error
88	Cryptographic failure	070	System error
89	Authentication failure	070	System error
91	Issuer or switch is inoperative	072	Destination not available
92	Unable to route at acquirer EM	073	Invalid destination
94	Duplicate transmission	068	Unable to process
96	System malfunction	070	System error

Any codes not listed are translated to the BASE24 response code 068 (external decline).

A.5 BASE24-pos TO EPS-Net REVERSAL CODE

BASE24-pos REVERSAL CODE		EPS-Net REVERSAL CODE	
00	Reason Unknown	96	System Malfunction
01	Timeout	68	Response Received Too Late
02	Command Reject	68	Response Received Too Late
03	Destination Not Available	96	System Malfunction
08	Customer Cancelled	17	Customer Cancellation
10	Hardware Error	96	System Malfunction
12	Original Amount Incorrect	06	Error
14	Suspicious Reversal Override	96	System Malfunction
15	Misdispense Reversal Override	96	System Malfunction
16	Duplicate Transaction	06	Error
17	Reconciliation Error	06	Error
18	Reserved	96	System Malfunction
19	System Error	96	System Malfunction
20	Suspect Reversal	17	Customer Cancellation
21	MAC Failure	06	Error
22	MAC Key Synchronisation Error	06	Error
23	Message Replay Error	06	Error
24	Invalid MAC	06	Error

Any codes not listed are translated to the EPS-Net reversal code 17 (customer cancellation).

A.6 EPS-Net TO BASE24-pos REVERSAL CODE

EPS-Net REVERSAL CODE		BASE24-pos REVERSAL CODE	
06	Error	23	Message Replay Error
17	Customer Cancellation	08	Customer Cancelled
32	Completed Partially	10	Hardware Error
68	Response Received Too Late	01	Timeout
96	System Malfunction	19	System Error

Any codes not listed are translated to the BASE24 reversal code 03 (Destination Not Available).

A.7 BASE24-atm TO EPS-Net REVERSAL CODE

BASE24-atm REVERSAL CODE		EPS-Net REVERSAL CODE	
01	Timeout	68	Response Received Too Late
02	Command Reject	68	Response Received Too Late
03	Destination Not Available	96	System Malfunction
08	Customer Cancelled	17	Customer Cancellation
10	Hardware Error	96	System Malfunction
12	Original Amount Incorrect	06	Error
14	Suspicious Reversal Override	96	System Malfunction
15	Misdispense Reversal Override	96	System Malfunction
16	Duplicate Transaction	06	Error
17	Reconciliation Error	06	Error
18	PLUS Add Cash Withdrawal Or Advance	96	System Malfunction
20	Suspect Reversal	17	Customer Cancellation
21	MAC Failure	06	Error
22	MAC Key Synchronisation Error	06	Error
23	Message Replay Error	06	Error
24	Invalid MAC	06	Error

Any codes not listed are translated to the EPS-Net reversal code 17 (customer cancellation).

For partial reversals, the EPS-Net reversal code is always set to 32 (completed partially).

A.8 EPS-Net TO BASE24-atm REVERSAL CODE

EPS-Net REVERSAL CODE		BASE24-atm REVERSAL CODE	
06	Error	23	Message Replay Error
17	Customer Cancellation	08	Customer Cancelled
32	Completed Partially	10	Hardware Error
68	Response Received Too Late	01	Timeout
96	System Malfunction	10	Hardware Error

Any codes not listed are translated to the BASE24 reversal code 03 (destination not available).

B FILE FORMATS

B.1 INTERCHANGE CONFIGURATION FILE (ICF)

This section gives a description of the fields included in the EPS-Net Interchange Interface specific segment of the ICF.

LEVEL	FIELD NAME AND DESCRIPTION	PICTURE
02	ACQ This field defines whether a station is an ACQUIRER.	PIC X OCCURS 2 TIMES
02	ISS This field defines whether a station is an ISSUER.	PIC X OCCURS 2 TIMES
02	AUTH-ONLY If Y then all transactions sent to MasterCard Europe will be for authorisation only.	PIC X
02	TEST This field indicates whether the EPS-Net Interchange Interface is in test mode. Valid values are: Y = In test mode. N = In production mode (default).	PIC X
02	EDCS-INST This field specifies the EPS-Net Institution identification assigned by MasterCard Europe.	PIC X(11)
02	USER-FLD1	PIC X
02	PROTOCOL This field is no longer used in processing.	TYPE BINARY 16
02	ILF-MAX-TRAN-UPDATE This field specifies the maximum number of transactions to be processed between ILF header record updates.	TYPE BINARY 16
02	EBCDIC This field identifies the data transmission format. Valid values are: Y = Data is transmitted in EBCDIC N = Data is transmitted in ASCII	PIC X(2)

LEVEL	FIELD NAME AND DESCRIPTION	PICTURE
02	VERSION-NUM This field specifies the version number of EPS-Net with which the EPS-Net Interchange Interface process is compatible. For the BASE24 EPS-Net Interface the field is set to 30.	PIC X(2)

B.2 EUROCARD INSTITUTION IDENTIFICATION FILE (ERIIF)

The Eurocard Institution Identification File (ERIIF) contains one record for every Terminal FIID processed by the EPS-Net Interchange Interface. Each record in the file contains an Acquiring Institution Identification Number (AID). The AID is used only for transactions acquired by the EPS-Net Interchange Interface.

Record size 34 bytes

Record structure Key-sequenced

Product usage BASE24-atm, BASE24-pos

Process usage BASE24 EPS-Net Interchange Interface

Standard location \$DATA.PRO1DATA

LCONF assign ERIIF

LEVEL	FIELD NAME AND DESCRIPTION	PICTURE
02	FIID This field contains the four-character key identifier for the financial institution.	PIC X(4)
02	ACQ-INST This field contains the AID assigned by MasterCard for the specified financial institution.	
04	NUMBER	PIC X(11)
02	CRNCY-IND This field contains an indicator to allow separate AIDs to be configured for different currencies.	PIC X
02	LAST-FM	
04	FM-TIMESTAMP	TYPE BINARY 48
04	FM-USER-GROUP	TYPE BINARY 16
04	FM-USER-NUMBER	TYPE BINARY 32
04	UPDATE-TYPE	PIC X
04	STA-NO	PIC X(4)
04	FILLER	PIC X

B.3 EPS-NET REQUEST LOG FILE (ENRLF)

The EPS-Net Request Log File (ENRLF) contains one record for every File Action request message processed by the EPS-Net Interchange Interface.

Record size 148 bytes

Record structure Relative

Product usage BASE24-fhm

Process usage BASE24 EPS-Net Interchange Interface

Standard location \$DATA.PRO1DATA

LCONF assign ENRLF

RECORD ENRLF FILE IS "ENRLF" RELATIVE.

The key used to access the file.

02 PRIKEY KEYTAG "PK".

This field contains the FIID of the switch to which the record belongs.

04 FIID PIC X(4).

This field contains the date the record was added to the ENRLF.

04 SWI-DAT PIC X(6).

This field contains the time the record was added to the ENRLF.

04 SWI-TIM PIC X(6).

This field identifies the type of File Action message sent to the switch.
Valid values:

1 = Add a new record

2 = Update an existing record

3 = Delete an existing record.

02 TXN-CDE PIC 9.

This field contains the system trace audit number associated with this transaction and does not change throughout the life of the transaction.

02 STAN PIC 9(6).

This field identifies the reason that the File Action message was sent to the switch. Valid values:

L = Lost card
S = Stolen card
U = Undelivered card
X = Counterfeit card
F = Fraudulent use of card
O = Other reasons.

02 ENTRY-RSN PIC X.

The cardholder number as encoded on Track 2 of the magnetic stripe of the card.

02 PAN PIC X(19).

The sequence number as encoded on Track 2 of the magnetic stripe of the card.

02 MBR-NUM PIC X(3).

The expiration date as encoded on Track-2 of the magnetic stripe of the card.

02 EXP-DAT PIC X(4).

Identifies the institution originating the transaction, as assigned by MasterCard.

02 TXN-ORIG-INST-ID-CDE PIC X(11).

This field contains the response code returned in the 1314 message.

02 RESP-CDE PIC X(3).

This field contains regional-specific data.

02 REGN-DATA OCCURS 7 TIMES.

This field indicates the presence of a purge date for the region.

04 IND PIC X.

This field contains the date a record should be purged for the region.

04 PURGE-DAT PIC X(6).

This field is reserved for future use.

02 USER-FLD1 PIC X(35).

END.

B.4 EPS-NET VERSION OF GENERIC BASE24 SWITCH TOKEN

The following Host-EM message fields are included in the token definition:

- SEM.TRACE-NUM (field DE 011)
- SEM.POS-ENTRY-MDE (field DE 022)
- SEM.FWD-INST-CDE (field DE 033)
- SEM.REF-NUM (field DE 037)
- SEM.RESP-CDE (field DE 039)
- SEM.EXP-DAT (field DE 014)
- SEM.PROC-CDE (field DE 003)

LEVEL	FIELD NAME AND DESCRIPTION	PICTURE
02	FIID The FIID is hard-coded as 'EURO'	PIC X(4)
02	VER-ID The version number of the token	PIC X(2)
02	STAN The System Trace Audit Number	PIC 9(6)
02	POS-ENTRY-MODE The point of sale entry mode	PIC 9(3)
02	FWD-INST-ID-CDE The forwarding institution ID number	PIC X(11)
02	RET-REF-NUM The retrieval reference number	PIC X(12)
02	RESP-CDE The response code	PIC X(2)
02	EXP-DAT The card expiration date	PIC 9(4)
02	PROC-CDE The processing code	PIC 9(6)

C LOG MESSAGES

Log messages are written to the Tandem Event Management System (EMS) logging facility.

The EUROLOGM file on the SW60EURO subvolume documents the log messages output by the EPS-Net Interchange Interface.

This EUROLOGM file documents the message number, severity code and the text of the message. Therefore, the format of the documentation is:

MMMM (X) TEXT

For example:

0490 (0) SAF cycle in progress

For each log message, additional information is given:

- the cause or reason the message is displayed
- the remedy or solution for the message, if relevant

Also, the following support/service personnel may need to be contacted:

- *Operations Personnel*

In-house operations staff who are responsible for initiating fault action for a particular problem.

- *HELP24*

ACI Worldwide personnel who are responsible for solving customer queries and problems.

ACI Worldwide Inc.

D GLOSSARY

AEM	Acquirer European Module
AHS	Acquirer Host System is the name given to a MasterCard Europe member's Data Processing Centre that is operated on behalf of a transaction acquirer.
AMS	Account Management System
AVS	Address Verification Service
BIN	Bank Identification Number
CAT	Cardholder Activated Terminal
CNP	Card Not Present
DE	Data Element
DES	Data Encryption Standard
ECCF	EPS-Net Common Clearing Format
EM	European Module is the name given to the end-points of the MasterCard Europe network. All MasterCard Europe members are connected to EPS-Net through an EM.
EMV	Europay/MasterCard/Visa
EPS-Net	EPS Network is the name given to the communications network over which MasterCard Europe transactions are transmitted to all users connected to the network.
GCMS	Global Clearing Management System
HSM	Hardware Security Module
IEM	Issuer European Module
IPM	Integrated Product Message
IHS	Issuer Host System is the name given to a MasterCard Europe member's Data Processing Centre that authorises transactions or forwards them to another system for authorisation.
ISO	International Organisation for Standardisation

KPE	PIN Block Encryption Key
LMK	Local Master Key
MCC	Merchant Category Code
MFK	Master File Key
MPE	Member Parameter Extract
NMM	Network Management Message
PCI	Payment Card Industry
PDS	Primary Data Sub-element
PIN	Personal Identification Number
SET	Secure Electronic Transaction
SPA	Secure Payment Architecture
UCAF	Universal Cardholder Authentication Field
ZMK	Zone Master Key